

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250278815

Kind Code

A1

Publication Date

September 04, 2025

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MOTION VECTOR REFINEMENT FOR TEMPORALLY AMORTIZED SUPERSAMPLING

Abstract

A residual network is used to predict a set of residual motion vectors that provide additional motion data for portions of the frame for which motion vectors are not provided, such as animated textures, mirrored/reflected objects, and/or moving objects without motion information.

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Appl. No.: 19/188633

Filed: April 24, 2025

Related U.S. Application Data

parent US continuation 17888253 20220815 PENDING child US 19188633

us-provisional-application US 63276168 20211105

Publication Classification

Int. Cl.: G06T3/4076 (20240101); G06T1/20 (20060101); G06T3/18 (20240101); G06T3/4046 (20240101); G06T15/50 (20110101)

U.S. Cl.:

CPC G06T3/4076 (20130101); G06T1/20 (20130101); G06T3/18 (20240101); G06T3/4046 (20130101); G06T15/503 (20130101);

Background/Summary

CROSS-REFERENCE [0001] This application is a continuation of and claims the benefit of and priority to U.S. application Ser. No. 17/888,253, entitled MOTION VECTOR REFINEMENT FOR TEMPORALLY AMORTIZED SUPERSAMPLING, by SungY e Kim, et al., filed Aug. 15, 2022, now allowed, which claims the benefit of and priority to from U.S. Provisional Application No. 63/276,168, filed Nov. 5, 2021, which is hereby incorporated herein by reference.

FIELD

[0002] This disclosure relates generally to graphics anti-aliasing via neural network operations performed via a matrix accelerator of a graphics processing unit.

BACKGROUND OF THE DISCLOSURE

[0003] Temporal Anti-aliasing (TAA) is an anti-aliasing technique in which the renderer jitters the camera every frame to sample different coordinates in screen space. The TAA stage accumulates these samples temporally to produce a supersampled image. The previously accumulated frame is warped using renderer generated velocity/motion vectors to align it with the current frame before accumulation. Although TAA is a widely used technique to generate temporally stable anti-aliased image, the warped sample history can be mismatched to the current pixel due to frame-to-frame changes in visibility and shading or errors in the motion vectors. This typically results in ghosting artifacts around moving object boundary.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The present invention is illustrated by way of example and not limitation in the figures of the accompanying drawings in which like references indicate similar elements, and in which:

[0005] FIG. 1 is a block diagram illustrating a computer system configured to implement one or more aspects of the embodiments described herein;

[0006] FIG. 2A-2D illustrate parallel processor components;

[0007] FIG. 3A-3C are block diagrams of graphics multiprocessors and multiprocessor-based GPUs;

[0008] FIG. 4A-4F illustrate an exemplary architecture in which a plurality of GPUs is communicatively coupled to a plurality of multi-core processors;

[0009] FIG. 5 illustrates a graphics processing pipeline;

[0010] FIG. 6 illustrates a machine learning software stack;

[0011] FIG. 7 illustrates a general-purpose graphics processing unit;

[0012] FIG. 8 illustrates a multi-GPU computing system;

[0013] FIG. 9A-9B illustrate layers of exemplary deep neural networks;

[0014] FIG. 10 illustrates an exemplary recurrent neural network;

[0015] FIG. 11 illustrates training and deployment of a deep neural network;

[0016] FIG. 12A is a block diagram illustrating distributed learning;

[0017] FIG. 12B is a block diagram illustrating a programmable network interface and data processing unit;

[0018] FIG. 13 illustrates an exemplary inferencing system on a chip (SOC) suitable for performing inferencing using a trained model;

[0019] FIG. 14 is a block diagram of a processing system;

[0020] FIG. 15A-15C illustrate computing systems and graphics processors;

[0021] FIG. 16A-16C illustrate block diagrams of additional graphics processor and compute accelerator architectures;

[0022] FIG. **17** is a block diagram of a graphics processing engine of a graphics processor;

[0023] FIG. **18A-18C** illustrate thread execution logic including an array of processing elements employed in a graphics processor core;

[0024] FIG. **19** illustrates a tile of a multi-tile processor, according to an embodiment;

[0025] FIG. **20** is a block diagram illustrating a graphics processor instruction formats;

[0026] FIG. **21** is a block diagram of an additional graphics processor architecture;

[0027] FIG. **22A-22B** illustrate a graphics processor command format and command sequence;

[0028] FIG. **23** illustrates exemplary graphics software architecture for a data processing system;

[0029] FIG. **24A** is a block diagram illustrating an IP core development system;

[0030] FIG. **24B** illustrates a cross-section side view of an integrated circuit package assembly;

[0031] FIG. **24C** illustrates a package assembly that includes multiple units of hardware logic chiplets connected to a substrate (e.g., base die);

[0032] FIG. **24D** illustrates a package assembly including interchangeable chiplets;

[0033] FIG. **25** is a block diagram illustrating an exemplary system on a chip integrated circuit;

[0034] FIG. **26A-26B** are block diagrams illustrating exemplary graphics processors for use within an SoC;

[0035] FIG. **27** is a block diagram of a data processing system, according to an embodiment;

[0036] FIG. **28A-28B** illustrate a matrix operation performed by an instruction pipeline, according to an embodiment;

[0037] FIG. **29** illustrates a systolic array including multiplier and adder circuits organized in a pipelined fashion;

[0038] FIG. **30A-30B** illustrates the use of a systolic array that can be configured to execute operations at an arbitrary systolic depth;

[0039] FIG. **31** illustrates a two-path matrix multiply accelerator in which each path has a depth of four stages;

[0040] FIG. **32** illustrates a four-path matrix multiply accelerator in which each path has a depth of two stages;

[0041] FIG. **33** illustrates a scalable sparse matrix multiply accelerator using systolic arrays with feedback inputs;

[0042] FIG. **34** shows a scalable sparse matrix multiply accelerator using systolic arrays with feedback inputs and outputs on each stage;

[0043] FIG. **35** illustrates a dual pipeline parallel systolic array for a matrix accelerator, according to an embodiment;

[0044] FIG. **36** illustrates a stage pair for a channel of a systolic array;

[0045] FIG. **37** illustrates a systolic array including partial sum loopback and circuitry to accelerate sparse matrix multiply;

[0046] FIG. **38A-38B** illustrate matrix acceleration circuitry including codecs to enable the reading of sparse data in a compressed format;

[0047] FIG. **39** illustrates a conventional renderer with Temporal Anti-aliasing (TAA);

[0048] FIG. **40** illustrates a renderer that replaces the TAA stage with a temporally amortized supersampling stage;

[0049] FIG. **41** illustrate components of the neural network model, according to an embodiment;

[0050] FIG. **42** illustrates the input block of the neural network model, according to an embodiment;

[0051] FIG. **43A-43B** illustrates output block variants for the neural network model, according to embodiments;

[0052] FIG. **44** illustrates a method to perform temporally amortized supersampling;

[0053] FIG. **45** illustrates exemplary rendering performance comparisons for multiple rendering techniques described herein;

[0054] FIG. **46** illustrates data flow for image reconstruction and upscaling, according to an

embodiment;

[0055] FIG. 47 illustrates an extended flow that includes a residual motion refinement network and warp loss optimization, according to an embodiment;

[0056] FIG. 48 illustrates a method of motion vector refinement via a residual motion refinement network, according to an embodiment; and

[0057] FIG. 49 is a block diagram of a computing device including a graphics processor, according to an embodiment.

DETAILED DESCRIPTION

[0058] A graphics processing unit (GPU) is communicatively coupled to host/processor cores to accelerate, for example, graphics operations, machine-learning operations, pattern analysis operations, and/or various general-purpose GPU (GPGPU) functions. The GPU may be communicatively coupled to the host processor/cores over a bus or another interconnect (e.g., a high-speed interconnect such as PCIe or NV Link). Alternatively, the GPU may be integrated on the same package or chip as the cores and communicatively coupled to the cores over an internal processor bus/interconnect (i.e., internal to the package or chip). Regardless of the manner in which the GPU is connected, the processor cores may allocate work to the GPU in the form of sequences of commands/instructions contained in a work descriptor. The GPU then uses dedicated circuitry/logic for efficiently processing these commands/instructions.

[0059] Current parallel graphics data processing includes systems and methods developed to perform specific operations on graphics data such as, for example, linear interpolation, tessellation, rasterization, texture mapping, depth testing, etc. Traditionally, graphics processors used fixed function computational units to process graphics data. However, more recently, portions of graphics processors have been made programmable, enabling such processors to support a wider variety of operations for processing vertex and fragment data.

[0060] To further increase performance, graphics processors typically implement processing techniques such as pipelining that attempt to process, in parallel, as much graphics data as possible throughout the different parts of the graphics pipeline. Parallel graphics processors with single instruction, multiple thread (SIMT) architectures are designed to maximize the amount of parallel processing in the graphics pipeline. In a SIMT architecture, groups of parallel threads attempt to execute program instructions synchronously together as often as possible to increase processing efficiency. A general overview of software and hardware for SIMT architectures can be found in Shane Cook, *CUDA Programming* Chapter 3, pages 37-51 (2013).

[0061] Temporal upsampling can be combined with TAA to upscale spatial resolution at the same time so that frames are rendered at lower spatial resolution to save render time. Post-process stages after the temporal anti-aliasing upsampling can then run at target display resolution. This allows the creation of sharper images than can be created using spatial-only upscaling techniques and effectively reduces render time than when rendering frames at native display resolution. However, such temporal anti-aliasing upsampling quality is much lower than using TAA for native resolution rendered frames. Described herein is a technique to use a mixed low precision convolutional neural network for temporally amortized supersampling to achieve a performance boost from rendering at lower resolution while also generating high quality images.

[0062] Additionally described herein is a technique to generate residual motion vectors for portions of a scene for which motion vectors are not generated by the renderer. Network inputs are typically warped by motion vectors generated analytically for directly visible surfaces only. Motion vectors are not generated for pixels associated with animated textures, mirrored/reflected objects, and/or moving objects without motion information. Embodiments described herein provide a residual motion refinement network that can be trained to generate residual motion vectors based on renderer generated motion vectors and a comparison between a previous and current frame.

[0063] In the following description, numerous specific details are set forth to provide a more thorough understanding. However, it will be apparent to one of skill in the art that the embodiments

described herein may be practiced without one or more of these specific details. In other instances, well-known features have not been described to avoid obscuring the details of the present embodiments.

System Overview

[0064] FIG. 1 is a block diagram illustrating a computing system **100** configured to implement one or more aspects of the embodiments described herein. The computing system **100** includes a processing subsystem **101** having one or more processor(s) **102** and a system memory **104** communicating via an interconnection path that may include a memory hub **105**. The memory hub **105** may be a separate component within a chipset component or may be integrated within the one or more processor(s) **102**. The memory hub **105** couples with an I/O subsystem **111** via a communication link **106**. The I/O subsystem **111** includes an I/O hub **107** that can enable the computing system **100** to receive input from one or more input device(s) **108**. Additionally, the I/O hub **107** can enable a display controller, which may be included in the one or more processor(s) **102**, to provide outputs to one or more display device(s) **110A**. In one embodiment the one or more display device(s) **110A** coupled with the I/O hub **107** can include a local, internal, or embedded display device.

[0065] The processing subsystem **101**, for example, includes one or more parallel processor(s) **112** coupled to memory hub **105** via a bus or other communication link **113**. The communication link **113** may be one of any number of standards-based communication link technologies or protocols, such as, but not limited to PCI Express, or may be a vendor specific communications interface or communications fabric. The one or more parallel processor(s) **112** may form a computationally focused parallel or vector processing system that can include a large number of processing cores and/or processing clusters, such as a many integrated core (MIC) processor. For example, the one or more parallel processor(s) **112** form a graphics processing subsystem that can output pixels to one of the one or more display device(s) **110A** coupled via the I/O hub **107**. The one or more parallel processor(s) **112** can also include a display controller and display interface (not shown) to enable a direct connection to one or more display device(s) **110B**.

[0066] Within the I/O subsystem **111**, a system storage unit **114** can connect to the I/O hub **107** to provide a storage mechanism for the computing system **100**. An I/O switch **116** can be used to provide an interface mechanism to enable connections between the I/O hub **107** and other components, such as a network adapter **118** and/or wireless network adapter **119** that may be integrated into the platform, and various other devices that can be added via one or more add-in device(s) **120**. The add-in device(s) **120** may also include, for example, one or more external graphics processor devices, graphics cards, and/or compute accelerators. The network adapter **118** can be an Ethernet adapter or another wired network adapter. The wireless network adapter **119** can include one or more of a Wi-Fi, Bluetooth, near field communication (NFC), or other network device that includes one or more wireless radios.

[0067] The computing system **100** can include other components not explicitly shown, including USB or other port connections, optical storage drives, video capture devices, and the like, which may also be connected to the I/O hub **107**. Communication paths interconnecting the various components in FIG. 1 may be implemented using any suitable protocols, such as PCI (Peripheral Component Interconnect) based protocols (e.g., PCI-Express), or any other bus or point-to-point communication interfaces and/or protocol(s), such as the NV Link high-speed interconnect, Compute Express Link™ (CXL™) (e.g., CXL.mem), Infinity Fabric (IF), Ethernet (IEEE 802.3), remote direct memory access (RDMA), InfiniBand, Internet Wide Area RDMA Protocol (iWARP), Transmission Control Protocol (TCP), User Datagram Protocol (UDP), quick UDP Internet Connections (QUIC), RDMA over Converged Ethernet (RoCE), Intel QuickPath Interconnect (QPI), Intel Ultra Path Interconnect (UPI), Intel On-Chip System Fabric (IOSF), OmniPath, HyperTransport, Advanced Microcontroller Bus Architecture (AMBA) interconnect, OpenCAPI, Gen-Z, Cache Coherent Interconnect for Accelerators (CCIX), 3GPP Long Term Evolution (LTE)

(4G), 3GPP 5G, and variations thereof, or wired or wireless interconnect protocols known in the art. In some examples, data can be copied or stored to virtualized storage nodes using a protocol such as non-volatile memory express (NVM e) over Fabrics (NVMe-oF) or NVMe.

[0068] The one or more parallel processor(s) **112** may incorporate circuitry optimized for graphics and video processing, including, for example, video output circuitry, and constitutes a graphics processing unit (GPU). Alternatively or additionally, the one or more parallel processor(s) **112** can incorporate circuitry optimized for general purpose processing, while preserving the underlying computational architecture, described in greater detail herein. Components of the computing system **100** may be integrated with one or more other system elements on a single integrated circuit. For example, the one or more parallel processor(s) **112**, memory hub **105**, processor(s) **102**, and I/O hub **107** can be integrated into a system on chip (SoC) integrated circuit. Alternatively, the components of the computing system **100** can be integrated into a single package to form a system in package (SIP) configuration. In one embodiment at least a portion of the components of the computing system **100** can be integrated into a multi-chip module (MCM), which can be interconnected with other multi-chip modules into a modular computing system.

[0069] It will be appreciated that the computing system **100** shown herein is illustrative and that variations and modifications are possible. The connection topology, including the number and arrangement of bridges, the number of processor(s) **102**, and the number of parallel processor(s) **112**, may be modified as desired. For instance, system memory **104** can be connected to the processor(s) **102** directly rather than through a bridge, while other devices communicate with system memory **104** via the memory hub **105** and the processor(s) **102**. In other alternative topologies, the parallel processor(s) **112** are connected to the I/O hub **107** or directly to one of the one or more processor(s) **102**, rather than to the memory hub **105**. In other embodiments, the I/O hub **107** and memory hub **105** may be integrated into a single chip. It is also possible that two or more sets of processor(s) **102** are attached via multiple sockets, which can couple with two or more instances of the parallel processor(s) **112**.

[0070] Some of the particular components shown herein are optional and may not be included in all implementations of the computing system **100**. For example, any number of add-in cards or peripherals may be supported, or some components may be eliminated. Furthermore, some architectures may use different terminology for components similar to those illustrated in FIG. **1**. For example, the memory hub **105** may be referred to as a Northbridge in some architectures, while the I/O hub **107** may be referred to as a Southbridge.

[0071] FIG. **2A** illustrates a parallel processor **200**. The parallel processor **200** may be a GPU, GPGPU or the like as described herein. The various components of the parallel processor **200** may be implemented using one or more integrated circuit devices, such as programmable processors, application specific integrated circuits (ASICs), or field programmable gate arrays (FPGA). The illustrated parallel processor **200** may be one or more of the parallel processor(s) **112** shown in FIG. **1**.

[0072] The parallel processor **200** includes a parallel processing unit **202**. The parallel processing unit includes an I/O unit **204** that enables communication with other devices, including other instances of the parallel processing unit **202**. The I/O unit **204** may be directly connected to other devices. For instance, the I/O unit **204** connects with other devices via the use of a hub or switch interface, such as memory hub **105**. The connections between the memory hub **105** and the I/O unit **204** form a communication link **113**. Within the parallel processing unit **202**, the I/O unit **204** connects with a host interface **206** and a memory crossbar **216**, where the host interface **206** receives commands directed to performing processing operations and the memory crossbar **216** receives commands directed to performing memory operations.

[0073] When the host interface **206** receives a command buffer via the I/O unit **204**, the host interface **206** can direct work operations to perform those commands to a front end **208**. In one embodiment the front end **208** couples with a scheduler **210**, which is configured to distribute

commands or other work items to a processing cluster array **212**. The scheduler **210** ensures that the processing cluster array **212** is properly configured and in a valid state before tasks are distributed to the processing clusters of the processing cluster array **212**. The scheduler **210** may be implemented via firmware logic executing on a microcontroller. The microcontroller implemented scheduler **210** is configurable to perform complex scheduling and work distribution operations at coarse and fine granularity, enabling rapid preemption and context switching of threads executing on the processing cluster array **212**. Preferably, the host software can prove workloads for scheduling on the processing cluster array **212** via one of multiple graphics processing doorbells. In other examples, polling for new workloads or interrupts can be used to identify or indicate availability of work to perform. The workloads can then be automatically distributed across the processing cluster array **212** by the scheduler **210** logic within the scheduler microcontroller.

[0074] The processing cluster array **212** can include up to “N” processing clusters (e.g., cluster **214A**, cluster **214B**, through cluster **214N**). Each cluster **214A-214N** of the processing cluster array **212** can execute a large number of concurrent threads. The scheduler **210** can allocate work to the clusters **214A-214N** of the processing cluster array **212** using various scheduling and/or work distribution algorithms, which may vary depending on the workload arising for each type of program or computation. The scheduling can be handled dynamically by the scheduler **210** or can be assisted in part by compiler logic during compilation of program logic configured for execution by the processing cluster array **212**. Optionally, different clusters **214A-214N** of the processing cluster array **212** can be allocated for processing different types of programs or for performing different types of computations.

[0075] The processing cluster array **212** can be configured to perform various types of parallel processing operations. For example, the processing cluster array **212** is configured to perform general-purpose parallel compute operations. For example, the processing cluster array **212** can include logic to execute processing tasks including filtering of video and/or audio data, performing modeling operations, including physics operations, and performing data transformations.

[0076] The processing cluster array **212** is configured to perform parallel graphics processing operations. In such embodiments in which the parallel processor **200** is configured to perform graphics processing operations, the processing cluster array **212** can include additional logic to support the execution of such graphics processing operations, including, but not limited to texture sampling logic to perform texture operations, as well as tessellation logic and other vertex processing logic. Additionally, the processing cluster array **212** can be configured to execute graphics processing related shader programs such as, but not limited to vertex shaders, tessellation shaders, geometry shaders, and pixel shaders. The parallel processing unit **202** can transfer data from system memory via the I/O unit **204** for processing. During processing the transferred data can be stored to on-chip memory (e.g., parallel processor memory **222**) during processing, then written back to system memory.

[0077] In embodiments in which the parallel processing unit **202** is used to perform graphics processing, the scheduler **210** may be configured to divide the processing workload into approximately equal sized tasks, to better enable distribution of the graphics processing operations to multiple clusters **214A-214N** of the processing cluster array **212**. In some of these embodiments, portions of the processing cluster array **212** can be configured to perform different types of processing. For example, a first portion may be configured to perform vertex shading and topology generation, a second portion may be configured to perform tessellation and geometry shading, and a third portion may be configured to perform pixel shading or other screen space operations, to produce a rendered image for display. Intermediate data produced by one or more of the clusters **214A-214N** may be stored in buffers to allow the intermediate data to be transmitted between clusters **214A-214N** for further processing.

[0078] During operation, the processing cluster array **212** can receive processing tasks to be executed via the scheduler **210**, which receives commands defining processing tasks from front end

208. For graphics processing operations, processing tasks can include indices of data to be processed, e.g., surface (patch) data, primitive data, vertex data, and/or pixel data, as well as state parameters and commands defining how the data is to be processed (e.g., what program is to be executed). The scheduler **210** may be configured to fetch the indices corresponding to the tasks or may receive the indices from the front end **208**. The front end **208** can be configured to ensure the processing cluster array **212** is configured to a valid state before the workload specified by incoming command buffers (e.g., batch-buffers, push buffers, etc.) is initiated.

[0079] Each of the one or more instances of the parallel processing unit **202** can couple with parallel processor memory **222**. The parallel processor memory **222** can be accessed via the memory crossbar **216**, which can receive memory requests from the processing cluster array **212** as well as the I/O unit **204**. The memory crossbar **216** can access the parallel processor memory **222** via a memory interface **218**. The memory interface **218** can include multiple partition units (e.g., partition unit **220A**, partition unit **220B**, through partition unit **220N**) that can each couple to a portion (e.g., memory unit) of parallel processor memory **222**. The number of partition units **220A-220N** may be configured to be equal to the number of memory units, such that a first partition unit **220A** has a corresponding first memory unit **224A**, a second partition unit **220B** has a corresponding second memory unit **224B**, and an Nth partition unit **220N** has a corresponding Nth memory unit **224N**. In other embodiments, the number of partition units **220A-220N** may not be equal to the number of memory devices.

[0080] The memory units **224A-224N** can include various types of memory devices, including dynamic random-access memory (DRAM) or graphics random access memory, such as synchronous graphics random access memory (SGRAM), including graphics double data rate (GDDR) memory. Optionally, the memory units **224A-224N** may also include 3D stacked memory, including but not limited to high bandwidth memory (HBM). Persons skilled in the art will appreciate that the specific implementation of the memory units **224A-224N** can vary and can be selected from one of various conventional designs. Render targets, such as frame buffers or texture maps may be stored across the memory units **224A-224N**, allowing partition units **220A-220N** to write portions of each render target in parallel to efficiently use the available bandwidth of parallel processor memory **222**. In some embodiments, a local instance of the parallel processor memory **222** may be excluded in favor of a unified memory design that utilizes system memory in conjunction with local cache memory.

[0081] Optionally, any one of the clusters **214A-214N** of the processing cluster array **212** has the ability to process data that will be written to any of the memory units **224A-224N** within parallel processor memory **222**. The memory crossbar **216** can be configured to transfer the output of each cluster **214A-214N** to any partition unit **220A-220N** or to another cluster **214A-214N**, which can perform additional processing operations on the output. Each cluster **214A-214N** can communicate with the memory interface **218** through the memory crossbar **216** to read from or write to various external memory devices. In one of the embodiments with the memory crossbar **216** the memory crossbar **216** has a connection to the memory interface **218** to communicate with the I/O unit **204**, as well as a connection to a local instance of the parallel processor memory **222**, enabling the processing units within the different processing clusters **214A-214N** to communicate with system memory or other memory that is not local to the parallel processing unit **202**. Generally, the memory crossbar **216** may, for example, be able to use virtual channels to separate traffic streams between the clusters **214A-214N** and the partition units **220A-220N**.

[0082] While a single instance of the parallel processing unit **202** is illustrated within the parallel processor **200**, any number of instances of the parallel processing unit **202** can be included. For example, multiple instances of the parallel processing unit **202** can be provided on a single add-in card, or multiple add-in cards can be interconnected. For example, the parallel processor **200** can be an add-in device, such as add-in device **120** of FIG. 1, which may be a graphics card such as a discrete graphics card that includes one or more GPUs, one or more memory devices, and device-

to-device or network or fabric interfaces. The different instances of the parallel processing unit **202** can be configured to inter-operate even if the different instances have different numbers of processing cores, different amounts of local parallel processor memory, and/or other configuration differences. Optionally, some instances of the parallel processing unit **202** can include higher precision floating point units relative to other instances. Systems incorporating one or more instances of the parallel processing unit **202** or the parallel processor **200** can be implemented in a variety of configurations and form factors, including but not limited to desktop, laptop, or handheld personal computers, servers, workstations, game consoles, and/or embedded systems. An orchestrator can form composite nodes for workload performance using one or more of: disaggregated processor resources, cache resources, memory resources, storage resources, and networking resources.

[0083] In one embodiment, the parallel processing unit **202** can be partitioned into multiple instances. Those multiple instances can be configured to execute workloads associated with different clients in an isolated manner, enabling a pre-determined quality of service to be provided for each client. For example, each cluster **214A-214N** can be compartmentalized and isolated from other clusters, allowing the processing cluster array **212** to be divided into multiple compute partitions or instances. In such configuration, workloads that execute on an isolated partition are protected from faults or errors associated with a different workload that executes on a different partition. The partition units **220A-220N** can be configured to enable a dedicated and/or isolated path to memory for the clusters **214A-214N** associated with the respective compute partitions. This datapath isolation enables the compute resources within a partition can communicate with one or more assigned memory units **224A-224N** without being subjected to inference by the activities of other partitions.

[0084] FIG. **2B** is a block diagram of a partition unit **220**. The partition unit **220** may be an instance of one of the partition units **220A-220N** of FIG. **2A**. As illustrated, the partition unit **220** includes an L2 cache **221**, a frame buffer interface **225**, and a ROP **226** (raster operations unit). The L2 cache **221** is a read/write cache that is configured to perform load and store operations received from the memory crossbar **216** and ROP **226**. Read misses and urgent write-back requests are output by L2 cache **221** to frame buffer interface **225** for processing. Updates can also be sent to the frame buffer via the frame buffer interface **225** for processing. In one embodiment the frame buffer interface **225** interfaces with one of the memory units in parallel processor memory, such as the memory units **224A-224N** of FIG. **2A** (e.g., within parallel processor memory **222**). The partition unit **220** may additionally or alternatively also interface with one of the memory units in parallel processor memory via a memory controller (not shown).

[0085] In graphics applications, the ROP **226** is a processing unit that performs raster operations such as stencil, z test, blending, and the like. The ROP **226** then outputs processed graphics data that is stored in graphics memory. In some embodiments the ROP **226** includes or couples with a CODEC **227** that includes compression logic to compress depth or color data that is written to memory or the L2 cache **221** and decompress depth or color data that is read from memory or the L2 cache **221**. The compression logic can be lossless compression logic that makes use of one or more of multiple compression algorithms. The type of compression that is performed by the CODEC **227** can vary based on the statistical characteristics of the data to be compressed. For example, in one embodiment, delta color compression is performed on depth and color data on a per-tile basis. In one embodiment the CODEC **227** includes compression and decompression logic that can compress and decompress compute data associated with machine learning operations. The CODEC **227** can, for example, compress sparse matrix data for sparse machine learning operations. The CODEC **227** can also compress sparse matrix data that is encoded in a sparse matrix format (e.g., coordinate list encoding (COO), compressed sparse row (CSR), compressed sparse column (CSC), etc.) to generate compressed and encoded sparse matrix data. The compressed and encoded sparse matrix data can be decompressed and/or decoded before being processed by processing

elements or the processing elements can be configured to consume compressed, encoded, or compressed and encoded data for processing.

[0086] The ROP **226** may be included within each processing cluster (e.g., cluster **214A-214N** of FIG. **2A**) instead of within the partition unit **220**. In such embodiment, read and write requests for pixel data are transmitted over the memory crossbar **216** instead of pixel fragment data. The processed graphics data may be displayed on a display device, such as one of the one or more display device(s) **110A-110B** of FIG. **1**, routed for further processing by the processor(s) **102**, or routed for further processing by one of the processing entities within the parallel processor **200** of FIG. **2A**.

[0087] FIG. **2C** is a block diagram of a processing cluster **214** within a parallel processing unit. For example, the processing cluster is an instance of one of the processing clusters **214A-214N** of FIG. **2A**. The processing cluster **214** can be configured to execute many threads in parallel, where the term “thread” refers to an instance of a particular program executing on a particular set of input data. Optionally, single-instruction, multiple-data (SIMD) instruction issue techniques may be used to support parallel execution of a large number of threads without providing multiple independent instruction units. Alternatively, single-instruction, multiple-thread (SIMT) techniques may be used to support parallel execution of a large number of generally synchronized threads, using a common instruction unit configured to issue instructions to a set of processing engines within each one of the processing clusters. Unlike a SIMD execution regime, where all processing engines typically execute identical instructions, SIMT execution allows different threads to more readily follow divergent execution paths through a given thread program. Persons skilled in the art will understand that a SIMD processing regime represents a functional subset of a SIMT processing regime.

[0088] Operation of the processing cluster **214** can be controlled via a pipeline manager **232** that distributes processing tasks to SIMT parallel processors. The pipeline manager **232** receives instructions from the scheduler **210** of FIG. **2A** and manages execution of those instructions via a graphics multiprocessor **234** and/or a texture unit **236**. The illustrated graphics multiprocessor **234** is an exemplary instance of a SIMT parallel processor. However, various types of SIMT parallel processors of differing architectures may be included within the processing cluster **214**. One or more instances of the graphics multiprocessor **234** can be included within a processing cluster **214**. The graphics multiprocessor **234** can process data and a data crossbar **240** can be used to distribute the processed data to one of multiple possible destinations, including other shader units. The pipeline manager **232** can facilitate the distribution of processed data by specifying destinations for processed data to be distributed via the data crossbar **240**.

[0089] Each graphics multiprocessor **234** within the processing cluster **214** can include an identical set of functional execution logic (e.g., arithmetic logic units, load-store units, etc.). The functional execution logic can be configured in a pipelined manner in which new instructions can be issued before previous instructions are complete. The functional execution logic supports a variety of operations including integer and floating-point arithmetic, comparison operations, Boolean operations, bit-shifting, and computation of various algebraic functions. The same functional-unit hardware could be leveraged to perform different operations and any combination of functional units may be present.

[0090] The instructions transmitted to the processing cluster **214** constitute a thread. A set of threads executing across the set of parallel processing engines is a thread group. A thread group executes the same program on different input data. Each thread within a thread group can be assigned to a different processing engine within a graphics multiprocessor **234**. A thread group may include fewer threads than the number of processing engines within the graphics multiprocessor **234**. When a thread group includes fewer threads than the number of processing engines, one or more of the processing engines may be idle during cycles in which that thread group is being processed. A thread group may also include more threads than the number of processing engines

within the graphics multiprocessor **234**. When the thread group includes more threads than the number of processing engines within the graphics multiprocessor **234**, processing can be performed over consecutive clock cycles. Optionally, multiple thread groups can be executed concurrently on the graphics multiprocessor **234**.

[0091] The graphics multiprocessor **234** may include an internal cache memory to perform load and store operations. Optionally, the graphics multiprocessor **234** can forego an internal cache and use a cache memory (e.g., level 1 (L1) cache **248**) within the processing cluster **214**. Each graphics multiprocessor **234** also has access to level 2 (L2) caches within the partition units (e.g., partition units **220A-220N** of FIG. 2A) that are shared among all processing clusters **214** and may be used to transfer data between threads. The graphics multiprocessor **234** may also access off-chip global memory, which can include one or more of local parallel processor memory and/or system memory. Any memory external to the parallel processing unit **202** may be used as global memory. Embodiments in which the processing cluster **214** includes multiple instances of the graphics multiprocessor **234** can share common instructions and data, which may be stored in the L1 cache **248**.

[0092] Each processing cluster **214** may include an MMU **245** (memory management unit) that is configured to map virtual addresses into physical addresses. In other embodiments, one or more instances of the MMU **245** may reside within the memory interface **218** of FIG. 2A. The MMU **245** includes a set of page table entries (PTEs) used to map a virtual address to a physical address of a tile and optionally a cache line index. The MMU **245** may include address translation lookaside buffers (TLB) or caches that may reside within the graphics multiprocessor **234** or the L1 cache **248** of processing cluster **214**. The physical address is processed to distribute surface data access locality to allow efficient request interleaving among partition units. The cache line index may be used to determine whether a request for a cache line is a hit or miss.

[0093] In graphics and computing applications, a processing cluster **214** may be configured such that each graphics multiprocessor **234** is coupled to a texture unit **236** for performing texture mapping operations, e.g., determining texture sample positions, reading texture data, and filtering the texture data. Texture data is read from an internal texture L1 cache (not shown) or in some embodiments from the L1 cache within graphics multiprocessor **234** and is fetched from an L2 cache, local parallel processor memory, or system memory, as needed. Each graphics multiprocessor **234** outputs processed tasks to the data crossbar **240** to provide the processed task to another processing cluster **214** for further processing or to store the processed task in an L2 cache, local parallel processor memory, or system memory via the memory crossbar **216**. A preROP **242** (pre-raster operations unit) is configured to receive data from graphics multiprocessor **234**, direct data to ROP units, which may be located with partition units as described herein (e.g., partition units **220A-220N** of FIG. 2A). The preROP **242** unit can perform optimizations for color blending, organize pixel color data, and perform address translations.

[0094] It will be appreciated that the core architecture described herein is illustrative and that variations and modifications are possible. Any number of processing units, e.g., graphics multiprocessor **234**, texture units **236**, preROPs **242**, etc., may be included within a processing cluster **214**. Further, while only one processing cluster **214** is shown, a parallel processing unit as described herein may include any number of instances of the processing cluster **214**. Optionally, each processing cluster **214** can be configured to operate independently of other processing clusters **214** using separate and distinct processing units, L1 caches, L2 caches, etc.

[0095] FIG. 2D shows an example of the graphics multiprocessor **234** in which the graphics multiprocessor **234** couples with the pipeline manager **232** of the processing cluster **214**. The graphics multiprocessor **234** has an execution pipeline including but not limited to an instruction cache **252**, an instruction unit **254**, an address mapping unit **256**, a register file **258**, one or more general purpose graphics processing unit (GPGPU) cores **262**, and one or more load/store units **266**. The GPGPU cores **262** and load/store units **266** are coupled with cache memory **272** and

shared memory **270** via a memory and cache interconnect **268**. The graphics multiprocessor **234** may additionally include tensor and/or ray-tracing cores **263** that include hardware logic to accelerate matrix and/or ray-tracing operations.

[0096] The instruction cache **252** may receive a stream of instructions to execute from the pipeline manager **232**. The instructions are cached in the instruction cache **252** and dispatched for execution by the instruction unit **254**. The instruction unit **254** can dispatch instructions as thread groups (e.g., warps), with each thread of the thread group assigned to a different execution unit within GPGPU core **262**. An instruction can access any of a local, shared, or global address space by specifying an address within a unified address space. The address mapping unit **256** can be used to translate addresses in the unified address space into a distinct memory address that can be accessed by the load/store units **266**.

[0097] The register file **258** provides a set of registers for the functional units of the graphics multiprocessor **234**. The register file **258** provides temporary storage for operands connected to the data paths of the functional units (e.g., GPGPU cores **262**, load/store units **266**) of the graphics multiprocessor **234**. The register file **258** may be divided between each of the functional units such that each functional unit is allocated a dedicated portion of the register file **258**. For example, the register file **258** may be divided between the different warps being executed by the graphics multiprocessor **234**.

[0098] The GPGPU cores **262** can each include floating point units (FPUs) and/or integer arithmetic logic units (ALUs) that are used to execute instructions of the graphics multiprocessor **234**. In some implementations, the GPGPU cores **262** can include hardware logic that may otherwise reside within the tensor and/or ray-tracing cores **263**. The GPGPU cores **262** can be similar in architecture or can differ in architecture. For example and in one embodiment, a first portion of the GPGPU cores **262** include a single precision FPU and an integer ALU while a second portion of the GPGPU cores include a double precision FPU. Optionally, the FPUs can implement the IEEE 754-2008 standard for floating point arithmetic or enable variable precision floating point arithmetic. The graphics multiprocessor **234** can additionally include one or more fixed function or special function units to perform specific functions such as copy rectangle or pixel blending operations. One or more of the GPGPU cores can also include fixed or special function logic.

[0099] The GPGPU cores **262** may include SIMD logic capable of performing a single instruction on multiple sets of data. Optionally, GPGPU cores **262** can physically execute SIMD4, SIMD8, and SIMD16 instructions and logically execute SIMD1, SIMD2, and SIMD32 instructions. The SIMD instructions for the GPGPU cores can be generated at compile time by a shader compiler or automatically generated when executing programs written and compiled for single program multiple data (SPMD) or SIMT architectures. Multiple threads of a program configured for the SIMT execution model can be executed via a single SIMD instruction. For example and in one embodiment, eight SIMT threads that perform the same or similar operations can be executed in parallel via a single SIMD8 logic unit.

[0100] The memory and cache interconnect **268** is an interconnect network that connects each of the functional units of the graphics multiprocessor **234** to the register file **258** and to the shared memory **270**. For example, the memory and cache interconnect **268** is a crossbar interconnect that allows the load/store unit **266** to implement load and store operations between the shared memory **270** and the register file **258**. The register file **258** can operate at the same frequency as the GPGPU cores **262**, thus data transfer between the GPGPU cores **262** and the register file **258** is very low latency. The shared memory **270** can be used to enable communication between threads that execute on the functional units within the graphics multiprocessor **234**. The cache memory **272** can be used as a data cache for example, to cache texture data communicated between the functional units and the texture unit **236**. The shared memory **270** can also be used as a program managed cache. The shared memory **270** and the cache memory **272** can couple with the data crossbar **240**

to enable communication with other components of the processing cluster. Threads executing on the GPGPU cores **262** can programmatically store data within the shared memory in addition to the automatically cached data that is stored within the cache memory **272**.

[0101] FIG. **3A-3C** illustrate additional graphics multiprocessors, according to embodiments. FIG. **3A-3B** illustrate graphics multiprocessors **325, 350**, which are related to the graphics multiprocessor **234** of FIG. **2C** and may be used in place of one of those. Therefore, the disclosure of any features in combination with the graphics multiprocessor **234** herein also discloses a corresponding combination with the graphics multiprocessor(s) **325, 350**, but is not limited to such. FIG. **3C** illustrates a graphics processing unit (GPU) **380** which includes dedicated sets of graphics processing resources arranged into multi-core groups **365A-365N**, which correspond to the graphics multiprocessors **325, 350**. The illustrated graphics multiprocessors **325, 350** and the multi-core groups **365A-365N** can be streaming multiprocessors (SM) capable of simultaneous execution of a large number of execution threads.

[0102] The graphics multiprocessor **325** of FIG. **3A** includes multiple additional instances of execution resource units relative to the graphics multiprocessor **234** of FIG. **2D**. For example, the graphics multiprocessor **325** can include multiple instances of the instruction unit **332A-332B**, register file **334A-334B**, and texture unit(s) **344A-344B**. The graphics multiprocessor **325** also includes multiple sets of graphics or compute execution units (e.g., GPGPU core **336A-336B**, tensor core **337A-337B**, ray-tracing core **338A-338B**) and multiple sets of load/store units **340A-340B**. The execution resource units have a common instruction cache **330**, texture and/or data cache memory **342**, and shared memory **346**.

[0103] The various components can communicate via an interconnect fabric **327**. The interconnect fabric **327** may include one or more crossbar switches to enable communication between the various components of the graphics multiprocessor **325**. The interconnect fabric **327** may be a separate, high-speed network fabric layer upon which each component of the graphics multiprocessor **325** is stacked. The components of the graphics multiprocessor **325** communicate with remote components via the interconnect fabric **327**. For example, the cores **336A-336B, 337A-337B**, and **338A-338B** can each communicate with shared memory **346** via the interconnect fabric **327**. The interconnect fabric **327** can arbitrate communication within the graphics multiprocessor **325** to ensure a fair bandwidth allocation between components.

[0104] The graphics multiprocessor **350** of FIG. **3B** includes multiple sets of execution resources **356A-356D**, where each set of execution resource includes multiple instruction units, register files, GPGPU cores, and load store units, as illustrated in FIG. **2D** and FIG. **3A**. The execution resources **356A-356D** can work in concert with texture unit(s) **360A-360D** for texture operations, while sharing an instruction cache **354**, and shared memory **353**. For example, the execution resources **356A-356D** can share an instruction cache **354** and shared memory **353**, as well as multiple instances of a texture and/or data cache memory **358A-358B**. The various components can communicate via an interconnect fabric **352** similar to the interconnect fabric **327** of FIG. **3A**.

[0105] Persons skilled in the art will understand that the architecture described in FIGS. **1, 2A-2D, and 3A-3B** are descriptive and not limiting as to the scope of the present embodiments. Thus, the techniques described herein may be implemented on any properly configured processing unit, including, without limitation, one or more mobile application processors, one or more desktop or server central processing units (CPUs) including multi-core CPUs, one or more parallel processing units, such as the parallel processing unit **202** of FIG. **2A**, as well as one or more graphics processors or special purpose processing units, without departure from the scope of the embodiments described herein.

[0106] The parallel processor or GPGPU as described herein may be communicatively coupled to host/processor cores to accelerate graphics operations, machine-learning operations, pattern analysis operations, and various general-purpose GPU (GPGPU) functions. The GPU may be communicatively coupled to the host processor/cores over a bus or other interconnect (e.g., a high-

speed interconnect such as PCIe, NV Link, or other known protocols, standardized protocols, or proprietary protocols). In other embodiments, the GPU may be integrated on the same package or chip as the cores and communicatively coupled to the cores over an internal processor bus/interconnect (i.e., internal to the package or chip). Regardless of the manner in which the GPU is connected, the processor cores may allocate work to the GPU in the form of sequences of commands/instructions contained in a work descriptor. The GPU then uses dedicated circuitry/logic for efficiently processing these commands/instructions.

[0107] FIG. 3C illustrates a graphics processing unit (GPU) **380** which includes dedicated sets of graphics processing resources arranged into multi-core groups **365A-365N**. While the details of only a single multi-core group **365A** are provided, it will be appreciated that the other multi-core groups **365B-365N** may be equipped with the same or similar sets of graphics processing resources. Details described with respect to the multi-core groups **365A-365N** may also apply to any graphics multiprocessor **234, 325, 350** described herein.

[0108] As illustrated, a multi-core group **365A** may include a set of graphics cores **370**, a set of tensor cores **371**, and a set of ray tracing cores **372**. A scheduler/dispatcher **368** schedules and dispatches the graphics threads for execution on the various cores **370, 371, 372**. A set of register files **369** store operand values used by the cores **370, 371, 372** when executing the graphics threads. These may include, for example, integer registers for storing integer values, floating point registers for storing floating point values, vector registers for storing packed data elements (integer and/or floating-point data elements) and tile registers for storing tensor/matrix values. The tile registers may be implemented as combined sets of vector registers.

[0109] One or more combined level 1 (L1) caches and shared memory units **373** store graphics data such as texture data, vertex data, pixel data, ray data, bounding volume data, etc., locally within each multi-core group **365A**. One or more texture units **374** can also be used to perform texturing operations, such as texture mapping and sampling. A Level 2 (L2) cache **375** shared by all or a subset of the multi-core groups **365A-365N** stores graphics data and/or instructions for multiple concurrent graphics threads. As illustrated, the L2 cache **375** may be shared across a plurality of multi-core groups **365A-365N**. One or more memory controllers **367** couple the GPU **380** to a memory **366** which may be a system memory (e.g., DRAM) and/or a dedicated graphics memory (e.g., GDDR6 memory).

[0110] Input/output (I/O) circuitry **363** couples the GPU **380** to one or more I/O devices **362** such as digital signal processors (DSPs), network controllers, or user input devices. An on-chip interconnect may be used to couple the I/O devices **362** to the GPU **380** and memory **366**. One or more I/O memory management units (IOMMUs) **364** of the I/O circuitry **363** couple the I/O devices **362** directly to the system memory **366**. Optionally, the IOMMU **364** manages multiple sets of page tables to map virtual addresses to physical addresses in system memory **366**. The I/O devices **362**, CPU(s) **361**, and GPU(s) **380** may then share the same virtual address space.

[0111] In one implementation of the IOMMU **364**, the IOM MU **364** supports virtualization. In this case, it may manage a first set of page tables to map guest/graphics virtual addresses to guest/graphics physical addresses and a second set of page tables to map the guest/graphics physical addresses to system/host physical addresses (e.g., within system memory **366**). The base addresses of each of the first and second sets of page tables may be stored in control registers and swapped out on a context switch (e.g., so that the new context is provided with access to the relevant set of page tables). While not illustrated in FIG. 3C, each of the cores **370, 371, 372** and/or multi-core groups **365A-365N** may include translation lookaside buffers (TLBs) to cache guest virtual to guest physical translations, guest physical to host physical translations, and guest virtual to host physical translations.

[0112] The CPU(s) **361**, GPUs **380**, and I/O devices **362** may be integrated on a single semiconductor chip and/or chip package. The illustrated memory **366** may be integrated on the same chip or may be coupled to the memory controllers **367** via an off-chip interface. In one

implementation, the memory **366** comprises GDDR6 memory which shares the same virtual address space as other physical system-level memories, although the underlying principles described herein are not limited to this specific implementation.

[0113] The tensor cores **371** may include a plurality of execution resources specifically designed to perform matrix operations, which are the fundamental compute operation used to perform deep learning operations. For example, simultaneous matrix multiplication operations may be used for neural network training and inferencing. The tensor cores **371** may perform matrix processing using a variety of operand precisions including single precision floating-point (e.g., 32 bits), half-precision floating point (e.g., 16 bits), integer words (16 bits), bytes (8 bits), and half-bytes (4 bits). For example, a neural network implementation extracts features of each rendered scene, potentially combining details from multiple frames, to construct a high-quality final image.

[0114] In deep learning implementations, parallel matrix multiplication work may be scheduled for execution on the tensor cores **371**. The training of neural networks, in particular, requires a significant number of matrix dot product operations. In order to process an inner-product formulation of an $N \times N \times N$ matrix multiply, the tensor cores **371** may include at least N dot-product processing elements. Before the matrix multiply begins, one entire matrix is loaded into tile registers and at least one column of a second matrix is loaded each cycle for N cycles. Each cycle, there are N dot products that are processed.

[0115] Matrix elements may be stored at different precisions depending on the particular implementation, including 16-bit words, 8-bit bytes (e.g., INT8) and 4-bit half-bytes (e.g., INT4). Different precision modes may be specified for the tensor cores **371** to ensure that the most efficient precision is used for different workloads (e.g., such as inferencing workloads which can tolerate quantization to bytes and half-bytes). Supported formats additionally include 64-bit floating point (FP64) and non-IEEE floating point formats such as the bfloat16 format (e.g., Brain floating point), a 16-bit floating point format with one sign bit, eight exponent bits, and eight significand bits, of which seven are explicitly stored. One embodiment includes support for a reduced precision tensor-float (TF32) mode, which performs computations using the range of FP32 (8-bits) and the precision of FP16 (10-bits). Reduced precision TF32 operations can be performed on FP32 inputs and produce FP32 outputs at higher performance relative to FP32 and increased precision relative to FP16. In one embodiment, one or more 8-bit floating point formats (FP8) are supported.

[0116] In one embodiment the tensor cores **371** support a sparse mode of operation for matrices in which the vast majority of values are zero. The tensor cores **371** include support for sparse input matrices that are encoded in a sparse matrix representation (e.g., coordinate list encoding (COO), compressed sparse row (CSR), compressed sparse column (CSC), etc.). The tensor cores **371** also include support for compressed sparse matrix representations in the event that the sparse matrix representation may be further compressed. Compressed, encoded, and/or compressed and encoded matrix data, along with associated compression and/or encoding metadata, can be read by the tensor cores **371** and the non-zero values can be extracted. For example, for a given input matrix A, a non-zero value can be loaded from the compressed and/or encoded representation of at least a portion of matrix A. Based on the location in matrix A for the non-zero value, which may be determined from index or coordinate metadata associated with the non-zero value, a corresponding value in input matrix B may be loaded. Depending on the operation to be performed (e.g., multiply), the load of the value from input matrix B may be bypassed if the corresponding value is a zero value. In one embodiment, the pairings of values for certain operations, such as multiply operations, may be pre-scanned by scheduler logic and only operations between non-zero inputs are scheduled. Depending on the dimensions of matrix A and matrix B and the operation to be performed, output matrix C may be dense or sparse. Where output matrix C is sparse and depending on the configuration of the tensor cores **371**, output matrix C may be output in a compressed format, a sparse encoding, or a compressed sparse encoding.

[0117] The ray tracing cores **372** may accelerate ray tracing operations for both real-time ray tracing and non-real-time ray tracing implementations. In particular, the ray tracing cores **372** may include ray traversal/intersection circuitry for performing ray traversal using bounding volume hierarchies (BVHs) and identifying intersections between rays and primitives enclosed within the BVH volumes. The ray tracing cores **372** may also include circuitry for performing depth testing and culling (e.g., using a Z buffer or similar arrangement). In one implementation, the ray tracing cores **372** perform traversal and intersection operations in concert with the image denoising techniques described herein, at least a portion of which may be executed on the tensor cores **371**. For example, the tensor cores **371** may implement a deep learning neural network to perform denoising of frames generated by the ray tracing cores **372**. However, the CPU(s) **361**, graphics cores **370**, and/or ray tracing cores **372** may also implement all or a portion of the denoising and/or deep learning algorithms.

[0118] In addition, as described above, a distributed approach to denoising may be employed in which the GPU **380** is in a computing device coupled to other computing devices over a network or high-speed interconnect. In this distributed approach, the interconnected computing devices may share neural network learning/training data to improve the speed with which the overall system learns to perform denoising for different types of image frames and/or different graphics applications.

[0119] The ray tracing cores **372** may process all BVH traversal and/or ray-primitive intersections, saving the graphics cores **370** from being overloaded with thousands of instructions per ray. For example, each ray tracing core **372** includes a first set of specialized circuitry for performing bounding box tests (e.g., for traversal operations) and/or a second set of specialized circuitry for performing the ray-triangle intersection tests (e.g., intersecting rays which have been traversed). Thus, for example, the multi-core group **365A** can simply launch a ray probe, and the ray tracing cores **372** independently perform ray traversal and intersection and return hit data (e.g., a hit, no hit, multiple hits, etc.) to the thread context. The other cores **370**, **371** are freed to perform other graphics or compute work while the ray tracing cores **372** perform the traversal and intersection operations.

[0120] Optionally, each ray tracing core **372** may include a traversal unit to perform BVH testing operations and/or an intersection unit which performs ray-primitive intersection tests. The intersection unit generates a “hit,” “no hit,” or “multiple hit” response, which it provides to the appropriate thread. During the traversal and intersection operations, the execution resources of the other cores (e.g., graphics cores **370** and tensor cores **371**) are freed to perform other forms of graphics work.

[0121] In one optional embodiment described below, a hybrid rasterization/ray tracing approach is used in which work is distributed between the graphics cores **370** and ray tracing cores **372**.

[0122] The ray tracing cores **372** (and/or other cores **370**, **371**) may include hardware support for a ray tracing instruction set such as Microsoft's DirectX Ray Tracing (DXR) which includes a DispatchRays command, as well as ray-generation, closest-hit, any-hit, and miss shaders, which enable the assignment of unique sets of shaders and textures for each object. Another ray tracing platform which may be supported by the ray tracing cores **372**, graphics cores **370** and tensor cores **371** is Vulkan API (e.g., Vulkan version 1.1.85 and later). Note, however, that the underlying principles described herein are not limited to any particular ray tracing ISA.

[0123] In general, the various cores **372**, **371**, **370** may support a ray tracing instruction set that includes instructions/functions for one or more of ray generation, closest hit, any hit, ray-primitive intersection, per-primitive and hierarchical bounding box construction, miss, visit, and exceptions. More specifically, a preferred embodiment includes ray tracing instructions to perform one or more of the following functions: [0124] Ray Generation—Ray generation instructions may be executed for each pixel, sample, or other user-defined work assignment. [0125] Closest Hit—A closest hit instruction may be executed to locate the closest intersection point of a ray with primitives within a

scene. [0126] Any Hit—An any hit instruction identifies multiple intersections between a ray and primitives within a scene, potentially to identify a new closest intersection point. [0127] Intersection—An intersection instruction performs a ray-primitive intersection test and outputs a result. [0128] Per-primitive Bounding box Construction—This instruction builds a bounding box around a given primitive or group of primitives (e.g., when building a new BVH or other acceleration data structure). [0129] Miss—Indicates that a ray misses all geometry within a scene, or specified region of a scene. [0130] Visit—Indicates the child volumes a ray will traverse. [0131] Exceptions—Includes various types of exception handlers (e.g., invoked for various error conditions).

[0132] In one embodiment the ray tracing cores **372** may be adapted to accelerate general-purpose compute operations that can be accelerated using computational techniques that are analogous to ray intersection tests. A compute framework can be provided that enables shader programs to be compiled into low level instructions and/or primitives that perform general-purpose compute operations via the ray tracing cores. Exemplary computational problems that can benefit from compute operations performed on the ray tracing cores **372** include computations involving beam, wave, ray, or particle propagation within a coordinate space. Interactions associated with that propagation can be computed relative to a geometry or mesh within the coordinate space. For example, computations associated with electromagnetic signal propagation through an environment can be accelerated via the use of instructions or primitives that are executed via the ray tracing cores. Diffraction and reflection of the signals by objects in the environment can be computed as direct ray-tracing analogies.

[0133] Ray tracing cores **372** can also be used to perform computations that are not directly analogous to ray tracing. For example, mesh projection, mesh refinement, and volume sampling computations can be accelerated using the ray tracing cores **372**. Generic coordinate space calculations, such as nearest neighbor calculations can also be performed. For example, the set of points near a given point can be discovered by defining a bounding box in the coordinate space around the point. BVH and ray probe logic within the ray tracing cores **372** can then be used to determine the set of point intersections within the bounding box. The intersections constitute the origin point and the nearest neighbors to that origin point. Computations that are performed using the ray tracing cores **372** can be performed in parallel with computations performed on the graphics cores **372** and tensor cores **371**. A shader compiler can be configured to compile a compute shader or other general-purpose graphics processing program into low level primitives that can be parallelized across the graphics cores **370**, tensor cores **371**, and ray tracing cores **372**.

Techniques for GPU to Host Processor Interconnection

[0134] FIG. **4A** illustrates an exemplary architecture in which a plurality of GPUs **410-413**, e.g., such as the parallel processors **200** shown in FIG. **2A**, are communicatively coupled to a plurality of multi-core processors **405-406** over high-speed links **440A-440D** (e.g., buses, point-to-point interconnects, etc.). The high-speed links **440A-440D** may support a communication throughput of 4 GB/s, 30 GB/s, 80 GB/s or higher, depending on the implementation. Various interconnect protocols may be used including, but not limited to, PCIe 4.0 or 5.0 and NV Link 2.0. However, the underlying principles described herein are not limited to any particular communication protocol or throughput.

[0135] Two or more of the GPUs **410-413** may be interconnected over high-speed links **442A-442B**, which may be implemented using the same or different protocols/links than those used for high-speed links **440A-440D**. Similarly, two or more of the multi-core processors **405-406** may be connected over high-speed link **443** which may be symmetric multi-processor (SMP) buses operating at 20 GB/s, 30 GB/s, 120 GB/s or lower or higher speeds. Alternatively, all communication between the various system components shown in FIG. **4A** may be accomplished using the same protocols/links (e.g., over a common interconnection fabric). As mentioned, however, the underlying principles described herein are not limited to any particular type of

interconnect technology.

[0136] Each of multi-core processor **405** and multi-core processor **406** may be communicatively coupled to a processor memory **401-402**, via memory interconnects **430A-430B**, respectively, and each GPU **410-413** is communicatively coupled to GPU memory **420-423** over GPU memory interconnects **450A-450D**, respectively. The memory interconnects **430A-430B** and **450A-450D** may utilize the same or different memory access technologies. By way of example, and not limitation, the processor memories **401-402** and GPU memories **420-423** may be volatile memories such as dynamic random-access memories (DRAMs) (including stacked DRAMs), Graphics DDR SDRAM (GDDR) (e.g., GDDR5, GDDR6), or High Bandwidth Memory (HBM) and/or may be non-volatile memories such as 3D X Point/Optane or Nano-Ram. For example, some portion of the memories may be volatile memory and another portion may be non-volatile memory (e.g., using a two-level memory (2L M) hierarchy). A memory subsystem as described herein may be compatible with a number of memory technologies, such as Double Data Rate versions released by JEDEC (Joint Electronic Device Engineering Council).

[0137] As described below, although the various processors **405-406** and GPUs **410-413** may be physically coupled to a particular memory **401-402**, **420-423**, respectively, a unified memory architecture may be implemented in which the same virtual system address space (also referred to as the “effective address” space) is distributed among all of the various physical memories. For example, processor memories **401-402** may each comprise 64 GB of the system memory address space and GPU memories **420-423** may each comprise 32 GB of the system memory address space (resulting in a total of 256 GB addressable memory in this example).

[0138] FIG. **4B** illustrates additional optional details for an interconnection between a multi-core processor **407** and a graphics acceleration module **446**. The graphics acceleration module **446** may include one or more GPU chips integrated on a line card which is coupled to the processor **407** via the high-speed link **440**. Alternatively, the graphics acceleration module **446** may be integrated on the same package or chip as the processor **407**.

[0139] The illustrated processor **407** includes a plurality of cores **460A-460D**, each with a translation lookaside buffer **461A-461D** and one or more caches **462A-462D**. The cores may include various other components for executing instructions and processing data which are not illustrated to avoid obscuring the underlying principles of the components described herein (e.g., instruction fetch units, branch prediction units, decoders, execution units, reorder buffers, etc.). The caches **462A-462D** may comprise level 1 (L1) and level 2 (L2) caches. In addition, one or more shared caches **456** may be included in the caching hierarchy and shared by sets of the cores **460A-460D**. For example, one embodiment of the processor **407** includes 24 cores, each with its own L1 cache, twelve shared L2 caches, and twelve shared L3 caches. In this embodiment, one of the L2 and L3 caches are shared by two adjacent cores. The processor **407** and the graphics accelerator integration module **446** connect with system memory **441**, which may include processor memories **401-402**.

[0140] Coherency is maintained for data and instructions stored in the various caches **462A-462D**, **456** and system memory **441** via inter-core communication over a coherence bus **464**. For example, each cache may have cache coherency logic/circuitry associated therewith to communicate to over the coherence bus **464** in response to detected reads or writes to particular cache lines. In one implementation, a cache snooping protocol is implemented over the coherence bus **464** to snoop cache accesses. Cache snooping/coherency techniques are well understood by those of skill in the art and will not be described in detail here to avoid obscuring the underlying principles described herein.

[0141] A proxy circuit **425** may be provided that communicatively couples the graphics acceleration module **446** to the coherence bus **464**, allowing the graphics acceleration module **446** to participate in the cache coherence protocol as a peer of the cores. In particular, an interface **435** provides connectivity to the proxy circuit **425** over high-speed link **440** (e.g., a PCIe bus, NV Link,

etc.) and an interface **437** connects the graphics acceleration module **446** to the high-speed link **440**.

[0142] In one implementation, an accelerator integration circuit **436** provides cache management, memory access, context management, and interrupt management services on behalf of a plurality of graphics processing engines **431, 432, N** of the graphics acceleration module **446**. The graphics processing engines **431, 432, N** may each comprise a separate graphics processing unit (GPU). Alternatively, the graphics processing engines **431, 432, N** may comprise different types of graphics processing engines within a GPU such as graphics execution units, media processing engines (e.g., video encoders/decoders), samplers, and blit engines. In other words, the graphics acceleration module may be a GPU with a plurality of graphics processing engines **431-432, N** or the graphics processing engines **431-432, N** may be individual GPUs integrated on a common package, line card, or chip. The graphics processing engines **431-432, N** may be configured with any graphics processor or compute accelerator architecture described herein.

[0143] The accelerator integration circuit **436** may include a memory management unit (MMU) **439** for performing various memory management functions such as virtual-to-physical memory translations (also referred to as effective-to-real memory translations) and memory access protocols for accessing system memory **441**. The MMU **439** may also include a translation lookaside buffer (TLB) (not shown) for caching the virtual/effective to physical/real address translations. In one implementation, a cache **438** stores commands and data for efficient access by the graphics processing engines **431, 432, N**. The data stored in cache **438** and graphics memories **433-434, M** may be kept coherent with the core caches **462A-462D, 456** and system memory **441**. As mentioned, this may be accomplished via proxy circuit **425** which takes part in the cache coherency mechanism on behalf of cache **438** and memories **433-434, M** (e.g., sending updates to the cache **438** related to modifications/accesses of cache lines on processor caches **462A-462D, 456** and receiving updates from the cache **438**).

[0144] A set of registers **445** store context data for threads executed by the graphics processing engines **431-432, N** and a context management circuit **448** manages the thread contexts. For example, the context management circuit **448** may perform save and restore operations to save and restore contexts of the various threads during contexts switches (e.g., where a first thread is saved and a second thread is restored so that the second thread can be executed by a graphics processing engine). For example, on a context switch, the context management circuit **448** may store current register values to a designated region in memory (e.g., identified by a context pointer). It may then restore the register values when returning to the context. An interrupt management circuit **447**, for example, may receive and process interrupts received from system devices.

[0145] In one implementation, virtual/effective addresses from a graphics processing engine **431** are translated to real/physical addresses in system memory **441** by the MMU **439**. Optionally, the accelerator integration circuit **436** supports multiple (e.g., 4, 8, 16) graphics accelerator modules **446** and/or other accelerator devices. The graphics accelerator module **446** may be dedicated to a single application executed on the processor **407** or may be shared between multiple applications. Optionally, a virtualized graphics execution environment is provided in which the resources of the graphics processing engines **431-432, N** are shared with multiple applications, virtual machines (VMs), or containers. The resources may be subdivided into “slices” which are allocated to different VMs and/or applications based on the processing requirements and priorities associated with the VMs and/or applications, or based on a pre-determined partitioning profile for a graphics accelerator module **446**. VMs and containers can be used interchangeably herein.

[0146] A virtual machine (VM) can be software that runs an operating system and one or more applications. A VM can be defined by specification, configuration files, virtual disk file, non-volatile random-access memory (NVRAM) setting file, and the log file and is backed by the physical resources of a host computing platform. A VM can include an operating system (OS) or application environment that is installed on software, which imitates dedicated hardware. The end

user has the same experience on a virtual machine as they would have on dedicated hardware. Specialized software, called a hypervisor, emulates the PC client or server's CPU, memory, hard disk, network, and other hardware resources completely, enabling virtual machines to share the resources. The hypervisor can emulate multiple virtual hardware platforms that are isolated from each other, allowing virtual machines to run Linux®, Windows® Server, VM ware ESXi, and other operating systems on the same underlying physical host.

[0147] A container can be a software package of applications, configurations, and dependencies so the applications run reliably on one computing environment to another. Containers can share an operating system installed on the server platform and run as isolated processes. A container can be a software package that contains everything the software needs to run such as system tools, libraries, and settings. Containers are not installed like traditional software programs, which allows them to be isolated from the other software and the operating system itself. The isolated nature of containers provides several benefits. First, the software in a container will run the same in different environments. For example, a container that includes PHP and MySQL can run identically on both a Linux® computer and a Windows® machine. Second, containers provide added security since the software will not affect the host operating system. While an installed application may alter system settings and modify resources, such as the Windows registry, a container can only modify settings within the container.

[0148] Thus, the accelerator integration circuit **436** acts as a bridge to the system for the graphics acceleration module **446** and provides address translation and system memory cache services. In one embodiment, to facilitate the bridging functionality, the accelerator integration circuit **436** may also include shared I/O **497** (e.g., PCIe, USB, or others) and hardware to enable system control of voltage, clocking, performance, thermals, and security. The shared I/O **497** may utilize separate physical connections or may traverse the high-speed link **440**. In addition, the accelerator integration circuit **436** may provide virtualization facilities for the host processor to manage virtualization of the graphics processing engines, interrupts, and memory management.

[0149] Because hardware resources of the graphics processing engines **431-432**, N are mapped explicitly to the real address space seen by the host processor **407**, any host processor can address these resources directly using an effective address value. One optional function of the accelerator integration circuit **436** is the physical separation of the graphics processing engines **431-432**, N so that they appear to the system as independent units.

[0150] One or more graphics memories **433-434**, M may be coupled to each of the graphics processing engines **431-432**, N, respectively. The graphics memories **433-434**, M store instructions and data being processed by each of the graphics processing engines **431-432**, N. The graphics memories **433-434**, M may be volatile memories such as DRAMs (including stacked DRAMs), GDDR memory (e.g., GDDR5, GDDR6), or HBM, and/or may be non-volatile memories such as 3D X Point/Optane, Samsung Z-NAND, or Nano-Ram.

[0151] To reduce data traffic over the high-speed link **440**, biasing techniques may be used to ensure that the data stored in graphics memories **433-434**, M is data which will be used most frequently by the graphics processing engines **431-432**, N and preferably not used by the cores **460A-460D** (at least not frequently). Similarly, the biasing mechanism attempts to keep data needed by the cores (and preferably not the graphics processing engines **431-432**, N) within the caches **462A-462D**, **456** of the cores and system memory **441**.

[0152] According to a variant shown in FIG. **4C** the accelerator integration circuit **436** is integrated within the processor **407**. The graphics processing engines **431-432**, N communicate directly over the high-speed link **440** to the accelerator integration circuit **436** via interface **437** and interface **435** (which, again, may be utilize any form of bus or interface protocol). The accelerator integration circuit **436** may perform the same operations as those described with respect to FIG. **4B**, but potentially at a higher throughput given its close proximity to the coherence bus **464** and caches **462A-462D**, **456**.

[0153] The embodiments described may support different programming models including a dedicated-process programming model (no graphics acceleration module virtualization) and shared programming models (with virtualization). The latter may include programming models which are controlled by the accelerator integration circuit **436** and programming models which are controlled by the graphics acceleration module **446**.

[0154] In the embodiments of the dedicated process model, graphics processing engines **431, 432, . . . N** may be dedicated to a single application or process under a single operating system. The single application can funnel other application requests to the graphics engines **431, 432, . . . N**, providing virtualization within a VM/partition.

[0155] In the dedicated-process programming models, the graphics processing engines **431, 432, N**, may be shared by multiple VM/application partitions. The shared models require a system hypervisor to virtualize the graphics processing engines **431-432, N** to allow access by each operating system. For single-partition systems without a hypervisor, the graphics processing engines **431-432, N** are owned by the operating system. In both cases, the operating system can virtualize the graphics processing engines **431-432, N** to provide access to each process or application.

[0156] For the shared programming model, the graphics acceleration module **446** or an individual graphics processing engine **431-432, N** selects a process element using a process handle. The process elements may be stored in system memory **441** and be addressable using the effective address to real address translation techniques described herein. The process handle may be an implementation-specific value provided to the host process when registering its context with the graphics processing engine **431-432, N** (that is, calling system software to add the process element to the process element linked list). The lower 16-bits of the process handle may be the offset of the process element within the process element linked list.

[0157] FIG. **4D** illustrates an exemplary accelerator integration slice **490**. As used herein, a “slice” comprises a specified portion of the processing resources of the accelerator integration circuit **436**. Application effective address space **482** within system memory **441** stores process elements **483**. The process elements **483** may be stored in response to GPU invocations **481** from applications **480** executed on the processor **407**. A process element **483** contains the process state for the corresponding application **480**. A work descriptor (WD) **484** contained in the process element **483** can be a single job requested by an application or may contain a pointer to a queue of jobs. In the latter case, the WD **484** is a pointer to the job request queue in the application's address space **482**.

[0158] The graphics acceleration module **446** and/or the individual graphics processing engines **431-432, N** can be shared by all or a subset of the processes in the system. For example, the technologies described herein may include an infrastructure for setting up the process state and sending a WD **484** to a graphics acceleration module **446** to start a job in a virtualized environment.

[0159] In one implementation, the dedicated-process programming model is implementation-specific. In this model, a single process owns the graphics acceleration module **446** or an individual graphics processing engine **431**. Because the graphics acceleration module **446** is owned by a single process, the hypervisor initializes the accelerator integration circuit **436** for the owning partition and the operating system initializes the accelerator integration circuit **436** for the owning process at the time when the graphics acceleration module **446** is assigned.

[0160] In operation, a WD fetch unit **491** in the accelerator integration slice **490** fetches the next WD **484** which includes an indication of the work to be done by one of the graphics processing engines of the graphics acceleration module **446**. Data from the WD **484** may be stored in registers **445** and used by the MMU **439**, interrupt management circuit **447** and/or context management circuit **448** as illustrated. For example, the MMU **439** may include segment/page walk circuitry for accessing segment/page tables **486** within the OS virtual address space **485**. The interrupt management circuit **447** may process interrupt events **492** received from the graphics acceleration module **446**. When performing graphics operations, an effective address **493** generated by a

graphics processing engine **431-432**, N is translated to a real address by the MMU **439**.
[0161] The same set of registers **445** may be duplicated for each graphics processing engine **431-432**, N and/or graphics acceleration module **446** and may be initialized by the hypervisor or operating system. Each of these duplicated registers may be included in an accelerator integration slice **490**. In one embodiment, each graphics processing engine **431-432**, N may be presented to the hypervisor **496** as a distinct graphics processor device. QoS settings can be configured for clients of a specific graphics processing engine **431-432**, N and data isolation between the clients of each engine can be enabled. Exemplary registers that may be initialized by the hypervisor are shown in Table 1.

TABLE-US-00001 TABLE 1 Hypervisor Initialized Registers 1 Slice Control Register 2 Real Address (RA) Scheduled Processes Area Pointer 3 Authority Mask Override Register 4 Interrupt Vector Table Entry Offset 5 Interrupt Vector Table Entry Limit 6 State Register 7 Logical Partition ID 8 Real address (RA) Hypervisor Accelerator Utilization Record Pointer 9 Storage Description Register

[0162] Exemplary registers that may be initialized by the operating system are shown in Table 2.
TABLE-US-00002 TABLE 2 Operating System Initialized Registers 1 Process and Thread Identification 2 Effective Address (EA) Context Save/Restore Pointer 3 Virtual Address (VA) Accelerator Utilization Record Pointer 4 Virtual Address (VA) Storage Segment Table Pointer 5 Authority Mask 6 Work descriptor

[0163] Each WD **484** may be specific to a particular graphics acceleration module **446** and/or graphics processing engine **431-432**, N. It contains all the information a graphics processing engine **431-432**, N requires to do its work, or it can be a pointer to a memory location where the application has set up a command queue of work to be completed.

[0164] FIG. 4E illustrates additional optional details of a shared model. It includes a hypervisor real address space **498** in which a process element list **499** is stored. The hypervisor real address space **498** is accessible via a hypervisor **496** which virtualizes the graphics acceleration module engines for the operating system **495**.

[0165] The shared programming models allow for all or a subset of processes from all or a subset of partitions in the system to use a graphics acceleration module **446**. There are two programming models where the graphics acceleration module **446** is shared by multiple processes and partitions: time-sliced shared and graphics directed shared.

[0166] In this model, the system hypervisor **496** owns the graphics acceleration module **446** and makes its function available to all operating systems **495**. For a graphics acceleration module **446** to support virtualization by the system hypervisor **496**, the graphics acceleration module **446** may adhere to the following requirements: 1) An application's job request must be autonomous (that is, the state does not need to be maintained between jobs), or the graphics acceleration module **446** must provide a context save and restore mechanism. 2) An application's job request is guaranteed by the graphics acceleration module **446** to complete in a specified amount of time, including any translation faults, or the graphics acceleration module **446** provides the ability to preempt the processing of the job. 3) The graphics acceleration module **446** must be guaranteed fairness between processes when operating in the directed shared programming model.

[0167] For the directed shared model, the application **480** may be required to make an operating system **495** system call with a graphics acceleration module **446** type, a work descriptor (WD), an authority mask register (AMR) value, and a context save/restore area pointer (CSRP). The graphics acceleration module **446** type describes the targeted acceleration function for the system call. The graphics acceleration module **446** type may be a system-specific value. The WD is formatted specifically for the graphics acceleration module **446** and can be in the form of a graphics acceleration module **446** command, an effective address pointer to a user-defined structure, an effective address pointer to a queue of commands, or any other data structure to describe the work to be done by the graphics acceleration module **446**. In one embodiment, the AMR value is the

AMR state to use for the current process. The value passed to the operating system is similar to an application setting the AMR. If the accelerator integration circuit **436** and graphics acceleration module **446** implementations do not support a User Authority Mask Override Register (UAMOR), the operating system may apply the current UAMOR value to the AMR value before passing the AMR in the hypervisor call. The hypervisor **496** may optionally apply the current Authority Mask Override Register (AMOR) value before placing the AMR into the process element **483**. The CSRP may be one of the registers **445** containing the effective address of an area in the application's address space **482** for the graphics acceleration module **446** to save and restore the context state. This pointer is optional if no state is required to be saved between jobs or when a job is preempted. The context save/restore area may be pinned system memory.

[0168] Upon receiving the system call, the operating system **495** may verify that the application **480** has registered and been given the authority to use the graphics acceleration module **446**. The operating system **495** then calls the hypervisor **496** with the information shown in Table 3.

TABLE-US-00003 TABLE 3 OS to Hypervisor Call Parameters 1 A work descriptor (WD) 2 An Authority Mask Register (AMR) value (potentially masked). 3 An effective address (EA) Context Save/Restore Area Pointer (CSRP) 4 A process ID (PID) and optional thread ID (TID) 5 A virtual address (VA) accelerator utilization record pointer (AURP) 6 The virtual address of the storage segment table pointer (SSTP) 7 A logical interrupt service number (LISN)

[0169] Upon receiving the hypervisor call, the hypervisor **496** verifies that the operating system **495** has registered and been given the authority to use the graphics acceleration module **446**. The hypervisor **496** then puts the process element **483** into the process element linked list for the corresponding graphics acceleration module **446** type. The process element may include the information shown in Table 4.

TABLE-US-00004 TABLE 4 Process Element Information 1 A work descriptor (WD) 2 An Authority Mask Register (AMR) value (potentially masked). 3 An effective address (EA) Context Save/Restore Area Pointer (CSRP) 4 A process ID (PID) and optional thread ID (TID) 5 A virtual address (VA) accelerator utilization record pointer (AURP) 6 The virtual address of the storage segment table pointer (SSTP) 7 A logical interrupt service number (LISN) 8 Interrupt vector table, derived from the hypervisor call parameters. 9 A state register (SR) value 10 A logical partition ID (LPID) 11 A real address (RA) hypervisor accelerator utilization record pointer 12 The Storage Descriptor Register (SDR)

[0170] The hypervisor may initialize a plurality of accelerator integration slice **490** registers **445**.

[0171] As illustrated in FIG. 4F, in one optional implementation a unified memory addressable via a common virtual memory address space used to access the physical processor memories **401-402** and GPU memories **420-423** is employed. In this implementation, operations executed on the GPUs **410-413** utilize the same virtual/effective memory address space to access the processors memories **401-402** and vice versa, thereby simplifying programmability. A first portion of the virtual/effective address space may be allocated to the processor memory **401**, a second portion to the second processor memory **402**, a third portion to the GPU memory **420**, and so on. The entire virtual/effective memory space (sometimes referred to as the effective address space) may thereby be distributed across each of the processor memories **401-402** and GPU memories **420-423**, allowing any processor or GPU to access any physical memory with a virtual address mapped to that memory.

[0172] Bias/coherence management circuitry **494A-494E** within one or more of the MMUs **439A-439E** may be provided that ensures cache coherence between the caches of the host processors (e.g., **405**) and the GPUs **410-413** and implements biasing techniques indicating the physical memories in which certain types of data should be stored. While multiple instances of bias/coherence management circuitry **494A-494E** are illustrated in FIG. 4F, the bias/coherence circuitry may be implemented within the MMU of one or more host processors **405** and/or within the accelerator integration circuit **436**.

[0173] The GPU-attached memory **420-423** may be mapped as part of system memory and accessed using shared virtual memory (SVM) technology, but without suffering the typical performance drawbacks associated with full system cache coherence. The ability to GPU-attached memory **420-423** to be accessed as system memory without onerous cache coherence overhead provides a beneficial operating environment for GPU offload. This arrangement allows the host processor **405** software to setup operands and access computation results, without the overhead of tradition I/O DMA data copies. Such traditional copies involve driver calls, interrupts and memory mapped I/O (M MIO) accesses that are all inefficient relative to simple memory accesses. At the same time, the ability to access GPU attached memory **420-423** without cache coherence overheads can be critical to the execution time of an offloaded computation. In cases with substantial streaming write memory traffic, for example, cache coherence overhead can significantly reduce the effective write bandwidth seen by a GPU **410-413**. The efficiency of operand setup, the efficiency of results access, and the efficiency of GPU computation all play a role in determining the effectiveness of GPU offload.

[0174] A selection between GPU bias and host processor bias may be driven by a bias tracker data structure. A bias table may be used, for example, which may be a page-granular structure (i.e., controlled at the granularity of a memory page) that includes 1 or 2 bits per GPU-attached memory page. The bias table may be implemented in a stolen memory range of one or more GPU-attached memories **420-423**, with or without a bias cache in the GPU **410-413** (e.g., to cache frequently/recently used entries of the bias table). Alternatively, the entire bias table may be maintained within the GPU.

[0175] In one implementation, the bias table entry associated with each access to the GPU-attached memory **420-423** is accessed prior the actual access to the GPU memory, causing the following operations. First, local requests from the GPU **410-413** that find their page in GPU bias are forwarded directly to a corresponding GPU memory **420-423**. Local requests from the GPU that find their page in host bias are forwarded to the processor **405** (e.g., over a high-speed link as discussed above). Optionally, requests from the processor **405** that find the requested page in host processor bias complete the request like a normal memory read. Alternatively, requests directed to a GPU-biased page may be forwarded to the GPU **410-413**. The GPU may then transition the page to a host processor bias if it is not currently using the page.

[0176] The bias state of a page can be changed either by a software-based mechanism, a hardware-assisted software-based mechanism, or, for a limited set of cases, a purely hardware-based mechanism.

[0177] One mechanism for changing the bias state employs an API call (e.g., OpenCL), which, in turn, calls the GPU's device driver which, in turn, sends a message (or enqueues a command descriptor) to the GPU directing it to change the bias state and, for some transitions, perform a cache flushing operation in the host. The cache flushing operation is required for a transition from host processor **405** bias to GPU bias but is not required for the opposite transition.

[0178] Cache coherency may be maintained by temporarily rendering GPU-biased pages uncacheable by the host processor **405**. To access these pages, the processor **405** may request access from the GPU **410** which may or may not grant access right away, depending on the implementation. Thus, to reduce communication between the host processor **405** and GPU **410** it is beneficial to ensure that GPU-biased pages are those which are required by the GPU but not the host processor **405** and vice versa.

Graphics Processing Pipeline

[0179] FIG. 5 illustrates a graphics processing pipeline **500**. A graphics multiprocessor, such as graphics multiprocessor **234** as in FIG. 2D, graphics multiprocessor **325** of FIG. 3A, graphics multiprocessor **350** of FIG. 3B can implement the illustrated graphics processing pipeline **500**. The graphics multiprocessor can be included within the parallel processing subsystems as described herein, such as the parallel processor **200** of FIG. 2A, which may be related to the parallel

processor(s) **112** of FIG. **1** and may be used in place of one of those. The various parallel processing systems can implement the graphics processing pipeline **500** via one or more instances of the parallel processing unit (e.g., parallel processing unit **202** of FIG. **2A**) as described herein. For example, a shader unit (e.g., graphics multiprocessor **234** of FIG. **2C**) may be configured to perform the functions of one or more of a vertex processing unit **504**, a tessellation control processing unit **508**, a tessellation evaluation processing unit **512**, a geometry processing unit **516**, and a fragment/pixel processing unit **524**. The functions of data assembler **502**, primitive assemblers **506**, **514**, **518**, tessellation unit **510**, rasterizer **522**, and raster operations unit **526** may also be performed by other processing engines within a processing cluster (e.g., processing cluster **214** of FIG. **2A**) and a corresponding partition unit (e.g., partition unit **220A-220N** of FIG. **2A**). The graphics processing pipeline **500** may also be implemented using dedicated processing units for one or more functions. It is also possible that one or more portions of the graphics processing pipeline **500** are performed by parallel processing logic within a general-purpose processor (e.g., CPU). Optionally, one or more portions of the graphics processing pipeline **500** can access on-chip memory (e.g., parallel processor memory **222** as in FIG. **2A**) via a memory interface **528**, which may be an instance of the memory interface **218** of FIG. **2A**. The graphics processor pipeline **500** may also be implemented via a multi-core group **365A** as in FIG. **3C**.

[0180] The data assembler **502** is a processing unit that may collect vertex data for surfaces and primitives. The data assembler **502** then outputs the vertex data, including the vertex attributes, to the vertex processing unit **504**. The vertex processing unit **504** is a programmable execution unit that executes vertex shader programs, lighting and transforming vertex data as specified by the vertex shader programs. The vertex processing unit **504** reads data that is stored in cache, local or system memory for use in processing the vertex data and may be programmed to transform the vertex data from an object-based coordinate representation to a world space coordinate space or a normalized device coordinate space.

[0181] A first instance of a primitive assembler **506** receives vertex attributes from the vertex processing unit **504**. The primitive assembler **506** readings stored vertex attributes as needed and constructs graphics primitives for processing by tessellation control processing unit **508**. The graphics primitives include triangles, line segments, points, patches, and so forth, as supported by various graphics processing application programming interfaces (APIs).

[0182] The tessellation control processing unit **508** treats the input vertices as control points for a geometric patch. The control points are transformed from an input representation from the patch (e.g., the patch's bases) to a representation that is suitable for use in surface evaluation by the tessellation evaluation processing unit **512**. The tessellation control processing unit **508** can also compute tessellation factors for edges of geometric patches. A tessellation factor applies to a single edge and quantifies a view-dependent level of detail associated with the edge. A tessellation unit **510** is configured to receive the tessellation factors for edges of a patch and to tessellate the patch into multiple geometric primitives such as line, triangle, or quadrilateral primitives, which are transmitted to a tessellation evaluation processing unit **512**. The tessellation evaluation processing unit **512** operates on parameterized coordinates of the subdivided patch to generate a surface representation and vertex attributes for each vertex associated with the geometric primitives.

[0183] A second instance of a primitive assembler **514** receives vertex attributes from the tessellation evaluation processing unit **512**, reading stored vertex attributes as needed, and constructs graphics primitives for processing by the geometry processing unit **516**. The geometry processing unit **516** is a programmable execution unit that executes geometry shader programs to transform graphics primitives received from primitive assembler **514** as specified by the geometry shader programs. The geometry processing unit **516** may be programmed to subdivide the graphics primitives into one or more new graphics primitives and calculate parameters used to rasterize the new graphics primitives.

[0184] The geometry processing unit **516** may be able to add or delete elements in the geometry

stream. The geometry processing unit **516** outputs the parameters and vertices specifying new graphics primitives to primitive assembler **518**. The primitive assembler **518** receives the parameters and vertices from the geometry processing unit **516** and constructs graphics primitives for processing by a viewport scale, cull, and clip unit **520**. The geometry processing unit **516** reads data that is stored in parallel processor memory or system memory for use in processing the geometry data. The viewport scale, cull, and clip unit **520** performs clipping, culling, and viewport scaling and outputs processed graphics primitives to a rasterizer **522**.

[0185] The rasterizer **522** can perform depth culling and other depth-based optimizations. The rasterizer **522** also performs scan conversion on the new graphics primitives to generate fragments and output those fragments and associated coverage data to the fragment/pixel processing unit **524**. The fragment/pixel processing unit **524** is a programmable execution unit that is configured to execute fragment shader programs or pixel shader programs. The fragment/pixel processing unit **524** transforming fragments or pixels received from rasterizer **522**, as specified by the fragment or pixel shader programs. For example, the fragment/pixel processing unit **524** may be programmed to perform operations included but not limited to texture mapping, shading, blending, texture correction and perspective correction to produce shaded fragments or pixels that are output to a raster operations unit **526**. The fragment/pixel processing unit **524** can read data that is stored in either the parallel processor memory or the system memory for use when processing the fragment data. Fragment or pixel shader programs may be configured to shade at sample, pixel, tile, or other granularities depending on the sampling rate configured for the processing units.

[0186] The raster operations unit **526** is a processing unit that performs raster operations including, but not limited to stencil, z-test, blending, and the like, and outputs pixel data as processed graphics data to be stored in graphics memory (e.g., parallel processor memory **222** as in FIG. 2A, and/or system memory **104** as in FIG. 1), to be displayed on the one or more display device(s) **110A-110B** or for further processing by one of the one or more processor(s) **102** or parallel processor(s) **112**. The raster operations unit **526** may be configured to compress z or color data that is written to memory and decompress z or color data that is read from memory.

Machine Learning Overview

[0187] The architecture described above can be applied to perform training and inference operations using machine learning models. Machine learning has been successful at solving many kinds of tasks. The computations that arise when training and using machine learning algorithms (e.g., neural networks) lend themselves naturally to efficient parallel implementations. Accordingly, parallel processors such as general-purpose graphics processing units (GPGPUs) have played a significant role in the practical implementation of deep neural networks. Parallel graphics processors with single instruction, multiple thread (SIMT) architectures are designed to maximize the amount of parallel processing in the graphics pipeline. In an SIMT architecture, groups of parallel threads attempt to execute program instructions synchronously together as often as possible to increase processing efficiency. The efficiency provided by parallel machine learning algorithm implementations allows the use of high-capacity networks and enables those networks to be trained on larger datasets.

[0188] A machine learning algorithm is an algorithm that can learn based on a set of data. For example, machine learning algorithms can be designed to model high-level abstractions within a data set. For example, image recognition algorithms can be used to determine which of several categories to which a given input belong; regression algorithms can output a numerical value given an input; and pattern recognition algorithms can be used to generate translated text or perform text to speech and/or speech recognition.

[0189] An exemplary type of machine learning algorithm is a neural network. There are many types of neural networks; a simple type of neural network is a feedforward network. A feedforward network may be implemented as an acyclic graph in which the nodes are arranged in layers. Typically, a feedforward network topology includes an input layer and an output layer that are

separated by at least one hidden layer. The hidden layer transforms input received by the input layer into a representation that is useful for generating output in the output layer. The network nodes are fully connected via edges to the nodes in adjacent layers, but there are no edges between nodes within each layer. Data received at the nodes of an input layer of a feedforward network are propagated (i.e., “fed forward”) to the nodes of the output layer via an activation function that calculates the states of the nodes of each successive layer in the network based on coefficients (“weights”) respectively associated with each of the edges connecting the layers. Depending on the specific model being represented by the algorithm being executed, the output from the neural network algorithm can take various forms.

[0190] Before a machine learning algorithm can be used to model a particular problem, the algorithm is trained using a training data set. Training a neural network involves selecting a network topology, using a set of training data representing a problem being modeled by the network, and adjusting the weights until the network model performs with a minimal error for all instances of the training data set. For example, during a supervised learning training process for a neural network, the output produced by the network in response to the input representing an instance in a training data set is compared to the “correct” labeled output for that instance, an error signal representing the difference between the output and the labeled output is calculated, and the weights associated with the connections are adjusted to minimize that error as the error signal is backward propagated through the layers of the network. The network is considered “trained” when the errors for each of the outputs generated from the instances of the training data set are minimized.

[0191] The accuracy of a machine learning algorithm can be affected significantly by the quality of the data set used to train the algorithm. The training process can be computationally intensive and may require a significant amount of time on a conventional general-purpose processor. Accordingly, parallel processing hardware is used to train many types of machine learning algorithms. This is particularly useful for optimizing the training of neural networks, as the computations performed in adjusting the coefficients in neural networks lend themselves naturally to parallel implementations. Specifically, many machine learning algorithms and software applications have been adapted to make use of the parallel processing hardware within general-purpose graphics processing devices.

[0192] FIG. 6 is a generalized diagram of a machine learning software stack **600**. A machine learning application **602** is any logic that can be configured to train a neural network using a training dataset or to use a trained deep neural network to implement machine intelligence. The machine learning application **602** can include training and inference functionality for a neural network and/or specialized software that can be used to train a neural network before deployment. The machine learning application **602** can implement any type of machine intelligence including but not limited to image recognition, mapping and localization, autonomous navigation, speech synthesis, medical imaging, or language translation. Example machine learning applications **602** include, but are not limited to, voice-based virtual assistants, image or facial recognition algorithms, autonomous navigation, and the software tools that are used to train the machine learning models used by the machine learning applications **602**.

[0193] Hardware acceleration for the machine learning application **602** can be enabled via a machine learning framework **604**. The machine learning framework **604** can provide a library of machine learning primitives. Machine learning primitives are basic operations that are commonly performed by machine learning algorithms. Without the machine learning framework **604**, developers of machine learning algorithms would be required to create and optimize the main computational logic associated with the machine learning algorithm, then re-optimize the computational logic as new parallel processors are developed. Instead, the machine learning application can be configured to perform the necessary computations using the primitives provided by the machine learning framework **604**. Exemplary primitives include tensor convolutions,

activation functions, and pooling, which are computational operations that are performed while training a convolutional neural network (CNN). The machine learning framework **604** can also provide primitives to implement basic linear algebra subprograms performed by many machine-learning algorithms, such as matrix and vector operations. Examples of a machine learning framework **604** include, but are not limited to, TensorFlow, TensorRT, PyTorch, MXNet, Caffe, and other high-level machine learning frameworks.

[0194] The machine learning framework **604** can process input data received from the machine learning application **602** and generate the appropriate input to a compute framework **606**. The compute framework **606** can abstract the underlying instructions provided to the GPGPU driver **608** to enable the machine learning framework **604** to take advantage of hardware acceleration via the GPGPU hardware **610** without requiring the machine learning framework **604** to have intimate knowledge of the architecture of the GPGPU hardware **610**. Additionally, the compute framework **606** can enable hardware acceleration for the machine learning framework **604** across a variety of types and generations of the GPGPU hardware **610**. Exemplary compute frameworks **606** include the CUDA compute framework and associated machine learning libraries, such as the CUDA Deep Neural Network (cuDNN) library. The machine learning software stack **600** can also include communication libraries or frameworks to facilitate multi-GPU and multi-node compute.

GPGPU Machine Learning Acceleration

[0195] FIG. 7 illustrates a general-purpose graphics processing unit, which may be the parallel processor **200** of FIG. 2A or the parallel processor(s) **112** of FIG. 1. The general-purpose processing unit (GPGPU **700**) may be configured to provide support for hardware acceleration of primitives provided by a machine learning framework to accelerate the processing the type of computational workloads associated with training deep neural networks. Additionally, the GPGPU **700** can be linked directly to other instances of the GPGPU to create a multi-GPU cluster to improve training speed for particularly deep neural networks. Primitives are also supported to accelerate inference operations for deployed neural networks.

[0196] The GPGPU **700** includes a host interface **702** to enable a connection with a host processor. The host interface **702** may be a PCI Express interface. However, the host interface can also be a vendor specific communications interface or communications fabric. The GPGPU **700** receives commands from the host processor and uses a global scheduler **704** to distribute execution threads associated with those commands to a set of processing clusters **706A-706H**. The processing clusters **706A-706H** share a cache memory **708**. The cache memory **708** can serve as a higher-level cache for cache memories within the processing clusters **706A-706H**. The illustrated processing clusters **706A-706H** may correspond with processing clusters **214A-214N** as in FIG. 2A.

[0197] The GPGPU **700** includes memory **714A-714B** coupled with the processing clusters **706A-706H** via a set of memory controllers **712A-712B**. The memory **714A-714B** can include various types of memory devices including dynamic random-access memory (DRAM) or graphics random access memory, such as synchronous graphics random access memory (SGRAM), including graphics double data rate (GDDR) memory. The memory **714A-714B** may also include 3D stacked memory, including but not limited to high bandwidth memory (HBM).

[0198] Each of the processing clusters **706A-706H** may include a set of graphics multiprocessors, such as the graphics multiprocessor **234** of FIG. 2D, graphics multiprocessor **325** of FIG. 3A, graphics multiprocessor **350** of FIG. 3B, or may include a multi-core group **365A-365N** as in FIG. 3C. The graphics multiprocessors of the compute cluster include multiple types of integer and floating-point logic units that can perform computational operations at a range of precisions including suited for machine learning computations. For example, at least a subset of the floating-point units in each of the processing clusters **706A-706H** can be configured to perform 16-bit or 32-bit floating point operations, while a different subset of the floating-point units can be configured to perform 64-bit floating point operations.

[0199] Multiple instances of the GPGPU **700** can be configured to operate as a compute cluster.

The communication mechanism used by the compute cluster for synchronization and data exchange varies across embodiments. For example, the multiple instances of the GPGPU **700** communicate over the host interface **702**. In one embodiment the GPGPU **700** includes an I/O hub **709** that couples the GPGPU **700** with a GPU link **710** that enables a direct connection to other instances of the GPGPU. The GPU link **710** may be coupled to a dedicated GPU-to-GPU bridge that enables communication and synchronization between multiple instances of the GPGPU **700**. Optionally, the GPU link **710** couples with a high-speed interconnect to transmit and receive data to other GPGPUs or parallel processors. The multiple instances of the GPGPU **700** may be located in separate data processing systems and communicate via a network device that is accessible via the host interface **702**. The GPU link **710** may be configured to enable a connection to a host processor in addition to or as an alternative to the host interface **702**.

[0200] While the illustrated configuration of the GPGPU **700** can be configured to train neural networks, an alternate configuration of the GPGPU **700** can be configured for deployment within a high performance or low power inferencing platform. In an inferencing configuration, the GPGPU **700** includes fewer of the processing clusters **706A-706H** relative to the training configuration. Additionally, memory technology associated with the memory **714A-714B** may differ between inferencing and training configurations. In one embodiment, the inferencing configuration of the GPGPU **700** can support inferencing specific instructions. For example, an inferencing configuration can provide support for one or more 8-bit integer or floating-point dot product instructions, which are commonly used during inferencing operations for deployed neural networks.

[0201] FIG. **8** illustrates a multi-GPU computing system **800**. The multi-GPU computing system **800** can include a processor **802** coupled to multiple GPGPUs **806A-806D** via a host interface switch **804**. The host interface switch **804** may be a PCI express switch device that couples the processor **802** to a PCI express bus over which the processor **802** can communicate with the set of GPGPUs **806A-806D**. Each of the multiple GPGPUs **806A-806D** can be an instance of the GPGPU **700** of FIG. **7**. The GPGPUs **806A-806D** can interconnect via a set of high-speed point to point GPU to GPU links **816**. The high-speed GPU to GPU links can connect to each of the GPGPUs **806A-806D** via a dedicated GPU link, such as the GPU link **710** as in FIG. **7**. The P2P GPU links **816** enable direct communication between each of the GPGPUs **806A-806D** without requiring communication over the host interface bus to which the processor **802** is connected. With GPU-to-GPU traffic directed to the P2P GPU links, the host interface bus remains available for system memory access or to communicate with other instances of the multi-GPU computing system **800**, for example, via one or more network devices. While in FIG. **8** the GPGPUs **806A-806D** connect to the processor **802** via the host interface switch **804**, the processor **802** may alternatively include direct support for the P2P GPU links **816** and connect directly to the GPGPUs **806A-806D**. In one embodiment the P2P GPU link **816** enable the multi-GPU computing system **800** to operate as a single logical GPU.

Machine Learning Neural Network Implementations

[0202] The computing architecture described herein can be configured to perform the types of parallel processing that is particularly suited for training and deploying neural networks for machine learning. A neural network can be generalized as a network of functions having a graph relationship. As is well-known in the art, there are a variety of types of neural network implementations used in machine learning. One exemplary type of neural network is the feedforward network, as previously described.

[0203] A second exemplary type of neural network is the Convolutional Neural Network (CNN). A CNN is a specialized feedforward neural network for processing data having a known, grid-like topology, such as image data. Accordingly, CNNs are commonly used for compute vision and image recognition applications, but they also may be used for other types of pattern recognition such as speech and language processing. The nodes in the CNN input layer are organized into a set

of “filters” (feature detectors inspired by the receptive fields found in the retina), and the output of each set of filters is propagated to nodes in successive layers of the network. The computations for a CNN include applying the convolution mathematical operation to each filter to produce the output of that filter. Convolution is a specialized kind of mathematical operation performed by two functions to produce a third function that is a modified version of one of the two original functions. In convolutional network terminology, the first function to the convolution can be referred to as the input, while the second function can be referred to as the convolution kernel. The output may be referred to as the feature map. For example, the input to a convolution layer can be a multidimensional array of data that defines the various color components of an input image. The convolution kernel can be a multidimensional array of parameters, where the parameters are adapted by the training process for the neural network.

[0204] Recurrent neural networks (RNNs) are a family of feedforward neural networks that include feedback connections between layers. RNNs enable modeling of sequential data by sharing parameter data across different parts of the neural network. The architecture for an RNN includes cycles. The cycles represent the influence of a present value of a variable on its own value at a future time, as at least a portion of the output data from the RNN is used as feedback for processing subsequent input in a sequence. This feature makes RNNs particularly useful for language processing due to the variable nature in which language data can be composed.

[0205] The figures described below present exemplary feedforward, CNN, and RNN networks, as well as describe a general process for respectively training and deploying each of those types of networks. It will be understood that these descriptions are exemplary and non-limiting as to any specific embodiment described herein and the concepts illustrated can be applied generally to deep neural networks and machine learning techniques in general.

[0206] The exemplary neural networks described above can be used to perform deep learning. Deep learning is machine learning using deep neural networks. The deep neural networks used in deep learning are artificial neural networks composed of multiple hidden layers, as opposed to shallow neural networks that include only a single hidden layer. Deeper neural networks are generally more computationally intensive to train. However, the additional hidden layers of the network enable multistep pattern recognition that results in reduced output error relative to shallow machine learning techniques.

[0207] Deep neural networks used in deep learning typically include a front-end network to perform feature recognition coupled to a back-end network which represents a mathematical model that can perform operations (e.g., object classification, speech recognition, etc.) based on the feature representation provided to the model. Deep learning enables machine learning to be performed without requiring hand crafted feature engineering to be performed for the model. Instead, deep neural networks can learn features based on statistical structure or correlation within the input data. The learned features can be provided to a mathematical model that can map detected features to an output. The mathematical model used by the network is generally specialized for the specific task to be performed, and different models will be used to perform different task.

[0208] Once the neural network is structured, a learning model can be applied to the network to train the network to perform specific tasks. The learning model describes how to adjust the weights within the model to reduce the output error of the network. Backpropagation of errors is a common method used to train neural networks. An input vector is presented to the network for processing. The output of the network is compared to the desired output using a loss function and an error value is calculated for each of the neurons in the output layer. The error values are then propagated backwards until each neuron has an associated error value which roughly represents its contribution to the original output. The network can then learn from those errors using an algorithm, such as the stochastic gradient descent algorithm, to update the weights of the of the neural network.

[0209] FIG. 9A-9B illustrate an exemplary convolutional neural network. FIG. 9A illustrates various layers within a CNN. As shown in FIG. 9A, an exemplary CNN used to model image

processing can receive input **902** describing the red, green, and blue (RGB) components of an input image. The input **902** can be processed by multiple convolutional layers (e.g., convolutional layer **904**, convolutional layer **906**). The output from the multiple convolutional layers may optionally be processed by a set of fully connected layers **908**. Neurons in a fully connected layer have full connections to all activations in the previous layer, as previously described for a feedforward network. The output from the fully connected layers **908** can be used to generate an output result from the network. The activations within the fully connected layers **908** can be computed using matrix multiplication instead of convolution. Not all CNN implementations make use of fully connected layers **908**. For example, in some implementations the convolutional layer **906** can generate output for the CNN.

[0210] The convolutional layers are sparsely connected, which differs from traditional neural network configuration found in the fully connected layers **908**. Traditional neural network layers are fully connected, such that every output unit interacts with every input unit. However, the convolutional layers are sparsely connected because the output of the convolution of a field is input (instead of the respective state value of each of the nodes in the field) to the nodes of the subsequent layer, as illustrated. The kernels associated with the convolutional layers perform convolution operations, the output of which is sent to the next layer. The dimensionality reduction performed within the convolutional layers is one aspect that enables the CNN to scale to process large images.

[0211] FIG. **9B** illustrates exemplary computation stages within a convolutional layer of a CNN. Input to a convolutional layer **912** of a CNN can be processed in three stages of a convolutional layer **914**. The three stages can include a convolution stage **916**, a detector stage **918**, and a pooling stage **920**. The convolutional layer **914** can then output data to a successive convolutional layer. The final convolutional layer of the network can generate output feature map data or provide input to a fully connected layer, for example, to generate a classification value for the input to the CNN.

[0212] In the convolution stage **916** performs several convolutions in parallel to produce a set of linear activations. The convolution stage **916** can include an affine transformation, which is any transformation that can be specified as a linear transformation plus a translation. Affine transformations include rotations, translations, scaling, and combinations of these transformations. The convolution stage computes the output of functions (e.g., neurons) that are connected to specific regions in the input, which can be determined as the local region associated with the neuron. The neurons compute a dot product between the weights of the neurons and the region in the local input to which the neurons are connected. The output from the convolution stage **916** defines a set of linear activations that are processed by successive stages of the convolutional layer **914**.

[0213] The linear activations can be processed by a detector stage **918**. In the detector stage **918**, each linear activation is processed by a non-linear activation function. The non-linear activation function increases the nonlinear properties of the overall network without affecting the receptive fields of the convolution layer. Several types of non-linear activation functions may be used. One particular type is the rectified linear unit (ReLU), which uses an activation function defined as $f(x)=\max(0, x)$, such that the activation is thresholded at zero.

[0214] The pooling stage **920** uses a pooling function that replaces the output of the convolutional layer **906** with a summary statistic of the nearby outputs. The pooling function can be used to introduce translation invariance into the neural network, such that small translations to the input do not change the pooled outputs. Invariance to local translation can be useful in scenarios where the presence of a feature in the input data is more important than the precise location of the feature. Various types of pooling functions can be used during the pooling stage **920**, including max pooling, average pooling, and 12-norm pooling. Additionally, some CNN implementations do not include a pooling stage. Instead, such implementations substitute an additional convolution stage having an increased stride relative to previous convolution stages.

[0215] The output from the convolutional layer **914** can then be processed by the next layer **922**. The next layer **922** can be an additional convolutional layer or one of the fully connected layers **908**. For example, the first convolutional layer **904** of FIG. **9A** can output to the second convolutional layer **906**, while the second convolutional layer can output to a first layer of the fully connected layers **908**.

[0216] FIG. **10** illustrates an exemplary recurrent neural network. In a recurrent neural network (RNN), the previous state of the network influences the output of the current state of the network. RNNs can be built in a variety of ways using a variety of functions. The use of RNNs generally revolves around using mathematical models to predict the future based on a prior sequence of inputs. For example, an RNN may be used to perform statistical language modeling to predict an upcoming word given a previous sequence of words. The RNN **1000** of FIG. **10** can be described as having an input layer **1002** that receives an input vector, hidden layers **1004** to implement a recurrent function, a feedback mechanism **1005** to enable a ‘memory’ of previous states, and an output layer **1006** to output a result. The RNN **1000** operates based on time-steps. The state of the RNN at a given time step is influenced based on the previous time step via the feedback mechanism **1005**. For a given time step, the state of the hidden layers **1004** is defined by the previous state and the input at the current time step. An initial input ($x_{sub.1}$) at a first-time step can be processed by the hidden layer **1004**. A second input ($x_{sub.2}$) can be processed by the hidden layer **1004** using state information that is determined during the processing of the initial input ($x_{sub.1}$). A given state can be computed as $s_{sub.t} = f(Ux_{sub.t} + Ws_{sub.t-1})$, where U and W are parameter matrices. The function f is generally a nonlinearity, such as the hyperbolic tangent function (Tanh) or a variant of the rectifier function $f(x) = \max(0, x)$. However, the specific mathematical function used in the hidden layers **1004** can vary depending on the specific implementation details of the RNN **1000**.

[0217] In addition to the basic CNN and RNN networks described, acceleration for variations on those networks may be enabled. One example RNN variant is the long short term memory (LSTM) RNN. LSTM RNNs are capable of learning long-term dependencies that may be necessary for processing longer sequences of language. A variant on the CNN is a convolutional deep belief network, which has a structure similar to a CNN and is trained in a manner similar to a deep belief network. A deep belief network (DBN) is a generative neural network that is composed of multiple layers of stochastic (random) variables. DBNs can be trained layer-by-layer using greedy unsupervised learning. The learned weights of the DBN can then be used to provide pre-train neural networks by determining an optimal initial set of weights for the neural network. In further embodiments, acceleration for reinforcement learning is enabled. In reinforcement learning, an artificial agent learns by interacting with its environment. The agent is configured to optimize certain objectives to maximize cumulative rewards.

[0218] FIG. **11** illustrates training and deployment of a deep neural network. Once a given network has been structured for a task the neural network is trained using a training dataset **1102**. Various training frameworks **1104** have been developed to enable hardware acceleration of the training process. For example, the machine learning framework **604** of FIG. **6** may be configured as a training framework **1104**. The training framework **1104** can hook into an untrained neural network **1106** and enable the untrained neural net to be trained using the parallel processing resources described herein to generate a trained neural network **1108**.

[0219] To start the training process the initial weights may be chosen randomly or by pre-training using a deep belief network. The training cycle then be performed in either a supervised or unsupervised manner.

[0220] Supervised learning is a learning method in which training is performed as a mediated operation, such as when the training dataset **1102** includes input paired with the desired output for the input, or where the training dataset includes input having known output and the output of the neural network is manually graded. The network processes the inputs and compares the resulting

outputs against a set of expected or desired outputs. Errors are then propagated back through the system. The training framework **1104** can adjust to adjust the weights that control the untrained neural network **1106**. The training framework **1104** can provide tools to monitor how well the untrained neural network **1106** is converging towards a model suitable to generating correct answers based on known input data. The training process occurs repeatedly as the weights of the network are adjusted to refine the output generated by the neural network. The training process can continue until the neural network reaches a statistically desired accuracy associated with a trained neural network **1108**. The trained neural network **1108** can then be deployed to implement any number of machine learning operations to generate an inference result **1114** based on input of new data **1112**.

[0221] Unsupervised learning is a learning method in which the network attempts to train itself using unlabeled data. Thus, for unsupervised learning the training dataset **1102** will include input data without any associated output data. The untrained neural network **1106** can learn groupings within the unlabeled input and can determine how individual inputs are related to the overall dataset. Unsupervised training can be used to generate a self-organizing map, which is a type of trained neural network **1108** capable of performing operations useful in reducing the dimensionality of data. Unsupervised training can also be used to perform anomaly detection, which allows the identification of data points in an input dataset that deviate from the normal patterns of the data.

[0222] Variations on supervised and unsupervised training may also be employed. Semi-supervised learning is a technique in which in the training dataset **1102** includes a mix of labeled and unlabeled data of the same distribution. Incremental learning is a variant of supervised learning in which input data is continuously used to further train the model. Incremental learning enables the trained neural network **1108** to adapt to the new data **1112** without forgetting the knowledge instilled within the network during initial training.

[0223] Whether supervised or unsupervised, the training process for particularly deep neural networks may be too computationally intensive for a single compute node. Instead of using a single compute node, a distributed network of computational nodes can be used to accelerate the training process.

[0224] FIG. **12A** is a block diagram illustrating distributed learning. Distributed learning is a training model that uses multiple distributed computing nodes to perform supervised or unsupervised training of a neural network. The distributed computational nodes can each include one or more host processors and one or more of the general-purpose processing nodes, such as the GPGPU **700** as in FIG. **7**. As illustrated, distributed learning can be performed with model parallelism **1202**, data parallelism **1204**, or a combination of model and data parallelism **1206**.

[0225] In model parallelism **1202**, different computational nodes in a distributed system can perform training computations for different parts of a single network. For example, each layer of a neural network can be trained by a different processing node of the distributed system. The benefits of model parallelism include the ability to scale to particularly large models. Splitting the computations associated with different layers of the neural network enables the training of very large neural networks in which the weights of all layers would not fit into the memory of a single computational node. In some instances, model parallelism can be particularly useful in performing unsupervised training of large neural networks.

[0226] In data parallelism **1204**, the different nodes of the distributed network have a complete instance of the model and each node receives a different portion of the data. The results from the different nodes are then combined. While different approaches to data parallelism are possible, data parallel training approaches all require a technique of combining results and synchronizing the model parameters between each node. Exemplary approaches to combining data include parameter averaging and update-based data parallelism. Parameter averaging trains each node on a subset of the training data and sets the global parameters (e.g., weights, biases) to the average of the parameters from each node. Parameter averaging uses a central parameter server that maintains the

parameter data. Update based data parallelism is similar to parameter averaging except that instead of transferring parameters from the nodes to the parameter server, the updates to the model are transferred. Additionally, update-based data parallelism can be performed in a decentralized manner, where the updates are compressed and transferred between nodes.

[0227] Combined model and data parallelism **1206** can be implemented, for example, in a distributed system in which each computational node includes multiple GPUs. Each node can have a complete instance of the model with separate GPUs within each node are used to train different portions of the model.

[0228] Distributed training has increased overhead relative to training on a single machine.

However, the parallel processors and GPGPUs described herein can each implement various techniques to reduce the overhead of distributed training, including techniques to enable high bandwidth GPU-to-GPU data transfer and accelerated remote data synchronization.

[0229] FIG. **12B** is a block diagram illustrating a programmable network interface **1210** and data processing unit. The programmable network interface **1210** is a programmable network engine that can be used to accelerate network-based compute tasks within a distributed environment. The programmable network interface **1210** can couple with a host system via host interface **1270**. The programmable network interface **1210** can be used to accelerate network or storage operations for CPUs or GPUs of the host system. The host system can be, for example, a node of a distributed learning system used to perform distributed training, for example, as shown in FIG. **12A**. The host system can also be a data center node within a data center.

[0230] In one embodiment, access to remote storage containing model data can be accelerated by the programmable network interface **1210**. For example, the programmable network interface **1210** can be configured to present remote storage devices as local storage devices to the host system. The programmable network interface **1210** can also accelerate remote direct memory access (RDMA) operations performed between GPUs of the host system with GPUs of remote systems. In one embodiment, the programmable network interface **1210** can enable storage functionality such as, but not limited to NVME-oF. The programmable network interface **1210** can also accelerate encryption, data integrity, compression, and other operations for remote storage on behalf of the host system, allowing remote storage to approach the latencies of storage devices that are directly attached to the host system.

[0231] The programmable network interface **1210** can also perform resource allocation and management on behalf of the host system. Storage security operations can be offloaded to the programmable network interface **1210** and performed in concert with the allocation and management of remote storage resources. Network-based operations to manage access to the remote storage that would otherwise be performed by a processor of the host system can instead be performed by the programmable network interface **1210**.

[0232] In one embodiment, network and/or data security operations can be offloaded from the host system to the programmable network interface **1210**. Data center security policies for a data center node can be handled by the programmable network interface **1210** instead of the processors of the host system. For example, the programmable network interface **1210** can detect and mitigate against an attempted network-based attack (e.g., DDoS) on the host system, preventing the attack from compromising the availability of the host system.

[0233] The programmable network interface **1210** can include a system on a chip (SoC **1220**) that executes an operating system via multiple processor cores **1222**. The processor cores **1222** can include general-purpose processor (e.g., CPU) cores. In one embodiment the processor cores **1222** can also include one or more GPU cores. The SoC **1220** can execute instructions stored in a memory device **1240**. A storage device **1250** can store local operating system data. The storage device **1250** and memory device **1240** can also be used to cache remote data for the host system. Network ports **1260A-1260B** enable a connection to a network or fabric and facilitate network access for the SoC **1220** and, via the host interface **1270**, for the host system. The programmable

network interface **1210** can also include an I/O interface **1275**, such as a USB interface. The I/O interface **1275** can be used to couple external devices to the programmable network interface **1210** or as a debug interface. The programmable network interface **1210** also includes a management interface **1230** that enables software on the host device to manage and configure the programmable network interface **1210** and/or SoC **1220**. In one embodiment the programmable network interface **1210** may also include one or more accelerators or GPUs **1245** to accept offload of parallel compute tasks from the SoC **1220**, host system, or remote systems coupled via the network ports **1260A-1260B**.

Exemplary Machine Learning Applications

[0234] Machine learning can be applied to solve a variety of technological problems, including but not limited to computer vision, autonomous driving and navigation, speech recognition, and language processing. Computer vision has traditionally been one of the most active research areas for machine learning applications. Applications of computer vision range from reproducing human visual abilities, such as recognizing faces, to creating new categories of visual abilities. For example, computer vision applications can be configured to recognize sound waves from the vibrations induced in objects visible in a video. Parallel processor accelerated machine learning enables computer vision applications to be trained using significantly larger training dataset than previously feasible and enables inferencing systems to be deployed using low power parallel processors.

[0235] Parallel processor accelerated machine learning has autonomous driving applications including lane and road sign recognition, obstacle avoidance, navigation, and driving control. Accelerated machine learning techniques can be used to train driving models based on datasets that define the appropriate responses to specific training input. The parallel processors described herein can enable rapid training of the increasingly complex neural networks used for autonomous driving solutions and enables the deployment of low power inferencing processors in a mobile platform suitable for integration into autonomous vehicles.

[0236] Parallel processor accelerated deep neural networks have enabled machine learning approaches to automatic speech recognition (ASR). ASR includes the creation of a function that computes the most probable linguistic sequence given an input acoustic sequence. Accelerated machine learning using deep neural networks have enabled the replacement of the hidden Markov models (HMMs) and Gaussian mixture models (GMMs) previously used for ASR.

[0237] Parallel processor accelerated machine learning can also be used to accelerate natural language processing. Automatic learning procedures can make use of statistical inference algorithms to produce models that are robust to erroneous or unfamiliar input. Exemplary natural language processor applications include automatic machine translation between human languages.

[0238] The parallel processing platforms used for machine learning can be divided into training platforms and deployment platforms. Training platforms are generally highly parallel and include optimizations to accelerate multi-GPU single node training and multi-node, multi-GPU training. Exemplary parallel processors suited for training include the GPGPU **700** of FIG. **7** and the multi-GPU computing system **800** of FIG. **8**. On the contrary, deployed machine learning platforms generally include lower power parallel processors suitable for use in products such as cameras, autonomous robots, and autonomous vehicles.

[0239] Additionally, machine learning techniques can be applied to accelerate or enhance graphics processing activities. For example, a machine learning model can be trained to recognize output generated by a GPU accelerated application and generate an upscaled version of that output. Such techniques can be applied to accelerate the generation of high-resolution images for a gaming application. Various other graphics pipeline activities can benefit from the use of machine learning. For example, machine learning models can be trained to perform tessellation operations on geometry data to increase the complexity of geometric models, allowing fine-detailed geometry to be automatically generated from geometry of relatively lower detail.

[0240] FIG. 13 illustrates an exemplary inferencing system on a chip (SOC) 1300 suitable for performing inferencing using a trained model. The SOC 1300 can integrate processing components including a media processor 1302, a vision processor 1304, a GPGPU 1306 and a multi-core processor 1308. The GPGPU 1306 may be a GPGPU as described herein, such as the GPGPU 700, and the multi-core processor 1308 may be a multi-core processor described herein, such as the multi-core processors 405-406. The SOC 1300 can additionally include on-chip memory 1305 that can enable a shared on-chip data pool that is accessible by each of the processing components. The processing components can be optimized for low power operation to enable deployment to a variety of machine learning platforms, including autonomous vehicles and autonomous robots. For example, one implementation of the SOC 1300 can be used as a portion of the main control system for an autonomous vehicle. Where the SOC 1300 is configured for use in autonomous vehicles the SOC is designed and configured for compliance with the relevant functional safety standards of the deployment jurisdiction.

[0241] During operation, the media processor 1302 and vision processor 1304 can work in concert to accelerate computer vision operations. The media processor 1302 can enable low latency decode of multiple high-resolution (e.g., 4K, 8K) video streams. The decoded video streams can be written to a buffer in the on-chip memory 1305. The vision processor 1304 can then parse the decoded video and perform preliminary processing operations on the frames of the decoded video in preparation of processing the frames using a trained image recognition model. For example, the vision processor 1304 can accelerate convolution operations for a CNN that is used to perform image recognition on the high-resolution video data, while back-end model computations are performed by the GPGPU 1306.

[0242] The multi-core processor 1308 can include control logic to assist with sequencing and synchronization of data transfers and shared memory operations performed by the media processor 1302 and the vision processor 1304. The multi-core processor 1308 can also function as an application processor to execute software applications that can make use of the inferencing compute capability of the GPGPU 1306. For example, at least a portion of the navigation and driving logic can be implemented in software executing on the multi-core processor 1308. Such software can directly issue computational workloads to the GPGPU 1306 or the computational workloads can be issued to the multi-core processor 1308, which can offload at least a portion of those operations to the GPGPU 1306.

[0243] The GPGPU 1306 can include compute clusters such as a low power configuration of the processing clusters 706A-706H within the GPGPU 700. The compute clusters within the GPGPU 1306 can support instruction that are specifically optimized to perform inferencing computations on a trained neural network. For example, the GPGPU 1306 can support instructions to perform low precision computations such as 8-bit and 4-bit integer vector operations.

Additional System Overview

[0244] FIG. 14 is a block diagram of a processing system 1400. The elements of FIG. 14 having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such. System 1400 may be used in a single processor desktop system, a multiprocessor workstation system, or a server system having a large number of processors 1402 or processor cores 1407. The system 1400 may be a processing platform incorporated within a system-on-a-chip (SoC) integrated circuit for use in mobile, handheld, or embedded devices such as within Internet-of-things (IoT) devices with wired or wireless connectivity to a local or wide area network.

[0245] The system 1400 may be a processing system having components that correspond with those of FIG. 1. For example, in different configurations, processor(s) 1402 or processor core(s) 1407 may correspond with processor(s) 102 of FIG. 1. Graphics processor(s) 1408 may correspond with parallel processor(s) 112 of FIG. 1. External graphics processor 1418 may be one of the add-in

device(s) **120** of FIG. 1.

[0246] The system **1400** can include, couple with, or be integrated within: a server-based gaming platform; a game console, including a game and media console; a mobile gaming console, a handheld game console, or an online game console. The system **1400** may be part of a mobile phone, smart phone, tablet computing device or mobile Internet-connected device such as a laptop with low internal storage capacity. Processing system **1400** can also include, couple with, or be integrated within: a wearable device, such as a smart watch wearable device; smart eyewear or clothing enhanced with augmented reality (AR) or virtual reality (VR) features to provide visual, audio or tactile outputs to supplement real world visual, audio or tactile experiences or otherwise provide text, audio, graphics, video, holographic images or video, or tactile feedback; other augmented reality (AR) device; or other virtual reality (VR) device. The processing system **1400** may include or be part of a television or set top box device. The system **1400** can include, couple with, or be integrated within a self-driving vehicle such as a bus, tractor trailer, car, motor or electric power cycle, plane or glider (or any combination thereof). The self-driving vehicle may use system **1400** to process the environment sensed around the vehicle.

[0247] The one or more processors **1402** may include one or more processor cores **1407** to process instructions which, when executed, perform operations for system or user software. The least one of the one or more processor cores **1407** may be configured to process a specific instruction set **1409**. The instruction set **1409** may facilitate Complex Instruction Set Computing (CISC), Reduced Instruction Set Computing (RISC), or computing via a Very Long Instruction Word (VLIW). One or more processor cores **1407** may process a different instruction set **1409**, which may include instructions to facilitate the emulation of other instruction sets. Processor core **1407** may also include other processing devices, such as a Digital Signal Processor (DSP).

[0248] The processor **1402** may include cache memory **1404**. Depending on the architecture, the processor **1402** can have a single internal cache or multiple levels of internal cache. In some embodiments, the cache memory is shared among various components of the processor **1402**. In some embodiments, the processor **1402** also uses an external cache (e.g., a Level-3 (L3) cache or Last Level Cache (LLC)) (not shown), which may be shared among processor cores **1407** using known cache coherency techniques. A register file **1406** can be additionally included in processor **1402** and may include different types of registers for storing different types of data (e.g., integer registers, floating point registers, status registers, and an instruction pointer register). Some registers may be general-purpose registers, while other registers may be specific to the design of the processor **1402**.

[0249] The one or more processor(s) **1402** may be coupled with one or more interface bus(es) **1410** to transmit communication signals such as address, data, or control signals between processor **1402** and other components in the system **1400**. The interface bus **1410**, in one of these embodiments, can be a processor bus, such as a version of the Direct Media Interface (DMI) bus. However, processor busses are not limited to the DMI bus, and may include one or more Peripheral Component Interconnect buses (e.g., PCI, PCI express), memory busses, or other types of interface busses. For example, the processor(s) **1402** may include an integrated memory controller **1416** and a platform controller hub **1430**. The memory controller **1416** facilitates communication between a memory device and other components of the system **1400**, while the platform controller hub (PCH) **1430** provides connections to I/O devices via a local I/O bus.

[0250] The memory device **1420** can be a dynamic random-access memory (DRAM) device, a static random-access memory (SRAM) device, flash memory device, phase-change memory device, or some other memory device having suitable performance to serve as process memory. The memory device **1420** can, for example, operate as system memory for the system **1400**, to store data **1422** and instructions **1421** for use when the one or more processors **1402** executes an application or process. Memory controller **1416** also couples with an optional external graphics processor **1418**, which may communicate with the one or more graphics processors **1408** in

processors **1402** to perform graphics and media operations. In some embodiments, graphics, media, and or compute operations may be assisted by an accelerator **1412** which is a coprocessor that can be configured to perform a specialized set of graphics, media, or compute operations. For example, the accelerator **1412** may be a matrix multiplication accelerator used to optimize machine learning or compute operations. The accelerator **1412** can be a ray-tracing accelerator that can be used to perform ray-tracing operations in concert with the graphics processor **1408**. In one embodiment, an external accelerator **1419** may be used in place of or in concert with the accelerator **1412**.

[0251] A display device **1411** may be provided that can connect to the processor(s) **1402**. The display device **1411** can be one or more of an internal display device, as in a mobile electronic device or a laptop device or an external display device attached via a display interface (e.g., DisplayPort, etc.). The display device **1411** can be a head mounted display (HM D) such as a stereoscopic display device for use in virtual reality (VR) applications or augmented reality (AR) applications.

[0252] The platform controller hub **1430** may enable peripherals to connect to memory device **1420** and processor **1402** via a high-speed I/O bus. The I/O peripherals include, but are not limited to, an audio controller **1446**, a network controller **1434**, a firmware interface **1428**, a wireless transceiver **1426**, touch sensors **1425**, a data storage device **1424** (e.g., non-volatile memory, volatile memory, hard disk drive, flash memory, NAND, 3D NAND, 3D X Point/Optane, etc.). The data storage device **1424** can connect via a storage interface (e.g., SATA) or via a peripheral bus, such as a Peripheral Component Interconnect bus (e.g., PCI, PCI express). The touch sensors **1425** can include touch screen sensors, pressure sensors, or fingerprint sensors. The wireless transceiver **1426** can be a Wi-Fi transceiver, a Bluetooth transceiver, or a mobile network transceiver such as a 3G, 4G, 5G, or Long-Term Evolution (LTE) transceiver. The firmware interface **1428** enables communication with system firmware, and can be, for example, a unified extensible firmware interface (UEFI). The network controller **1434** can enable a network connection to a wired network. In some embodiments, a high-performance network controller (not shown) couples with the interface bus **1410**. The audio controller **1446** may be a multi-channel high-definition audio controller. In some of these embodiments the system **1400** includes an optional legacy I/O controller **1440** for coupling legacy (e.g., Personal System 2 (PS/2)) devices to the system. The platform controller hub **1430** can also connect to one or more Universal Serial Bus (USB) controllers **1442** connect input devices, such as keyboard and mouse **1443** combinations, a camera **1444**, or other USB input devices.

[0253] It will be appreciated that the system **1400** shown is exemplary and not limiting, as other types of data processing systems that are differently configured may also be used. For example, an instance of the memory controller **1416** and platform controller hub **1430** may be integrated into a discrete external graphics processor, such as the external graphics processor **1418**. The platform controller hub **1430** and/or memory controller **1416** may be external to the one or more processor(s) **1402**. For example, the system **1400** can include an external memory controller **1416** and platform controller hub **1430**, which may be configured as a memory controller hub and peripheral controller hub within a system chipset that is in communication with the processor(s) **1402**.

[0254] For example, circuit boards (“sleds”) can be used on which components such as CPUs, memory, and other components are placed are designed for increased thermal performance. Processing components such as the processors may be located on a top side of a sled while near memory, such as DIM Ms, are located on a bottom side of the sled. As a result of the enhanced airflow provided by this design, the components may operate at higher frequencies and power levels than in typical systems, thereby increasing performance. Furthermore, the sleds are configured to blindly mate with power and data communication cables in a rack, thereby enhancing their ability to be quickly removed, upgraded, reinstalled, and/or replaced. Similarly, individual components located on the sleds, such as processors, accelerators, memory, and data storage drives,

are configured to be easily upgraded due to their increased spacing from each other. In the illustrative embodiment, the components additionally include hardware attestation features to prove their authenticity.

[0255] A data center can utilize a single network architecture (“fabric”) that supports multiple other network architectures including Ethernet and Omni-Path. The sleds can be coupled to switches via optical fibers, which provide higher bandwidth and lower latency than typical twisted pair cabling (e.g., Category 5, Category 5e, Category 6, etc.). Due to the high bandwidth, low latency interconnections and network architecture, the data center may, in use, pool resources, such as memory, accelerators (e.g., GPUs, graphics accelerators, FPGAs, ASICs, neural network and/or artificial intelligence accelerators, etc.), and data storage drives that are physically disaggregated, and provide them to compute resources (e.g., processors) on an as needed basis, enabling the compute resources to access the pooled resources as if they were local.

[0256] A power supply or source can provide voltage and/or current to system **1400** or any component or system described herein. In one example, the power supply includes an AC to DC (alternating current to direct current) adapter to plug into a wall outlet. Such AC power can be renewable energy (e.g., solar power) power source. In one example, the power source includes a DC power source, such as an external AC to DC converter. A power source or power supply may also include wireless charging hardware to charge via proximity to a charging field. The power source can include an internal battery, alternating current supply, motion-based power supply, solar power supply, or fuel cell source.

[0257] FIG. **15A-15C** illustrate computing systems and graphics processors. The elements of FIG. **15A-15C** having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such.

[0258] FIG. **15A** is a block diagram of a processor **1500**, which may be a variant of one of the processors **1402** and may be used in place of one of those. Therefore, the disclosure of any features in combination with the processor **1500** herein also discloses a corresponding combination with the processor(s) **1402** but is not limited to such. The processor **1500** may have one or more processor cores **1502A-1502N**, an integrated memory controller **1514**, and an integrated graphics processor **1508**. Where an integrated graphics processor **1508** is excluded, the system that includes the processor will include a graphics processor device within a system chipset or coupled via a system bus. Processor **1500** can include additional cores up to and including additional core **1502N** represented by the dashed lined boxes. Each of processor cores **1502A-1502N** includes one or more internal cache units **1504A-1504N**. In some embodiments each processor core **1502A-1502N** also has access to one or more shared cache units **1506**. The internal cache units **1504A-1504N** and shared cache units **1506** represent a cache memory hierarchy within the processor **1500**. The cache memory hierarchy may include at least one level of instruction and data cache within each processor core and one or more levels of shared mid-level cache, such as a Level 2 (L2), Level 3 (L3), Level 4 (L4), or other levels of cache, where the highest level of cache before external memory is classified as the LLC. In some embodiments, cache coherency logic maintains coherency between the various cache units **1506** and **1504A-1504N**.

[0259] The processor **1500** may also include a set of one or more bus controller units **1516** and a system agent core **1510**. The one or more bus controller units **1516** manage a set of peripheral buses, such as one or more PCI or PCI express busses. System agent core **1510** provides management functionality for the various processor components. The system agent core **1510** may include one or more integrated memory controllers **1514** to manage access to various external memory devices (not shown).

[0260] For example, one or more of the processor cores **1502A-1502N** may include support for simultaneous multi-threading. The system agent core **1510** includes components for coordinating

and operating cores **1502A-1502N** during multi-threaded processing. System agent core **1510** may additionally include a power control unit (PCU), which includes logic and components to regulate the power state of processor cores **1502A-1502N** and graphics processor **1508**.

[0261] The processor **1500** may additionally include graphics processor **1508** to execute graphics processing operations. In some of these embodiments, the graphics processor **1508** couples with the set of shared cache units **1506**, and the system agent core **1510**, including the one or more integrated memory controllers **1514**. The system agent core **1510** may also include a display controller **1511** to drive graphics processor output to one or more coupled displays. The display controller **1511** may also be a separate module coupled with the graphics processor via at least one interconnect or may be integrated within the graphics processor **1508**.

[0262] A ring-based interconnect **1512** may be used to couple the internal components of the processor **1500**. However, an alternative interconnect unit may be used, such as a point-to-point interconnect, a switched interconnect, or other techniques, including techniques well known in the art. In some of these embodiments with a ring-based interconnect **1512**, the graphics processor **1508** couples with the ring-based interconnect **1512** via an I/O link **1513**.

[0263] The exemplary I/O link **1513** represents at least one of multiple varieties of I/O interconnects, including an on package I/O interconnect which facilitates communication between various processor components and a high-performance memory module **1518**, such as an eDRAM module or a high-bandwidth memory (HBM) module. Optionally, each of the processor cores **1502A-1502N** and graphics processor **1508** can use the high-performance memory module **1518** as a shared Last Level Cache.

[0264] The processor cores **1502A-1502N** may, for example, be homogenous cores executing the same instruction set architecture. Alternatively, the processor cores **1502A-1502N** are heterogeneous in terms of instruction set architecture (ISA), where one or more of processor cores **1502A-1502N** execute a first instruction set, while at least one of the other cores executes a subset of the first instruction set or a different instruction set. The processor cores **1502A-1502N** may be heterogeneous in terms of microarchitecture, where one or more cores having a relatively higher power consumption couple with one or more power cores having a lower power consumption. As another example, the processor cores **1502A-1502N** are heterogeneous in terms of computational capability. Additionally, processor **1500** can be implemented on one or more chips or as an SoC integrated circuit having the illustrated components, in addition to other components.

[0265] FIG. **15B** is a block diagram of hardware logic of a graphics processor core block **1519**, according to some embodiments described herein. In some embodiments, elements of FIG. **15B** having the same reference numbers (or names) as the elements of any other figure herein may operate or function in a manner similar to that described elsewhere herein. In one embodiment, the graphics processor core block **1519** is exemplary of one partition of a graphics processor. The graphics processor core block **1519** can be included within the integrated graphics processor **1508** of FIG. **15A** or a discrete graphics processor, parallel processor, and/or compute accelerator. A graphics processor as described herein may include multiple graphics core blocks based on target power and performance envelopes. Each graphics processor core block **1519** can include a function block **1530** coupled with multiple graphics cores **1521A-1521F** that include modular blocks of fixed function logic and general-purpose programmable logic. The graphics processor core block **1519** also includes shared/cache memory **1536** that is accessible by all graphics cores **1521A-1521F**, rasterizer logic **1537**, and additional fixed function logic **1538**.

[0266] In some embodiments, the function block **1530** includes a geometry/fixed function pipeline **1531** that can be shared by all graphics cores in the graphics processor core block **1519**. In various embodiments, the geometry/fixed function pipeline **1531** includes a 3D geometry pipeline a video front-end unit, a thread spawner and global thread dispatcher, and a unified return buffer manager, which manages unified return buffers. In one embodiment the function block **1530** also includes a graphics SoC interface **1532**, a graphics microcontroller **1533**, and a media pipeline **1534**. The

graphics SoC interface **1532** provides an interface between the graphics processor core block **1519** and other core blocks within a graphics processor or compute accelerator SoC. The graphics microcontroller **1533** is a programmable sub-processor that is configurable to manage various functions of the graphics processor core block **1519**, including thread dispatch, scheduling, and pre-emption. The media pipeline **1534** includes logic to facilitate the decoding, encoding, pre-processing, and/or post-processing of multimedia data, including image and video data. The media pipeline **1534** implement media operations via requests to compute or sampling logic within the graphics cores **1521-1521F**. One or more pixel backends **1535** can also be included within the function block **1530**. The pixel backends **1535** include a cache memory to store pixel color values and can perform blend operations and lossless color compression of rendered pixel data.

[0267] In one embodiment the graphics SoC interface **1532** enables the graphics processor core block **1519** to communicate with general-purpose application processor cores (e.g., CPUs) and/or other components within an SoC or a system host CPU that is coupled with the SoC via a peripheral interface. The graphics SoC interface **1532** also enables communication with off-chip memory hierarchy elements such as a shared last level cache memory, system RAM, and/or embedded on-chip or on-package DRAM. The SoC interface **1532** can also enable communication with fixed function devices within the SoC, such as camera imaging pipelines, and enables the use of and/or implements global memory atomics that may be shared between the graphics processor core block **1519** and CPUs within the SoC. The graphics SoC interface **1532** can also implement power management controls for the graphics processor core block **1519** and enable an interface between a clock domain of the graphics processor core block **1519** and other clock domains within the SoC. In one embodiment the graphics SoC interface **1532** enables receipt of command buffers from a command streamer and global thread dispatcher that are configured to provide commands and instructions to each of one or more graphics cores within a graphics processor. The commands and instructions can be dispatched to the media pipeline **1534** when media operations are to be performed, the geometry and fixed function pipeline **1531** when graphics processing operations are to be performed. When compute operations are to be performed, compute dispatch logic can dispatch the commands to the graphics cores **1521A-1521F**, bypassing the geometry and media pipelines.

[0268] The graphics microcontroller **1533** can be configured to perform various scheduling and management tasks for the graphics processor core block **1519**. In one embodiment the graphics microcontroller **1533** can perform graphics and/or compute workload scheduling on the various vector engines **1522A-1522F**, **1524A-1524F** and matrix engines **1523A-1523F**, **1525A-1525F** within the graphics cores **1521A-1521F**. In this scheduling model, host software executing on a CPU core of an SoC including the graphics processor core block **1519** can submit workloads one of multiple graphics processor doorbells, which invokes a scheduling operation on the appropriate graphics engine. Scheduling operations include determining which workload to run next, submitting a workload to a command streamer, pre-empting existing workloads running on an engine, monitoring progress of a workload, and notifying host software when a workload is complete. In one embodiment the graphics microcontroller **1533** can also facilitate low-power or idle states for the graphics processor core block **1519**, providing the graphics processor core block **1519** with the ability to save and restore registers within the graphics processor core block **1519** across low-power state transitions independently from the operating system and/or graphics driver software on the system.

[0269] The graphics processor core block **1519** may have greater than or fewer than the illustrated graphics cores **1521A-1521F**, up to N modular graphics cores. For each set of N graphics cores, the graphics processor core block **1519** can also include shared/cache memory **1536**, which can be configured as shared memory or cache memory, rasterizer logic **1537**, and additional fixed function logic **1538** to accelerate various graphics and compute processing operations.

[0270] Within each graphics cores **1521A-1521F** is set of execution resources that may be used to

perform graphics, media, and compute operations in response to requests by graphics pipeline, media pipeline, or shader programs. The graphics cores **1521A-1521F** include multiple vector engines **1522A-1522F**, **1524A-1524F**, matrix acceleration units **1523A-1523F**, **1525A-1525D**, cache/shared local memory (SLM), a sampler **1526A-1526F**, and a ray tracing unit **1527A-1527F**. [0271] The vector engines **1522A-1522F**, **1524A-1524F** are general-purpose graphics processing units capable of performing floating-point and integer/fixed-point logic operations in service of a graphics, media, or compute operation, including graphics, media, or compute/GPGPU programs. The vector engines **1522A-1522F**, **1524A-1524F** can operate at variable vector widths using SIMD, SIMT, or SIMT+SIMD execution modes. The matrix acceleration units **1523A-1523F**, **1525A-1525D** include matrix-matrix and matrix-vector acceleration logic that improves performance on matrix operations, particularly low and mixed precision (e.g., INT8, FP16, BF16, FP8) matrix operations used for machine learning. In one embodiment, each of the matrix acceleration units **1523A-1523F**, **1525A-1525D** includes one or more systolic arrays of processing elements that can perform concurrent matrix multiply or dot product operations on matrix elements.

[0272] The sampler **1526A-1526F** can read media or texture data into memory and can sample data differently based on a configured sampler state and the texture/media format that is being read. Threads executing on the vector engines **1522A-1522F**, **1524A-1524F** or matrix acceleration units **1523A-1523F**, **1525A-1525D** can make use of the cache/SLM **1528A-1528F** within each of the graphics cores **1521A-1521F**. The cache/SLM **1528A-1528F** can be configured as cache memory or as a pool of shared memory that is local to each of the respective graphics cores **1521A-1521F**. The ray tracing units **1527A-1527F** within the graphics cores **1521A-1521F** include ray traversal/intersection circuitry for performing ray traversal using bounding volume hierarchies (BVHs) and identifying intersections between rays and primitives enclosed within the BVH volumes. In one embodiment the ray tracing units **1527A-1527F** include circuitry for performing depth testing and culling (e.g., using a depth buffer or similar arrangement). In one implementation, the ray tracing units **1527A-1527F** perform traversal and intersection operations in concert with image denoising, at least a portion of which may be performed using an associated matrix acceleration unit **1523A-1523F**, **1525A-1525D**.

[0273] FIG. **15C** is a block diagram of general-purpose graphics processing unit (GPGPU) **1570** that can be configured as a graphics processor, e.g., the graphics processor **1508**, and/or compute accelerator, according to embodiments described herein. The GPGPU **1570** can interconnect with host processors (e.g., one or more CPU(s) **1546**) and memory **1571**, **1572** via one or more system and/or memory busses. Memory **1571** may be system memory that can be shared with the one or more CPU(s) **1546**, while memory **1572** is device memory that is dedicated to the GPGPU **1570**. For example, components within the GPGPU **1570** and memory **1572** may be mapped into memory addresses that are accessible to the one or more CPU(s) **1546**. Access to memory **1571** and **1572** may be facilitated via a memory controller **1568**. The memory controller **1568** may include an internal direct memory access (DMA) controller **1569** or can include logic to perform operations that would otherwise be performed by a DMA controller.

[0274] The GPGPU **1570** includes multiple cache memories, including an L2 cache **1553**, L1 cache **1554**, an instruction cache **1555**, and shared memory **1556**, at least a portion of which may also be partitioned as a cache memory. The GPGPU **1570** also includes multiple compute units **1560A-1560N**. Each compute unit **1560A-1560N** includes a set of vector registers **1561**, scalar registers **1562**, vector logic units **1563**, and scalar logic units **1564**. The compute units **1560A-1560N** can also include local shared memory **1565** and a program counter **1566**. The compute units **1560A-1560N** can couple with a constant cache **1567**, which can be used to store constant data, which is data that will not change during the run of kernel or shader program that executes on the GPGPU **1570**. The constant cache **1567** may be a scalar data cache and cached data can be fetched directly into the scalar registers **1562**.

[0275] During operation, the one or more CPU(s) **1546** can write commands into registers or

memory in the GPGPU **1570** that has been mapped into an accessible address space. The command processors **1557** can read the commands from registers or memory and determine how those commands will be processed within the GPGPU **1570**. A thread dispatcher **1558** can then be used to dispatch threads to the compute units **1560A-1560N** to perform those commands. Each compute unit **1560A-1560N** can execute threads independently of the other compute units. Additionally, each compute unit **1560A-1560N** can be independently configured for conditional computation and can conditionally output the results of computation to memory. The command processors **1557** can interrupt the one or more CPU(s) **1546** when the submitted commands are complete.

[0276] FIG. **16A-16C** illustrate block diagrams of additional graphics processor and compute accelerator architectures provided by embodiments described herein, e.g., in accordance with FIG. **15A-15C**. The elements of FIG. **16A-16C** having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such.

[0277] FIG. **16A** is a block diagram of a graphics processor **1600**, which may be a discrete graphics processing unit, or may be a graphics processor integrated with a plurality of processing cores, or other semiconductor devices such as, but not limited to, memory devices or network interfaces. The graphics processor **1600** may be a variant of the graphics processor **1508** and may be used in place of the graphics processor **1508**. Therefore, the disclosure of any features in combination with the graphics processor **1508** herein also discloses a corresponding combination with the graphics processor **1600** but is not limited to such. The graphics processor may communicate via a memory mapped I/O interface to registers on the graphics processor and with commands placed into the processor memory. Graphics processor **1600** may include a memory interface **1614** to access memory. Memory interface **1614** can be an interface to local memory, one or more internal caches, one or more shared external caches, and/or to system memory.

[0278] Optionally, graphics processor **1600** also includes a display controller **1602** to drive display output data to a display device **1618**. Display controller **1602** includes hardware for one or more overlay planes for the display and composition of multiple layers of video or user interface elements. The display device **1618** can be an internal or external display device. In one embodiment the display device **1618** is a head mounted display device, such as a virtual reality (VR) display device or an augmented reality (AR) display device. Graphics processor **1600** may include a video codec engine **1606** to encode, decode, or transcode media to, from, or between one or more media encoding formats, including, but not limited to Moving Picture Experts Group (MPEG) formats such as MPEG-2, Advanced Video Coding (AVC) formats such as H.264/MPEG-4 AVC, H.265/HEVC, Alliance for Open Media (AOMedia) VP8, VP9, as well as the Society of Motion Picture & Television Engineers (SMPTE) 421M/VC-1, and Joint Photographic Experts Group (JPEG) formats such as JPEG, and Motion JPEG (MJPEG) formats.

[0279] Graphics processor **1600** may include a block image transfer (BLIT) engine **1603** to perform two-dimensional (2D) rasterizer operations including, for example, bit-boundary block transfers. However, alternatively, 2D graphics operations may be performed using one or more components of graphics processing engine (GPE **1610**). In some embodiments, GPE **1610** is a compute engine for performing graphics operations, including three-dimensional (3D) graphics operations and media operations.

[0280] GPE **1610** may include a 3D pipeline **1612** for performing 3D operations, such as rendering three-dimensional images and scenes using processing functions that act upon 3D primitive shapes (e.g., rectangle, triangle, etc.). The 3D pipeline **1612** includes programmable and fixed function elements that perform various tasks within the element and/or spawn execution threads to a 3D/Media subsystem **1615**. While 3D pipeline **1612** can be used to perform media operations, an embodiment of GPE **1610** also includes a media pipeline **1616** that is specifically used to perform media operations, such as video post-processing and image enhancement.

[0281] Media pipeline **1616** may include fixed function or programmable logic units to perform one or more specialized media operations, such as video decode acceleration, video de-interlacing, and video encode acceleration in place of, or on behalf of video codec engine **1606**. Media pipeline **1616** may additionally include a thread spawning unit to spawn threads for execution on 3D/Media subsystem **1615**. The spawned threads perform computations for the media operations on one or more graphics execution units included in 3D/Media subsystem **1615**.

[0282] The 3D/Media subsystem **1615** may include logic for executing threads spawned by 3D pipeline **1612** and media pipeline **1616**. The pipelines may send thread execution requests to 3D/Media subsystem **1615**, which includes thread dispatch logic for arbitrating and dispatching the various requests to available thread execution resources. The execution resources include an array of graphics execution resources to process the 3D and media threads. The 3D/Media subsystem **1615** may include one or more internal caches for thread instructions and data. Additionally, the 3D/Media subsystem **1615** may also include shared memory, including registers and addressable memory, to share data between threads and to store output data.

[0283] FIG. **16B** illustrates a graphics processor **1620**, being a variant of the graphics processor **1600** and may be used in place of the graphics processor **1600** and vice versa. Therefore, the disclosure of any features in combination with the graphics processor **1600** herein also discloses a corresponding combination with the graphics processor **1620** but is not limited to such. The graphics processor **1620** has a tiled architecture, according to embodiments described herein. The graphics processor **1620** may include a graphics processing engine cluster **1622** having multiple instances of the GPE **1610** of FIG. **16A** within a graphics engine tile **1610A-1610D**. Each graphics engine tile **1610A-1610D** can be interconnected via a set of tile interconnects **1623A-1623F**. Each graphics engine tile **1610A-1610D** can also be connected to a memory module or memory device **1626A-1626D** via memory interconnects **1625A-1625D**. The memory devices **1626A-1626D** can use any graphics memory technology. For example, the memory devices **1626A-1626D** may be graphics double data rate (GDDR) memory. The memory devices **1626A-1626D** may be high-bandwidth memory (HBM) modules that can be on-die with their respective graphics engine tile **1610A-1610D**. The memory devices **1626A-1626D** may be stacked memory devices that can be stacked on top of their respective graphics engine tile **1610A-1610D**. Each graphics engine tile **1610A-1610D** and associated memory **1626A-1626D** may reside on separate chiplets, which are bonded to a base die or base substrate, as described in further detail in FIG. **24B-24D**.

[0284] The graphics processor **1620** may be configured with a non-uniform memory access (NUMA) system in which memory devices **1626A-1626D** are coupled with associated graphics engine tiles **1610A-1610D**. A given memory device may be accessed by graphics engine tiles other than the tile to which it is directly connected. However, access latency to the memory devices **1626A-1626D** may be lowest when accessing a local tile. In one embodiment, a cache coherent NUMA (ccNUMA) system is enabled that uses the tile interconnects **1623A-1623F** to enable communication between cache controllers within the graphics engine tiles **1610A-1610D** to keep a consistent memory image when more than one cache stores the same memory location.

[0285] The graphics processing engine cluster **1622** can connect with an on-chip or on-package fabric interconnect **1624**. In one embodiment the fabric interconnect **1624** includes a network processor, network on a chip (NoC), or another switching processor to enable the fabric interconnect **1624** to act as a packet switched fabric interconnect that switches data packets between components of the graphics processor **1620**. The fabric interconnect **1624** can enable communication between graphics engine tiles **1610A-1610D** and components such as the video codec engine **1606** and one or more copy engines **1604**. The copy engines **1604** can be used to move data out of, into, and between the memory devices **1626A-1626D** and memory that is external to the graphics processor **1620** (e.g., system memory). The fabric interconnect **1624** can also be used to interconnect the graphics engine tiles **1610A-1610D**. The graphics processor **1620** may optionally include a display controller **1602** to enable a connection with an external display

device **1618**. The graphics processor may also be configured as a graphics or compute accelerator. In the accelerator configuration, the display controller **1602** and display device **1618** may be omitted.

[0286] The graphics processor **1620** can connect to a host system via a host interface **1628**. The host interface **1628** can enable communication between the graphics processor **1620**, system memory, and/or other system components. The host interface **1628** can be, for example, a PCI express bus or another type of host system interface. For example, the host interface **1628** may be an NV Link or NV Switch interface. The host interface **1628** and fabric interconnect **1624** can cooperate to enable multiple instances of the graphics processor **1620** to act as single logical device. Cooperation between the host interface **1628** and fabric interconnect **1624** can also enable the individual graphics engine tiles **1610A-1610D** to be presented to the host system as distinct logical graphics devices.

[0287] FIG. **16C** illustrates a compute accelerator **1630**, according to embodiments described herein. The compute accelerator **1630** can include architectural similarities with the graphics processor **1620** of FIG. **16B** and is optimized for compute acceleration. A compute engine cluster **1632** can include a set of compute engine tiles **1640A-1640D** that include execution logic that is optimized for parallel or vector-based general-purpose compute operations. The compute engine tiles **1640A-1640D** may not include fixed function graphics processing logic, although in some embodiments one or more of the compute engine tiles **1640A-1640D** can include logic to perform media acceleration. The compute engine tiles **1640A-1640D** can connect to memory **1626A-1626D** via memory interconnects **1625A-1625D**. The memory **1626A-1626D** and memory interconnects **1625A-1625D** may be similar technology as in graphics processor **1620** or can be different. The graphics compute engine tiles **1640A-1640D** can also be interconnected via a set of tile interconnects **1623A-1623F** and may be connected with and/or interconnected by a fabric interconnect **1624**. In one embodiment the compute accelerator **1630** includes a large L3 cache **1636** that can be configured as a device-wide cache. The compute accelerator **1630** can also connect to a host processor and memory via a host interface **1628** in a similar manner as the graphics processor **1620** of FIG. **16B**.

[0288] The compute accelerator **1630** can also include an integrated network interface **1642**. In one embodiment the integrated network interface **1642** includes a network processor and controller logic that enables the compute engine cluster **1632** to communicate over a physical layer interconnect **1644** without requiring data to traverse memory of a host system. In one embodiment, one of the compute engine tiles **1640A-1640D** is replaced by network processor logic and data to be transmitted or received via the physical layer interconnect **1644** may be transmitted directly to or from memory **1626A-1626D**. Multiple instances of the compute accelerator **1630** may be joined via the physical layer interconnect **1644** into a single logical device. Alternatively, the various compute engine tiles **1640A-1640D** may be presented as distinct network accessible compute accelerator devices.

Graphics Processing Engine

[0289] FIG. **17** is a block diagram of a graphics processing engine of a graphics processor in accordance with some embodiments. The graphics processing engine (GPE **1710**) may be a version of the GPE **1610** shown in FIG. **16A** and may also represent a graphics engine tile **1610A-1610D** of FIG. **16B**. The elements of FIG. **17** having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such. For example, the 3D pipeline **1612** and media pipeline **1616** of FIG. **16A** are also illustrated in FIG. **17**. The media pipeline **1616** is optional in some embodiments of the GPE **1710** and may not be explicitly included within the GPE **1710**. For example and in at least one embodiment, a separate media and/or image processor is coupled to the GPE **1710**.

[0290] GPE **1710** may couple with or include a command streamer **1703**, which provides a command stream to the 3D pipeline **1612** and/or media pipelines **1616**. Alternatively or additionally, the command streamer **1703** may be directly coupled to a unified return buffer (URB **1718**). The URB **1718** may be communicatively coupled to a graphics core cluster **1714**. Optionally, the command streamer **1703** is coupled with memory, which can be system memory, or one or more of internal cache memory and shared cache memory. The command streamer **1703** may receive commands from the memory and sends the commands to 3D pipeline **1612** and/or media pipeline **1616**. The commands are directives fetched from a ring buffer, which stores commands for the 3D pipeline **1612** and media pipeline **1616**. The ring buffer can additionally include batch command buffers storing batches of multiple commands. The commands for the 3D pipeline **1612** can also include references to data stored in memory, such as but not limited to vertex and geometry data for the 3D pipeline **1612** and/or image data and memory objects for the media pipeline **1616**. The 3D pipeline **1612** and media pipeline **1616** process the commands and data by performing operations via logic within the respective pipelines or by dispatching one or more execution threads to the graphics core cluster **1714**. The graphics core cluster **1714** may include one or more blocks of graphics cores (e.g., graphics core block **1715A**, graphics core block **1715B**), each block including one or more graphics cores. Each graphics core includes a set of graphics execution resources that includes general-purpose and graphics specific execution logic to perform graphics and compute operations, as well as fixed function texture processing and/or machine learning and artificial intelligence acceleration logic.

[0291] In various embodiments the 3D pipeline **1612** can include fixed function and programmable logic to process one or more shader programs, such as vertex shaders, geometry shaders, pixel shaders, fragment shaders, compute shaders, or other shader programs, by processing the instructions and dispatching execution threads to the graphics core cluster **1714**. The graphics core cluster **1714** provides a unified block of execution resources for use in processing these shader programs. Multi-purpose execution logic within the graphics core block **1715A-1715B** of the graphics core cluster **1714** includes support for various 3D API shader languages and can execute multiple simultaneous execution threads associated with multiple shaders.

[0292] The graphics core cluster **1714** may include execution logic to perform media functions, such as video and/or image processing. The execution resources may include general-purpose logic that is programmable to perform parallel general-purpose computational operations, in addition to graphics processing operations. The general-purpose logic can perform processing operations in parallel or in conjunction with general-purpose logic within the processor core(s) **1407** of FIG. **14** or core **1502A-1502N** as in FIG. **15A**.

[0293] Output data generated by threads executing on the graphics core cluster **1714** can output data to memory in the URB **1718**. The URB **1718** can store data for multiple threads. The URB **1718** may be used to send data between different threads executing on the graphics core cluster **1714**. The URB **1718** may additionally be used for synchronization between threads on the graphics core cluster **1714** and fixed function logic within the shared function logic **1720**.

[0294] Optionally, the graphics core cluster **1714** may be scalable, such that the array includes a variable number of graphics cores, each having a variable number of execution resources based on the target power and performance level of GPE **1710**. The execution resources may be dynamically scalable, such that execution resources may be enabled or disabled as needed.

[0295] The graphics core cluster **1714** couples with shared function logic **1720** that includes multiple resources that are shared between the graphics cores in the graphics core array. The shared functions within the shared function logic **1720** are hardware logic units that provide specialized supplemental functionality to the graphics core cluster **1714**. In various embodiments, shared function logic **1720** includes but is not limited to sampler **1721**, math **1722**, and inter-thread communication (ITC) **1723** logic. Additionally, one or more cache(s) **1725** within the shared function logic **1720** may be implemented.

[0296] A shared function is implemented at least in a case where the demand for a given specialized function is insufficient for inclusion within the graphics core cluster **1714**. Instead, a single instantiation of that specialized function is implemented as a stand-alone entity in the shared function logic **1720** and shared among the execution resources within the graphics core cluster **1714**. The precise set of functions that are shared between the graphics core cluster **1714** and included within the graphics core cluster **1714** varies across embodiments. Specific shared functions within the shared function logic **1720** that are used extensively by the graphics core cluster **1714** may be included within shared function logic **1716** within the graphics core cluster **1714**. Optionally, the shared function logic **1716** within the graphics core cluster **1714** can include some or all logic within the shared function logic **1720**. All logic elements within the shared function logic **1720** may be duplicated within the shared function logic **1716** of the graphics core cluster **1714**. Alternatively, the shared function logic **1720** is excluded in favor of the shared function logic **1716** within the graphics core cluster **1714**.

Graphics Processing Resources

[0297] FIG. **18A-18C** illustrate execution logic including an array of processing elements employed in a graphics processor, according to embodiments described herein. FIG. **18A** illustrates graphics core cluster, according to an embodiment. FIG. **18B** illustrates a vector engine of a graphics core, according to an embodiment. FIG. **18C** illustrates a matrix engine of a graphics core, according to an embodiment. Elements of FIG. **18A-18C** having the same reference numbers as the elements of any other figure herein may operate or function in any manner similar to that described elsewhere herein but are not limited as such. For example, the elements of FIG. **18A-18C** can be considered in the context of the graphics processor core block **1519** of FIG. **15B**, and/or the graphics core blocks **1715A-1715B** of FIG. **17**. In one embodiment, the elements of FIG. **18A-18C** have similar functionality to equivalent components of the graphics processor **1508** of FIG. **15A** or the GPGPU **1570** of FIG. **15C**.

[0298] As shown in FIG. **18A**, in one embodiment the graphics core cluster **1714** includes a graphics core block **1715**, which may be graphics core block **1715A** or graphics core block **1715B** of FIG. **17**. The graphics core block **1715** can include any number of graphics cores (e.g., graphics core **1815A**, graphics core **1815B**, through graphics core **1815N**). Multiple instances of the graphics core block **1715** may be included. In one embodiment the elements of the graphics cores **1815A-1815N** have similar or equivalent functionality as the elements of the graphics cores **1521A-1521F** of FIG. **15B**. In such embodiment, the graphics cores **1815A-1815N** each include circuitry including but not limited to vector engines **1802A-1802N**, matrix engines **1803A-1803N**, memory load/store units **1804A-1804N**, instruction caches **1805A-1805N**, data caches/shared local memory **1806A-1806N**, ray tracing units **1808A-1808N**, samplers **1810A-1810N**. The circuitry of the graphics cores **1815A-1815N** can additionally include fixed function logic **1812A-1812N**. The number of vector engines **1802A-1802N** and matrix engines **1803A-1803N** within the graphics cores **1815A-1815N** of a design can vary based on the workload, performance, and power targets for the design.

[0299] With reference to graphics core **1815A**, the vector engine **1802A** and matrix engine **1803A** are configurable to perform parallel compute operations on data in a variety of integer and floating-point data formats based on instructions associated with shader programs. Each vector engine **1802A** and matrix engine **1803A** can act as a programmable general-purpose computational unit that is capable of executing multiple simultaneous hardware threads while processing multiple data elements in parallel for each thread. The vector engine **1802A** and matrix engine **1803A** support the processing of variable width vectors at various SIMD widths, including but not limited to SIMD8, SIMD16, and SIMD32. Input data elements can be stored as a packed data type in a register and the vector engine **1802A** and matrix engine **1803A** can process the various elements based on the data size of the elements. For example, when operating on a 256-bit wide vector, the 256 bits of the vector are stored in a register and the vector is processed as four separate 64-bit packed data

elements (Quad-Word (QW) size data elements), eight separate 32-bit packed data elements (Double Word (DW) size data elements), sixteen separate 16-bit packed data elements (Word (W) size data elements), or thirty-two separate 8-bit data elements (byte (B) size data elements). However, different vector widths and register sizes are possible. In one embodiment, the vector engine **1802A** and matrix engine **1803A** are also configurable for SIMT operation on warps or thread groups of various sizes (e.g., 8, 16, or 32 threads).

[0300] Continuing with graphics core **1815A**, the memory load/store unit **1804A** services memory access requests that are issued by the vector engine **1802A**, matrix engine **1803A**, and/or other components of the graphics core **1815A** that have access to memory. The memory access request can be processed by the memory load/store unit **1804A** to load or store the requested data to or from cache or memory into a register file associated with the vector engine **1802A** and/or matrix engine **1803A**. The memory load/store unit **1804A** can also perform prefetching operations. With additional reference to FIG. **19**, in one embodiment, the memory load/store unit **1804A** is configured to provide SIMT scatter/gather prefetching or block prefetching for data stored in memory **1910**, from memory that is local to other tiles via the tile interconnect **1908**, or from system memory. Prefetching can be performed to a specific L1 cache (e.g., data cache/shared local memory **1806A**), the L2 cache **1904** or the L3 cache **1906**. In one embodiment, a prefetch to the L3 cache **1906** automatically results in the data being stored in the L2 cache **1904**.

[0301] The instruction cache **1805A** stores instructions to be executed by the graphics core **1815A**. In one embodiment, the graphics core **1815A** also includes instruction fetch and prefetch circuitry that fetches or prefetches instructions into the instruction cache **1805A**. The graphics core **1815A** also includes instruction decode logic to decode instructions within the instruction cache **1805A**. The data cache/shared local memory **1806A** can be configured as a data cache that is managed by a cache controller that implements a cache replacement policy and/or configured as explicitly managed shared memory. The ray tracing unit **1808A** includes circuitry to accelerate ray tracing operations. The sampler **1810A** provides texture sampling for 3D operations and media sampling for media operations. The fixed function logic **1812A** includes fixed function circuitry that is shared between the various instances of the vector engine **1802A** and matrix engine **1803A**. Graphics cores **1815B-1815N** can operate in a similar manner as graphics core **1815A**.

[0302] Functionality of the instruction caches **1805A-1805N**, data caches/shared local memory **1806A-1806N**, ray tracing units **1808A-1808N**, samplers **1810A-1810N**, and fixed function logic **1812A-1812N** corresponds with equivalent functionality in the graphics processor architectures described herein. For example, the instruction caches **1805A-1805N** can operate in a similar manner as instruction cache **1555** of FIG. **15C**. The data caches/shared local memory **1806A-1806N**, ray tracing units **1808A-1808N**, and samplers **1810A-1810N** can operate in a similar manner as the cache/SLM **1528A-1528F**, ray tracing units **1527A-1527F**, and samplers **1526A-1526F** of FIG. **15B**. The fixed function logic **1812A-1812N** can include elements of the geometry/fixed function pipeline **1531** and/or additional fixed function logic **1538** of FIG. **15B**. In one embodiment, the ray tracing units **1808A-1808N** include circuitry to perform ray tracing acceleration operations performed by the ray tracing cores **372** of FIG. **3C**.

[0303] As shown in FIG. **18B**, in one embodiment the vector engine **1802** includes an instruction fetch unit **1837**, a general register file array (GRF) **1824**, an architectural register file array (ARF) **1826**, a thread arbiter **1822**, a send unit **1830**, a branch unit **1832**, a set of SIMD floating point units (FPUs) **1834**, and in one embodiment a set of integer SIMD ALUs **1835**. The GRF **1824** and ARF **1826** includes the set of general register files and architecture register files associated with each hardware thread that may be active in the vector engine **1802**. In one embodiment, per thread architectural state is maintained in the ARF **1826**, while data used during thread execution is stored in the GRF **1824**. The execution state of each thread, including the instruction pointers for each thread, can be held in thread-specific registers in the ARF **1826**. Register renaming may be used to dynamically allocate registers to hardware threads.

[0304] In one embodiment the vector engine **1802** has an architecture that is a combination of Simultaneous Multi-Threading (SMT) and fine-grained Interleaved Multi-Threading (IMT). The architecture has a modular configuration that can be fine-tuned at design time based on a target number of simultaneous threads and number of registers per graphics core, where graphics core resources are divided across logic used to execute multiple simultaneous threads. The number of logical threads that may be executed by the vector engine **1802** is not limited to the number of hardware threads, and multiple logical threads can be assigned to each hardware thread.

[0305] In one embodiment, the vector engine **1802** can co-issue multiple instructions, which may each be different instructions. The thread arbiter **1822** can dispatch the instructions to one of the send unit **1830**, branch unit **1832**, or SIMD FPU(s) **1834** for execution. Each execution thread can access **128** general-purpose registers within the GRF **1824**, where each register can store 32 bytes, accessible as a variable width vector of 32-bit data elements. In one embodiment, each thread has access to 4 K bytes within the GRF **1824**, although embodiments are not so limited, and greater or fewer register resources may be provided in other embodiments. In one embodiment the vector engine **1802** is partitioned into seven hardware threads that can independently perform computational operations, although the number of threads per vector engine **1802** can also vary according to embodiments. For example, in one embodiment up to 16 hardware threads are supported. In an embodiment in which seven threads may access 4 K bytes, the GRF **1824** can store a total of 28 K bytes. Where 16 threads may access 4K bytes, the GRF **1824** can store a total of 64K bytes. Flexible addressing modes can permit registers to be addressed together to build effectively wider registers or to represent strided rectangular block data structures.

[0306] In one embodiment, memory operations, sampler operations, and other longer-latency system communications are dispatched via “send” instructions that are executed by the message passing send unit **1830**. In one embodiment, branch instructions are dispatched to a dedicated branch unit **1832** to facilitate SIMD divergence and eventual convergence.

[0307] In one embodiment the vector engine **1802** includes one or more SIMD floating point units (FPU(s)) **1834** to perform floating-point operations. In one embodiment, the FPU(s) **1834** also support integer computation. In one embodiment the FPU(s) **1834** can execute up to M number of 32-bit floating-point (or integer) operations, or execute up to 2M 16-bit integer or 16-bit floating-point operations. In one embodiment, at least one of the FPU(s) provides extended math capability to support high-throughput transcendental math functions and double precision 64-bit floating-point. In some embodiments, a set of 8-bit integer SIMD ALUs **1835** are also present and may be specifically optimized to perform operations associated with machine learning computations. In one embodiment, the SIMD ALUs are replaced by an additional set of SIMD FPUs **1834** that are configurable to perform integer and floating-point operations. In one embodiment, the SIMD FPUs **1834** and SIMD ALUs **1835** are configurable to execute SIMT programs. In one embodiment, combined SIMD+SIMT operation is supported.

[0308] In one embodiment, arrays of multiple instances of the vector engine **1802** can be instantiated in a graphics core. For scalability, product architects can choose the exact number of vector engines per graphics core grouping. In one embodiment the vector engine **1802** can execute instructions across a plurality of execution channels. In a further embodiment, each thread executed on the vector engine **1802** is executed on a different channel.

[0309] As shown in FIG. **18C**, in one embodiment the matrix engine **1803** includes an array of processing elements that are configured to perform tensor operations including vector/matrix and matrix/matrix operations, such as but not limited to matrix multiply and/or dot product operations. The matrix engine **1803** is configured with M rows and N columns of processing elements (PE **1852AA**-PE **1852MN**) that include multiplier and adder circuits organized in a pipelined fashion. In one embodiment, the processing elements **1852AA**-PE **1852MN** make up the physical pipeline stages of an N wide and M deep systolic array that can be used to perform vector/matrix or matrix/matrix operations in a data-parallel manner, including matrix multiply, fused multiply-add,

dot product or other general matrix-matrix multiplication (GEM M) operations. In one embodiment the matrix engine **1803** supports 16-bit and 8-bit floating point operations, as well as 8-bit, 4-bit, 2-bit, and binary integer operations. The matrix engine **1803** can also be configured to accelerate specific machine learning operations. In such embodiments, the matrix engine **1803** can be configured with support for the bfloat (brain floating point) 16-bit floating point format or a tensor float 32-bit floating point format (TF32) that have different numbers of mantissa and exponent bits relative to Institute of Electrical and Electronics Engineers (IEEE) 754 formats.

[0310] In one embodiment, during each cycle, each stage can add the result of operations performed at that stage to the output of the previous stage. In other embodiments, the pattern of data movement between the processing elements **1852AA-1852MN** after a set of computational cycles can vary based on the instruction or macro-operation being performed. For example, in one embodiment partial sum loopback is enabled and the processing elements may instead add the output of a current cycle with output generated in the previous cycle. In one embodiment, the final stage of the systolic array can be configured with a loopback to the initial stage of the systolic array. In such embodiment, the number of physical pipeline stages may be decoupled from the number of logical pipeline stages that are supported by the matrix engine **1803**. For example, where the processing elements **1852AA-1852MN** are configured as a systolic array of M physical stages, a loopback from stage M to the initial pipeline stage can enable the processing elements **1852AA-1852MN** to operate as a systolic array of, for example, 2M, 3M, 4M, etc., logical pipeline stages.

[0311] In one embodiment, the matrix engine **1803** includes memory **1841A-1841N**, **1842A-1842M** to store input data in the form of row and column data for input matrices. Memory **1842A-1842M** is configurable to store row elements (A0-Am) of a first input matrix and memory **1841A-1841N** is configurable to store column elements (B0-Bn) of a second input matrix. The row and column elements are provided as input to the processing elements **1852AA-1852MN** for processing. In one embodiment, row and column elements of the input matrices can be stored in a systolic register file **1840** within the matrix engine **1803** before those elements are provided to the memory **1841A-1841N**, **1842A-1842M**. In one embodiment, the systolic register file **1840** is excluded and the memory **1841A-1841N**, **1842A-1842M** is loaded from registers in an associated vector engine (e.g., GRF **1824** of vector engine **1802** of FIG. **18B**) or other memory of the graphics core that includes the matrix engine **1803** (e.g., data cache/shared local memory **1806A** for matrix engine **1803A** of FIG. **18A**). Results generated by the processing elements **1852AA-1852MN** are then output to an output buffer and/or written to a register file (e.g., systolic register file **1840**, GRF **1824**, data cache/shared local memory **1806A-1806N**) for further processing by other functional units of the graphics processor or for output to memory.

[0312] In some embodiments, the matrix engine **1803** is configured with support for input sparsity, where multiplication operations for sparse regions of input data can be bypassed by skipping multiply operations that have a zero-value operand. In one embodiment, the processing elements **1852AA-1852MN** are configured to skip the performance of certain operations that have zero value input. In one embodiment, sparsity within input matrices can be detected and operations having known zero output values can be bypassed before being submitted to the processing elements **1852AA-1852MN**. The loading of zero value operands into the processing elements can be bypassed and the processing elements **1852AA-1852MN** can be configured to perform multiplications on the non-zero value input elements. The matrix engine **1803** can also be configured with support for output sparsity, such that operations with results that are pre-determined to be zero are bypassed. For input sparsity and/or output sparsity, in one embodiment, metadata is provided to the processing elements **1852AA-1852MN** to indicate, for a processing cycle, which processing elements and/or data channels are to be active during that cycle.

[0313] In one embodiment, the matrix engine **1803** includes hardware to enable operations on sparse data having a compressed representation of a sparse matrix that stores non-zero values and metadata that identifies the positions of the non-zero values within the matrix. Exemplary

compressed representations include but are not limited to compressed tensor representations such as compressed sparse row (CSR), compressed sparse column (CSC), compressed sparse fiber (CSF) representations. Support for compressed representations enable operations to be performed on input in a compressed tensor format without requiring the compressed representation to be decompressed or decoded. In such embodiment, operations can be performed only on non-zero input values and the resulting non-zero output values can be mapped into an output matrix. In some embodiments, hardware support is also provided for machine-specific lossless data compression formats that are used when transmitting data within hardware or across system busses. Such data may be retained in a compressed format for sparse input data and the matrix engine **1803** can use the compression metadata for the compressed data to enable operations to be performed on only non-zero values, or to enable blocks of zero data input to be bypassed for multiply operations. [0314] In various embodiments, input data can be provided by a programmer in a compressed tensor representation, or a codec can compress input data into the compressed tensor representation or another sparse data encoding. In addition to support for compressed tensor representations, streaming compression of sparse input data can be performed before the data is provided to the processing elements **1852AA-1852MN**. In one embodiment, compression is performed on data written to a cache memory associated with the graphics core cluster **1714**, with the compression being performed with an encoding that is supported by the matrix engine **1803**. In one embodiment, the matrix engine **1803** includes support for input having structured sparsity in which a pre-determined level or pattern of sparsity is imposed on input data. This data may be compressed to a known compression ratio, with the compressed data being processed by the processing elements **1852AA-1852MN** according to metadata associated with the compressed data.

[0315] FIG. **19** illustrates a tile **1900** of a multi-tile processor, according to an embodiment. In one embodiment, the tile **1900** is representative of one of the graphics engine tiles **1610A-1610D** of FIG. **16B** or compute engine tiles **1640A-1640D** of FIG. **16C**. The tile **1900** of the multi-tile graphics processor includes an array of graphics core clusters (e.g., graphics core cluster **1714A**, graphics core cluster **1714B**, through graphics core cluster **1714N**), with each graphics core cluster having an array of graphics cores **515A-515N**. The tile **1900** also includes a global dispatcher **1902** to dispatch threads to processing resources of the tile **1900**.

[0316] The tile **1900** can include or couple with an L3 cache **1906** and memory **1910**. In various embodiments, the L3 cache **1906** may be excluded or the tile **1900** can include additional levels of cache, such as an L4 cache. In one embodiment, each instance of the tile **1900** in the multi-tile graphics processor has an associated memory **1910**, such as in FIG. **16B** and FIG. **16C**. In one embodiment, a multi-tile processor can be configured as a multi-chip module in which the L3 cache **1906** and/or memory **1910** reside on separate chiplets than the graphics core clusters **1714A-1714N**. In this context, a chiplet is an at least partially packaged integrated circuit that includes distinct units of logic that can be assembled with other chiplets into a larger package. For example, the L3 cache **1906** can be included in a dedicated cache chiplet or can reside on the same chiplet as the graphics core clusters **1714A-1714N**. In one embodiment, the L3 cache **1906** can be included in an active base die or active interposer, as illustrated in FIG. **24C**.

[0317] A memory fabric **1903** enables communication among the graphics core clusters **1714A-1714N**, L3 cache **1906**, and memory **1910**. An L2 cache **1904** couples with the memory fabric **1903** and is configurable to cache transactions performed via the memory fabric **1903**. A tile interconnect **1908** enables communication with other tiles on the graphics processors and may be one of tile interconnects **1623A-1623F** of FIGS. **16B** and **16C**. In embodiments in which the L3 cache **1906** is excluded from the tile **1900**, the L2 cache **1904** may be configured as a combined L2/L3 cache. The memory fabric **1903** is configurable to route data to the L3 cache **1906** or memory controllers associated with the memory **1910** based on the presence or absence of the L3 cache **1906** in a specific implementation. The L3 cache **1906** can be configured as a per-tile cache that is dedicated to processing resources of the tile **1900** or may be a partition of a GPU-wide L3 cache.

[0318] FIG. **20** is a block diagram illustrating graphics processor instruction formats **2000**. The graphics processor execution resources support an instruction set having instructions in multiple formats. The solid lined boxes illustrate the components that are generally included in an execution unit instruction, while the dashed lines include components that are optional or that are only included in a sub-set of the instructions. In some embodiments the graphics processor instruction formats **2000** described and illustrated are macro-instructions, in that they are instructions supplied to the execution unit, as opposed to micro-operations resulting from instruction decode once the instruction is processed. Thus, a single instruction may cause hardware to perform multiple micro-operations

[0319] The graphics processor execution resources as described herein may natively support instructions in a 128-bit instruction format **2010**. A 64-bit compacted instruction format **2030** is available for some instructions based on the selected instruction, instruction options, and number of operands. The native 128-bit instruction format **2010** provides access to all instruction options, while some options and operations are restricted in the 64-bit compacted instruction format **2030**. The native instructions available in the 64-bit compacted instruction format **2030** vary by embodiment. The instruction is compacted in part using a set of index values in an index field **2013**. The execution unit hardware references a set of compaction tables based on the index values and uses the compaction table outputs to reconstruct a native instruction in the 128-bit instruction format **2010**. Other sizes and formats of instruction can be used.

[0320] For each format, instruction opcode **2012** defines the operation that the execution unit is to perform. The execution resources execute each instruction in parallel across the multiple data elements of each operand. For example, in response to an add instruction the execution unit performs a simultaneous add operation across each color channel representing a texture element or picture element. By default, the execution unit performs each instruction across all data channels of the operands. Instruction control field **2014** may enable control over certain execution options, such as channels selection (e.g., predication) and data channel order (e.g., swizzle). For instructions in the 128-bit instruction format **2010** an exec-size field **2016** limits the number of data channels that will be executed in parallel. An exec-size field **2016** may not be available for use in the 64-bit compacted instruction format **2030**.

[0321] Some execution unit instructions have up to three operands including two source operands, src0 **2020**, src1 **2022**, and one destination operand (dest **2018**). Other instructions, such as, for example, data manipulation instructions, dot product instructions, multiply-add instructions, or multiply-accumulate instructions, can have a third source operand (e.g., SRC2 **2024**). The instruction opcode **2012** determines the number of source operands. An instruction's last source operand can be an immediate (e.g., hard-coded) value passed with the instruction. The execution resources may also support multiple destination instructions, where one or more of the destinations is implied or implicit based on the instruction and/or the specified destination.

[0322] The 128-bit instruction format **2010** may include an access/address mode field **2026** specifying, for example, whether direct register addressing mode or indirect register addressing mode is used. When direct register addressing mode is used, the register address of one or more operands is directly provided by bits in the instruction.

[0323] The 128-bit instruction format **2010** may also include an access/address mode field **2026**, which specifies an address mode and/or an access mode for the instruction. The access mode may be used to define a data access alignment for the instruction. Access modes including a 16-byte aligned access mode and a 1-byte aligned access mode may be supported, where the byte alignment of the access mode determines the access alignment of the instruction operands. For example, when in a first mode, the instruction may use byte-aligned addressing for source and destination operands and when in a second mode, the instruction may use 16-byte-aligned addressing for all source and destination operands.

[0324] The address mode portion of the access/address mode field **2026** may determine whether the

instruction is to use direct or indirect addressing. When direct register addressing mode is used bits in the instruction directly provide the register address of one or more operands. When indirect register addressing mode is used, the register address of one or more operands may be computed based on an address register value and an address immediate field in the instruction.

[0325] Instructions may be grouped based on opcode **2012** bit-fields to simplify Opcode decode **2040**. For an 8-bit opcode, bits **4**, **5**, and **6** allow the execution unit to determine the type of opcode. The precise opcode grouping shown is merely an example. A move and logic opcode group **2042** may include data movement and logic instructions (e.g., move (mov), compare (cmp)). Move and logic group **2042** may share the five least significant bits (LSB), where move (mov) instructions are in the form of 0000xxxxb and logic instructions are in the form of 0001xxxxb. A flow control instruction group **2044** (e.g., call, jump (jmp)) includes instructions in the form of 0010xxxxb (e.g., 0x20). A miscellaneous instruction group **2046** includes a mix of instructions, including synchronization instructions (e.g., wait, send) in the form of 0011xxxxb (e.g., 0x30). A parallel math instruction group **2048** includes component-wise arithmetic instructions (e.g., add, multiply (mul)) in the form of 0100xxxxb (e.g., 0x40). The parallel math instruction group **2048** performs the arithmetic operations in parallel across data channels. The vector math group **2050** includes arithmetic instructions (e.g., dp4) in the form of 0101xxxxb (e.g., 0x50). The vector math group performs arithmetic such as dot product calculations on vector operands. The illustrated opcode decode **2040**, in one embodiment, can be used to determine which portion of an execution unit will be used to execute a decoded instruction. For example, some instructions may be designated as systolic instructions that will be performed by a systolic array. Other instructions, such as ray-tracing instructions (not shown) can be routed to a ray-tracing core or ray-tracing logic within a slice or partition of execution logic.

Graphics Pipeline

[0326] FIG. **21** is a block diagram of graphics processor **2100**, according to another embodiment. The elements of FIG. **21** having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such.

[0327] The graphics processor **2100** may include different types of graphics processing pipelines, such as a geometry pipeline **2120**, a media pipeline **2130**, a display engine **2140**, thread execution logic **2150**, and a render output pipeline **2170**. Graphics processor **2100** may be a graphics processor within a multi-core processing system that includes one or more general-purpose processing cores. The graphics processor may be controlled by register writes to one or more control registers (not shown) or via commands issued to graphics processor **2100** via a ring interconnect **2102**. Ring interconnect **2102** may couple graphics processor **2100** to other processing components, such as other graphics processors or general-purpose processors. Commands from ring interconnect **2102** are interpreted by a command streamer **2103**, which supplies instructions to individual components of the geometry pipeline **2120** or the media pipeline **2130**.

[0328] Command streamer **2103** may direct the operation of a vertex fetcher **2105** that reads vertex data from memory and executes vertex-processing commands provided by command streamer **2103**. The vertex fetcher **2105** may provide vertex data to a vertex shader **2107**, which performs coordinate space transformation and lighting operations to each vertex. Vertex fetcher **2105** and vertex shader **2107** may execute vertex-processing instructions by dispatching execution threads to graphics cores **2152A-2152B** via a thread dispatcher **2131**.

[0329] The graphics cores **2152A-2152B** may be an array of vector processors having an instruction set for performing graphics and media operations. The graphics cores **2152A-2152B** may have an attached L1 cache **2151** that is specific for each array or shared between the arrays. The cache can be configured as a data cache, an instruction cache, or a single cache that is partitioned to contain data and instructions in different partitions.

[0330] A geometry pipeline **2120** may include tessellation components to perform hardware-accelerated tessellation of 3D objects. A programmable hull shader **2111** may configure the tessellation operations. A programmable domain shader **2117** may provide back-end evaluation of tessellation output. A tessellator **2113** may operate at the direction of hull shader **2111** and contain special purpose logic to generate a set of detailed geometric objects based on a coarse geometric model that is provided as input to geometry pipeline **2120**. In addition, if tessellation is not used, tessellation components (e.g., hull shader **2111**, tessellator **2113**, and domain shader **2117**) can be bypassed. The tessellation components can operate based on data received from the vertex shader **2107**.

[0331] Complete geometric objects may be processed by a geometry shader **2119** via one or more threads dispatched to graphics cores **2152A-2152B**, or can proceed directly to the clipper **2129**. The geometry shader may operate on entire geometric objects, rather than vertices or patches of vertices as in previous stages of the graphics pipeline. If the tessellation is disabled, the geometry shader **2119** receives input from the vertex shader **2107**. The geometry shader **2119** may be programmable by a geometry shader program to perform geometry tessellation if the tessellation units are disabled.

[0332] Before rasterization, a clipper **2129** processes vertex data. The clipper **2129** may be a fixed function clipper or a programmable clipper having clipping and geometry shader functions. A rasterizer and depth test component **2173** in the render output pipeline **2170** may dispatch pixel shaders to convert the geometric objects into per pixel representations. The pixel shader logic may be included in thread execution logic **2150**. Optionally, an application can bypass the rasterizer and depth test component **2173** and access un-rasterized vertex data via a stream out unit **2123**.

[0333] The graphics processor **2100** has an interconnect bus, interconnect fabric, or some other interconnect mechanism that allows data and message passing amongst the major components of the processor. In some embodiments, graphics cores **2152A-2152B** and associated logic units (e.g., L1 cache **2151**, sampler **2154**, texture cache **2158**, etc.) interconnect via a data port **2156** to perform memory access and communicate with render output pipeline components of the processor. A sampler **2154**, caches **2151**, **2158** and graphics cores **2152A-2152B** each may have separate memory access paths. Optionally, the texture cache **2158** can also be configured as a sampler cache.

[0334] The render output pipeline **2170** may contain a rasterizer and depth test component **2173** that converts vertex-based objects into an associated pixel-based representation. The rasterizer logic may include a windower/masker unit to perform fixed function triangle and line rasterization. An associated render cache **2178** and depth cache **2179** are also available in some embodiments. A pixel operations component **2177** performs pixel-based operations on the data, though in some instances, pixel operations associated with 2D operations (e.g., bit block image transfers with blending) are performed by the 2D engine **2141** or substituted at display time by the display controller **2143** using overlay display planes. A shared L3 cache **2175** may be available to all graphics components, allowing the sharing of data without the use of main system memory.

[0335] The media pipeline **2130** may include a media engine **2137** and a video front-end **2134**. Video front-end **2134** may receive pipeline commands from the command streamer **2103**. The media pipeline **2130** may include a separate command streamer. Video front-end **2134** may process media commands before sending the command to the media engine **2137**. Media engine **2137** may include thread spawning functionality to spawn threads for dispatch to thread execution logic **2150** via thread dispatcher **2131**.

[0336] The graphics processor **2100** may include a display engine **2140**. This display engine **2140** may be external to processor **2100** and may couple with the graphics processor via the ring interconnect **2102**, or some other interconnect bus or fabric. Display engine **2140** may include a 2D engine **2141** and a display controller **2143**. Display engine **2140** may contain special purpose logic capable of operating independently of the 3D pipeline. Display controller **2143** may couple with a

display device (not shown), which may be a system integrated display device, as in a laptop computer, or an external display device attached via a display device connector.

[0337] The geometry pipeline **2120** and media pipeline **2130** maybe configurable to perform operations based on multiple graphics and media programming interfaces and are not specific to any one application programming interface (API). A driver software for the graphics processor may translate API calls that are specific to a particular graphics or media library into commands that can be processed by the graphics processor. Support may be provided for the Open Graphics Library (OpenGL), Open Computing Language (OpenCL), and/or Vulkan graphics and compute API, all from the Khronos Group. Support may also be provided for the Direct3D library from the Microsoft Corporation. A combination of these libraries may be supported. Support may also be provided for the Open Source Computer Vision Library (OpenCV). A future API with a compatible 3D pipeline would also be supported if a mapping can be made from the pipeline of the future API to the pipeline of the graphics processor.

Graphics Pipeline Programming

[0338] FIG. **22A** is a block diagram illustrating a graphics processor command format **2200** used for programming graphics processing pipelines, such as, for example, the pipelines described herein in conjunction with FIG. **16A**, **17**, **21**. FIG. **22B** is a block diagram illustrating a graphics processor command sequence **2210** according to an embodiment. The solid lined boxes in FIG. **22A** illustrate the components that are generally included in a graphics command while the dashed lines include components that are optional or that are only included in a sub-set of the graphics commands. The exemplary graphics processor command format **2200** of FIG. **22A** includes fields to identify a client **2202**, a command operation code (opcode) **2204**, and a data field **2206** for the command. A sub-opcode **2205** and a command size **2208** are also included in some commands.

[0339] Client **2202** may specify the client unit of the graphics device that processes the command data. A graphics processor command parser may examine the client field of each command to condition the further processing of the command and route the command data to the appropriate client unit. The graphics processor client units may include a memory interface unit, a render unit, a 2D unit, a 3D unit, and a media unit. Each client unit may have a corresponding processing pipeline that processes the commands. Once the command is received by the client unit, the client unit reads the opcode **2204** and, if present, sub-opcode **2205** to determine the operation to perform. The client unit performs the command using information in data field **2206**. For some commands an explicit command size **2208** is expected to specify the size of the command. The command parser may automatically determine the size of at least some of the commands based on the command opcode. Commands may be aligned via multiples of a double word. Other command formats can also be used.

[0340] The flow diagram in FIG. **22B** illustrates an exemplary graphics processor command sequence **2210**. Software or firmware of a data processing system that features an exemplary graphics processor may use a version of the command sequence shown to set up, execute, and terminate a set of graphics operations. A sample command sequence is shown and described for purposes of example only and is not limited to these specific commands or to this command sequence. Moreover, the commands may be issued as batch of commands in a command sequence, such that the graphics processor will process the sequence of commands in at least partially concurrence.

[0341] The graphics processor command sequence **2210** may begin with a pipeline flush command **2212** to cause any active graphics pipeline to complete the currently pending commands for the pipeline. Optionally, the 3D pipeline **2222** and the media pipeline **2224** may not operate concurrently. The pipeline flush is performed to cause the active graphics pipeline to complete any pending commands. In response to a pipeline flush, the command parser for the graphics processor will pause command processing until the active drawing engines complete pending operations and the relevant read caches are invalidated. Optionally, any data in the render cache that is marked

‘dirty’ can be flushed to memory. Pipeline flush command **2212** can be used for pipeline synchronization or before placing the graphics processor into a low power state.

[0342] A pipeline select command **2213** may be used when a command sequence requires the graphics processor to explicitly switch between pipelines. A pipeline select command **2213** may be required only once within an execution context before issuing pipeline commands unless the context is to issue commands for both pipelines. A pipeline flush command **2212** may be required immediately before a pipeline switch via the pipeline select command **2213**.

[0343] A pipeline control command **2214** may configure a graphics pipeline for operation and may be used to program the 3D pipeline **2222** and the media pipeline **2224**. The pipeline control command **2214** may configure the pipeline state for the active pipeline. The pipeline control command **2214** may be used for pipeline synchronization and to clear data from one or more cache memories within the active pipeline before processing a batch of commands.

[0344] Commands related to the return buffer state **2216** may be used to configure a set of return buffers for the respective pipelines to write data. Some pipeline operations require the allocation, selection, or configuration of one or more return buffers into which the operations write intermediate data during processing. The graphics processor may also use one or more return buffers to store output data and to perform cross thread communication. The return buffer state **2216** may include selecting the size and number of return buffers to use for a set of pipeline operations.

[0345] The remaining commands in the command sequence differ based on the active pipeline for operations. Based on a pipeline determination **2220**, the command sequence is tailored to the 3D pipeline **2222** beginning with the 3D pipeline state **2230** or the media pipeline **2224** beginning at the media pipeline state **2240**.

[0346] The commands to configure the 3D pipeline state **2230** include 3D state setting commands for vertex buffer state, vertex element state, constant color state, depth buffer state, and other state variables that are to be configured before 3D primitive commands are processed. The values of these commands are determined at least in part based on the particular 3D API in use. The 3D pipeline state **2230** commands may also be able to selectively disable or bypass certain pipeline elements if those elements will not be used.

[0347] A 3D primitive **2232** command may be used to submit 3D primitives to be processed by the 3D pipeline. Commands and associated parameters that are passed to the graphics processor via the 3D primitive **2232** command are forwarded to the vertex fetch function in the graphics pipeline. The vertex fetch function uses the 3D primitive **2232** command data to generate vertex data structures. The vertex data structures are stored in one or more return buffers. The 3D primitive **2232** command may be used to perform vertex operations on 3D primitives via vertex shaders. To process vertex shaders, 3D pipeline **2222** dispatches shader execution threads to graphics processor execution resources.

[0348] The 3D pipeline **2222** may be triggered via an execute **2234** command or event. A register may write trigger command executions. An execution may be triggered via a ‘go’ or ‘kick’ command in the command sequence. Command execution may be triggered using a pipeline synchronization command to flush the command sequence through the graphics pipeline. The 3D pipeline will perform geometry processing for the 3D primitives. Once operations are complete, the resulting geometric objects are rasterized and the pixel engine colors the resulting pixels. Additional commands to control pixel shading and pixel back-end operations may also be included for those operations.

[0349] The graphics processor command sequence **2210** may follow the media pipeline **2224** path when performing media operations. In general, the specific use and manner of programming for the media pipeline **2224** depends on the media or compute operations to be performed. Specific media decode operations may be offloaded to the media pipeline during media decode. The media pipeline can also be bypassed and media decode can be performed in whole or in part using

resources provided by one or more general-purpose processing cores. The media pipeline may also include elements for general-purpose graphics processor unit (GPGPU) operations, where the graphics processor is used to perform SIMD vector operations using computational shader programs that are not explicitly related to the rendering of graphics primitives.

[0350] Media pipeline **2224** may be configured in a similar manner as the 3D pipeline **2222**. A set of commands to configure the media pipeline state **2240** are dispatched or placed into a command queue before the media object commands **2242**. Commands for the media pipeline state **2240** may include data to configure the media pipeline elements that will be used to process the media objects. This includes data to configure the video decode and video encode logic within the media pipeline, such as encode or decode format. Commands for the media pipeline state **2240** may also support the use of one or more pointers to “indirect” state elements that contain a batch of state settings.

[0351] Media object commands **2242** may supply pointers to media objects for processing by the media pipeline. The media objects include memory buffers containing video data to be processed. Optionally, all media pipeline states must be valid before issuing a media object command **2242**. Once the pipeline state is configured and media object commands **2242** are queued, the media pipeline **2224** is triggered via an execute command **2244** or an equivalent execute event (e.g., register write). Output from media pipeline **2224** may then be post processed by operations provided by the 3D pipeline **2222** or the media pipeline **2224**. GPGPU operations may be configured and executed in a similar manner as media operations.

Graphics Software Architecture

[0352] FIG. **23** illustrates an exemplary graphics software architecture for a data processing system **2300**. Such a software architecture may include a 3D graphics application **2310**, an operating system **2320**, and at least one processor **2330**. Processor **2330** may include a graphics processor **2332** and one or more general-purpose processor core(s) **2334**. The processor **2330** may be a variant of the processor **1402** or any other of the processors described herein. The processor **2330** may be used in place of the processor **1402** or any other of the processors described herein. Therefore, the disclosure of any features in combination with the processor **1402** or any other of the processors described herein also discloses a corresponding combination with the graphics processor **2332** but is not limited to such. Moreover, the elements of FIG. **23** having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such. The graphics application **2310** and operating system **2320** are each executed in the system memory **2350** of the data processing system.

[0353] 3D graphics application **2310** may contain one or more shader programs including shader instructions **2312**. The shader language instructions may be in a high-level shader language, such as the High-Level Shader Language (HLSL) of Direct3D, the OpenGL Shader Language (GLSL), and so forth. The application may also include executable instructions **2314** in a machine language suitable for execution by the general-purpose processor core **2334**. The application may also include graphics objects **2316** defined by vertex data.

[0354] The operating system **2320** may be a Microsoft® Windows® operating system from the Microsoft Corporation, a proprietary UNIX-like operating system, or an open source UNIX-like operating system using a variant of the Linux kernel. The operating system **2320** can support a graphics API **2322** such as the Direct3D API, the OpenGL API, or the Vulkan API. When the Direct3D API is in use, the operating system **2320** uses a front-end shader compiler **2324** to compile any shader instructions **2312** in HLSL into a lower-level shader language. The compilation may be a just-in-time (JIT) compilation or the application can perform shader pre-compilation. High-level shaders may be compiled into low-level shaders during the compilation of the 3D graphics application **2310**. The shader instructions **2312** may be provided in an intermediate form,

such as a version of the Standard Portable Intermediate Representation (SPIR) used by the Vulkan API.

[0355] User mode graphics driver **2326** may contain a back-end shader compiler **2327** to convert the shader instructions **2312** into a hardware specific representation. When the OpenGL API is in use, shader instructions **2312** in the GLSL high-level language are passed to a user mode graphics driver **2326** for compilation. The user mode graphics driver **2326** may use operating system kernel mode functions **2328** to communicate with a kernel mode graphics driver **2329**. The kernel mode graphics driver **2329** may communicate with graphics processor **2332** to dispatch commands and instructions.

IP Core Implementations

[0356] One or more aspects may be implemented by representative code stored on a machine-readable medium which represents and/or defines logic within an integrated circuit such as a processor. For example, the machine-readable medium may include instructions which represent various logic within the processor. When read by a machine, the instructions may cause the machine to fabricate the logic to perform the techniques described herein. Such representations, known as “IP cores,” are reusable units of logic for an integrated circuit that may be stored on a tangible, machine-readable medium as a hardware model that describes the structure of the integrated circuit. The hardware model may be supplied to various customers or manufacturing facilities, which load the hardware model on fabrication machines that manufacture the integrated circuit. The integrated circuit may be fabricated such that the circuit performs operations described in association with any of the embodiments described herein.

[0357] FIG. **24A** is a block diagram illustrating an IP core development system **2400** that may be used to manufacture an integrated circuit to perform operations according to an embodiment. The IP core development system **2400** may be used to generate modular, re-usable designs that can be incorporated into a larger design or used to construct an entire integrated circuit (e.g., an SOC integrated circuit). A design facility **2430** can generate a software simulation **2410** of an IP core design in a high-level programming language (e.g., C/C++). The software simulation **2410** can be used to design, test, and verify the behavior of the IP core using a simulation model **2412**. The simulation model **2412** may include functional, behavioral, and/or timing simulations. A register transfer level (RTL) design **2415** can then be created or synthesized from the simulation model **2412**. The RTL design **2415** is an abstraction of the behavior of the integrated circuit that models the flow of digital signals between hardware registers, including the associated logic performed using the modeled digital signals. In addition to an RTL design **2415**, lower-level designs at the logic level or transistor level may also be created, designed, or synthesized. Thus, the particular details of the initial design and simulation may vary.

[0358] The RTL design **2415** or equivalent may be further synthesized by the design facility into a hardware model **2420**, which may be in a hardware description language (HDL), or some other representation of physical design data. The HDL may be further simulated or tested to verify the IP core design. The IP core design can be stored for delivery to a 3.sup.rd party fabrication facility **2465** using non-volatile memory **2440** (e.g., hard disk, flash memory, or any non-volatile storage medium). Alternatively, the IP core design may be transmitted (e.g., via the Internet) over a wired connection **2450** or wireless connection **2460**. The fabrication facility **2465** may then fabricate an integrated circuit that is based at least in part on the IP core design. The fabricated integrated circuit can be configured to perform operations in accordance with at least one embodiment described herein.

[0359] FIG. **24B** illustrates a cross-section side view of an integrated circuit package assembly **2470**. The integrated circuit package assembly **2470** illustrates an implementation of one or more processor or accelerator devices as described herein. The package assembly **2470** includes multiple units of hardware logic **2472**, **2474** connected to a substrate **2480**. The logic **2472**, **2474** may be implemented at least partly in configurable logic or fixed-functionality logic hardware and can

include one or more portions of any of the processor core(s), graphics processor(s), or other accelerator devices described herein. Each unit of logic **2472**, **2474** can be implemented within a semiconductor die and coupled with the substrate **2480** via an interconnect structure **2473**. The interconnect structure **2473** may be configured to route electrical signals between the logic **2472**, **2474** and the substrate **2480**, and can include interconnects such as, but not limited to bumps or pillars. The interconnect structure **2473** may be configured to route electrical signals such as, for example, input/output (I/O) signals and/or power or ground signals associated with the operation of the logic **2472**, **2474**. Optionally, the substrate **2480** may be an epoxy-based laminate substrate. The substrate **2480** may also include other suitable types of substrates. The package assembly **2470** can be connected to other electrical devices via a package interconnect **2483**. The package interconnect **2483** may be coupled to a surface of the substrate **2480** to route electrical signals to other electrical devices, such as a motherboard, other chipset, or multi-chip module.

[0360] The units of logic **2472**, **2474** may be electrically coupled with a bridge **2482** that is configured to route electrical signals between the logic **2472**, **2474**. The bridge **2482** may be a dense interconnect structure that provides a route for electrical signals. The bridge **2482** may include a bridge substrate composed of glass or a suitable semiconductor material. Electrical routing features can be formed on the bridge substrate to provide a chip-to-chip connection between the logic **2472**, **2474**.

[0361] Although two units of logic **2472**, **2474** and a bridge **2482** are illustrated, embodiments described herein may include more or fewer logic units on one or more dies. The one or more dies may be connected by zero or more bridges, as the bridge **2482** may be excluded when the logic is included on a single die. Alternatively, multiple dies or units of logic can be connected by one or more bridges. Additionally, multiple logic units, dies, and bridges can be connected together in other possible configurations, including three-dimensional configurations.

[0362] FIG. **24C** illustrates a package assembly **2490** that includes multiple units of hardware logic chiplets connected to a substrate **2480** (e.g., base die). A graphics processing unit, parallel processor, and/or compute accelerator as described herein can be composed from diverse silicon chiplets that are separately manufactured. In this context, a chiplet is an at least partially packaged integrated circuit that includes distinct units of logic that can be assembled with other chiplets into a larger package. A diverse set of chiplets with different IP core logic can be assembled into a single device. Additionally, the chiplets can be integrated into a base die or base chiplet using active interposer technology. The concepts described herein enable the interconnection and communication between the different forms of IP within the GPU. IP cores can be manufactured using different process technologies and composed during manufacturing, which avoids the complexity of converging multiple IPs, especially on a large SoC with several flavors IPs, to the same manufacturing process. Enabling the use of multiple process technologies improves the time to market and provides a cost-effective way to create multiple product SKUs. Additionally, the disaggregated IPs are more amenable to being power gated independently, components that are not in use on a given workload can be powered off, reducing overall power consumption.

[0363] In various embodiments a package assembly **2490** can include fewer or greater number of components and chiplets that are interconnected by a fabric **2485** or one or more bridges **2487**. The chiplets within the package assembly **2490** may have a 2.5D arrangement using Chip-on-Wafer-on-Substrate stacking in which multiple dies are stacked side-by-side on a silicon interposer that includes through-silicon vias (TSV s) to couple the chiplets with the substrate **2480**, which includes electrical connections to the package interconnect **2483**.

[0364] In one embodiment, silicon interposer is an active interposer **2489** that includes embedded logic in addition to TSVs. In such embodiment, the chiplets within the package assembly **2490** are arranged using 3D face to face die stacking on top of the active interposer **2489**. The active interposer **2489** can include hardware logic for I/O **2491**, cache memory **2492**, and other hardware logic **2493**, in addition to interconnect fabric **2485** and a silicon bridge **2487**. The fabric **2485**

enables communication between the various logic chiplets **2472**, **2474** and the logic **2491**, **2493** within the active interposer **2489**. The fabric **2485** may be an NoC interconnect or another form of packet switched fabric that switches data packets between components of the package assembly. For complex assemblies, the fabric **2485** may be a dedicated chiplet enables communication between the various hardware logic of the package assembly **2490**.

[0365] Bridge structures **2487** within the active interposer **2489** may be used to facilitate a point-to-point interconnect between, for example, logic or I/O chiplets **2474** and memory chiplets **2475**. In some implementations, bridge structures **2487** may also be embedded within the substrate **2480**.

[0366] The hardware logic chiplets can include special purpose hardware logic chiplets **2472**, logic or I/O chiplets **2474**, and/or memory chiplets **2475**. The hardware logic chiplets **2472** and logic or I/O chiplets **2474** may be implemented at least partly in configurable logic or fixed-functionality logic hardware and can include one or more portions of any of the processor core(s), graphics processor(s), parallel processors, or other accelerator devices described herein. The memory chiplets **2475** can be DRAM (e.g., GDDR, HBM) memory or cache (SRAM) memory. Cache memory **2492** within the active interposer **2489** (or substrate **2480**) can act as a global cache for the package assembly **2490**, part of a distributed global cache, or as a dedicated cache for the fabric **2485**.

[0367] Each chiplet can be fabricated as separate semiconductor die and coupled with a base die that is embedded within or coupled with the substrate **2480**. The coupling with the substrate **2480** can be performed via an interconnect structure **2473**. The interconnect structure **2473** may be configured to route electrical signals between the various chiplets and logic within the substrate **2480**. The interconnect structure **2473** can include interconnects such as, but not limited to bumps or pillars. In some embodiments, the interconnect structure **2473** may be configured to route electrical signals such as, for example, input/output (I/O) signals and/or power or ground signals associated with the operation of the logic, I/O and memory chiplets. In one embodiment, an additional interconnect structure couples the active interposer **2489** with the substrate **2480**.

[0368] The substrate **2480** may be an epoxy-based laminate substrate, however, it is not limited to that and the substrate **2480** may also include other suitable types of substrates. The package assembly **2490** can be connected to other electrical devices via a package interconnect **2483**. The package interconnect **2483** may be coupled to a surface of the substrate **2480** to route electrical signals to other electrical devices, such as a motherboard, other chipset, or multi-chip module.

[0369] A logic or I/O chiplet **2474** and a memory chiplet **2475** may be electrically coupled via a bridge **2487** that is configured to route electrical signals between the logic or I/O chiplet **2474** and a memory chiplet **2475**. The bridge **2487** may be a dense interconnect structure that provides a route for electrical signals. The bridge **2487** may include a bridge substrate composed of glass or a suitable semiconductor material. Electrical routing features can be formed on the bridge substrate to provide a chip-to-chip connection between the logic or I/O chiplet **2474** and a memory chiplet **2475**. The bridge **2487** may also be referred to as a silicon bridge or an interconnect bridge. For example, the bridge **2487** is an Embedded Multi-die Interconnect Bridge (EMIB). Alternatively, the bridge **2487** may simply be a direct connection from one chiplet to another chiplet.

[0370] FIG. **24D** illustrates a package assembly **2494** including interchangeable chiplets **2495**, according to an embodiment. The interchangeable chiplets **2495** can be assembled into standardized slots on one or more base chiplets **2496**, **2498**. The base chiplets **2496**, **2498** can be coupled via a bridge interconnect **2497**, which can be similar to the other bridge interconnects described herein and may be, for example, an EMIB. Memory chiplets can also be connected to logic or I/O chiplets via a bridge interconnect. I/O and logic chiplets can communicate via an interconnect fabric. The base chiplets can each support one or more slots in a standardized format for one of logic or I/O or memory/cache.

[0371] SRAM and power delivery circuits may be fabricated into one or more of the base chiplets **2496**, **2498**, which can be fabricated using a different process technology relative to the

interchangeable chiplets **2495** that are stacked on top of the base chiplets. For example, the base chiplets **2496**, **2498** can be fabricated using a larger process technology, while the interchangeable chiplets can be manufactured using a smaller process technology. One or more of the interchangeable chiplets **2495** may be memory (e.g., DRAM) chiplets. Different memory densities can be selected for the package assembly **2494** based on the power, and/or performance targeted for the product that uses the package assembly **2494**. Additionally, logic chiplets with a different number of type of functional units can be selected at time of assembly based on the power, and/or performance targeted for the product. Additionally, chiplets containing IP logic cores of differing types can be inserted into the interchangeable chiplet slots, enabling hybrid processor designs that can mix and match different technology IP blocks.

Exemplary System on a Chip Integrated Circuit

[0372] FIG. **25-26B** illustrate exemplary integrated circuits and associated graphics processors that may be fabricated using one or more IP cores. In addition to what is illustrated, other logic and circuits may be included, including additional graphics processors/cores, peripheral interface controllers, or general-purpose processor cores. The elements of FIG. **25-26B** having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such.

[0373] FIG. **25** is a block diagram illustrating an exemplary system on a chip integrated circuit **2500** that may be fabricated using one or more IP cores. Exemplary integrated circuit **2500** includes one or more application processor(s) **2505** (e.g., CPUs), at least one graphics processor **2510**, which may be a variant of the graphics processor **1408**, **1508**, **2510**, or of any graphics processor described herein and may be used in place of any graphics processor described. Therefore, the disclosure of any features in combination with a graphics processor herein also discloses a corresponding combination with the graphics processor **2510** but is not limited to such. The integrated circuit **2500** may additionally include an image processor **2515** and/or a video processor **2520**, any of which may be a modular IP core from the same or multiple different design facilities. Integrated circuit **2500** may include peripheral or bus logic including a USB controller **2525**, UART controller **2530**, an SPI/SDIO controller **2535**, and an I.sup.2S/I.sup.2C controller **2540**. Additionally, the integrated circuit can include a display device **2545** coupled to one or more of a high-definition multimedia interface (HDMI) controller **2550** and a mobile industry processor interface (MIPI) display interface **2555**. Storage may be provided by a flash memory subsystem **2560** including flash memory and a flash memory controller. Memory interface may be provided via a memory controller **2565** for access to SDRAM or SRAM memory devices. Some integrated circuits additionally include an embedded security engine **2570**.

[0374] FIG. **26A-26B** are block diagrams illustrating exemplary graphics processors for use within an SoC, according to embodiments described herein. The graphics processors may be variants of the graphics processor **1408**, **1508**, **2510**, or any other graphics processor described herein. The graphics processors may be used in place of the graphics processor **1408**, **1508**, **2510**, or any other of the graphics processors described herein. Therefore, the disclosure of any features in combination with the graphics processor **1408**, **1508**, **2510**, or any other of the graphics processors described herein also discloses a corresponding combination with the graphics processors of FIG. **26A-26B** but is not limited to such. FIG. **26A** illustrates an exemplary graphics processor **2610** of a system on a chip integrated circuit that may be fabricated using one or more IP cores, according to an embodiment. FIG. **26B** illustrates an additional exemplary graphics processor **2640** of a system on a chip integrated circuit that may be fabricated using one or more IP cores, according to an embodiment. Graphics processor **2610** of FIG. **26A** is an example of a low power graphics processor core. Graphics processor **2640** of FIG. **26B** is an example of a higher performance graphics processor core. For example, each of graphics processor **2610** and graphics processor **2640** can be a variant of the graphics processor **2510** of FIG. **25**, as mentioned at the outset of this

paragraph.

[0375] As shown in FIG. 26A, graphics processor **2610** includes a vertex processor **2605** and one or more fragment processor(s) **2615A-2615N** (e.g., **2615A**, **2615B**, **2615C**, **2615D**, through **2615N-1**, and **2615N**). Graphics processor **2610** can execute different shader programs via separate logic, such that the vertex processor **2605** is optimized to execute operations for vertex shader programs, while the one or more fragment processor(s) **2615A-2615N** execute fragment (e.g., pixel) shading operations for fragment or pixel shader programs. The vertex processor **2605** performs the vertex processing stage of the 3D graphics pipeline and generates primitives and vertex data. The fragment processor(s) **2615A-2615N** use the primitive and vertex data generated by the vertex processor **2605** to produce a framebuffer that is displayed on a display device. The fragment processor(s) **2615A-2615N** may be optimized to execute fragment shader programs as provided for in the OpenGL API, which may be used to perform similar operations as a pixel shader program as provided for in the Direct 3D API.

[0376] Graphics processor **2610** additionally includes one or more memory management units (MMUs) **2620A-2620B**, cache(s) **2625A-2625B**, and circuit interconnect(s) **2630A-2630B**. The one or more MMU(s) **2620A-2620B** provide for virtual to physical address mapping for the graphics processor **2610**, including for the vertex processor **2605** and/or fragment processor(s) **2615A-2615N**, which may reference vertex or image/texture data stored in memory, in addition to vertex or image/texture data stored in the one or more cache(s) **2625A-2625B**. The one or more MMU(s) **2620A-2620B** may be synchronized with other MMUs within the system, including one or more MMUs associated with the one or more application processor(s) **2505**, image processor **2515**, and/or video processor **2520** of FIG. 25, such that each processor **2505-2520** can participate in a shared or unified virtual memory system. Components of graphics processor **2610** may correspond with components of other graphics processors described herein. The one or more MMU(s) **2620A-2620B** may correspond with MMU **245** of FIG. 2C. Vertex processor **2605** and fragment processor **2615A-2615N** may correspond with graphics multiprocessor **234**. The one or more circuit interconnect(s) **2630A-2630B** enable graphics processor **2610** to interface with other IP cores within the SoC, either via an internal bus of the SoC or via a direct connection, according to embodiments. The one or more circuit interconnect(s) **2630A-2630B** may correspond with the data crossbar **240** of FIG. 2C. Further correspondence may be found between analogous components of the graphics processor **2610** and the various graphics processor architectures described herein.

[0377] As shown FIG. 26B, graphics processor **2640** includes the one or more MMU(s) **2620A-2620B**, cache(s) **2625A-2625B**, and circuit interconnect(s) **2630A-2630B** of the graphics processor **2610** of FIG. 26A. Graphics processor **2640** includes one or more shader cores **2655A-2655N** (e.g., **2655A**, **2655B**, **2655C**, **2655D**, **2655E**, **2655F**, through **2655N-1**, and **2655N**), which provides for a unified shader core architecture in which a single core or type or core can execute all types of programmable shader code, including shader program code to implement vertex shaders, fragment shaders, and/or compute shaders. The exact number of shader cores present can vary among embodiments and implementations. Additionally, graphics processor **2640** includes an inter-core task manager **2645**, which acts as a thread dispatcher to dispatch execution threads to one or more shader cores **2655A-2655N** and a tiling unit **2658** to accelerate tiling operations for tile-based rendering, in which rendering operations for a scene are subdivided in image space, for example to exploit local spatial coherence within a scene or to optimize use of internal caches. Shader cores **2655A-2655N** may correspond with, for example, graphics multiprocessor **234** as in FIG. 2D, or graphics multiprocessors **325**, **350** of FIGS. 3A and 3B respectively, or multi-core group **365A** of FIG. 3C.

Tensor Acceleration Logic for Graphics and Machine Learning Workloads

[0378] FIG. 27 is a block diagram of a data processing system **2700**, according to an embodiment. The data processing system **2700** is a heterogeneous processing system having a processor **2702**,

unified memory **2710**, and a GPGPU **2720** including machine learning acceleration logic. The processor **2702** and the GPGPU **2720** can be any of the processors and GPGPU/parallel processors as described herein. For example, with additional reference to FIG. **1**, processor **2702** can be a variant of and/or share an architecture with a processor of the illustrated one or more processor(s) **102** and the GPGPU **2720** can be a variant of and/or share an architecture with a parallel processor of the illustrated one or more parallel processor(s) **112**. With additional reference to FIG. **14**, processor **2702** can be a variant of and/or share an architecture with one of the illustrated processor(s) **1402** and the GPGPU **2720** can be a variant of and/or share an architecture with one of the illustrated graphics processor(s) **1408**.

[0379] The processor **2702** can execute instructions for a compiler **2715** stored in system memory **2712**. The compiler **2715** executes on the processor **2702** to compile source code **2714A** into compiled code **2714B**. The compiled code **2714B** can include instructions that may be executed by the processor **2702** and/or instructions that may be executed by the GPGPU **2720**. Compilation of instructions to be executed by the GPGPU can be facilitated using shader or compute program compilers, such as a back-end shader compiler **2327** and/or front-end shader compiler **2324** as in FIG. **23**. During compilation, the compiler **2715** can perform operations to insert metadata, including hints as to the level of data parallelism present in the compiled code **2714B** and/or hints regarding the data locality associated with threads to be dispatched based on the compiled code **2714B**. The compiler **2715** can include the information necessary to perform such operations or the operations can be performed with the assistance of a runtime library **2716**. The runtime library **2716** can also assist the compiler **2715** in the compilation of the source code **2714A** and can also include instructions that are linked at runtime with the compiled code **2714B** to facilitate execution of the compiled instructions on the GPGPU **2720**. The compiler **2715** can also facilitate register allocation for variables via a register allocator (RA) and generate load and store instructions to move data for variables between memory and the register assigned for the variable.

[0380] The unified memory **2710** represents a unified address space that may be accessed by the processor **2702** and the GPGPU **2720**. The unified memory can include system memory **2712** as well as GPGPU memory **2718**. The GPGPU memory **2718** is memory within an address space of the GPGPU **2720** and can include some or all of system memory **2712**. In one embodiment, compiled code **2714B** stored in system memory **2712** can be mapped into GPGPU memory **2718** for access by the GPGPU **2720**. The GPGPU memory **2718** also includes GPGPU local memory **2728** of the GPGPU **2720**. The GPGPU local memory **2728** can include, for example, HBM or GDDR memory.

[0381] The GPGPU **2720** includes multiple compute blocks **2724A-2724N**, which can include one or more of a variety of processing resources described herein. The processing resources can be or include a variety of different computational resources such as, for example, execution units, compute units, streaming multiprocessors, graphics multiprocessors, or multi-core groups. In one embodiment the GPGPU **2720** additionally includes a tensor accelerator **2723** (e.g., matrix accelerator), which can include one or more special function compute units that are designed to accelerate a subset of matrix operations (e.g., dot product, etc.). The tensor accelerator **2723** may also be referred to as a tensor accelerator or tensor core. In one embodiment, logic components within the tensor accelerator **2723** may be distributed across the processing resources of the multiple compute blocks **2724A-2724N**.

[0382] The GPGPU **2720** can also include a set of resources that can be shared by the compute blocks **2724A-2724N** and the tensor accelerator **2723**, including but not limited to a set of registers **2725**, a power and performance module **2726**, and a cache **2727**. In one embodiment the registers **2725** include directly and indirectly accessible registers, where the indirectly accessible registers are optimized for use by the tensor accelerator **2723**. The power and performance module **2726** can be configured to adjust power delivery and clock frequencies for the compute blocks **2724A-2724N** to power gate idle components within the compute blocks **2724A-2724N**. In various embodiments

the cache **2727** can include an instruction cache and/or a lower-level data cache.

[0383] The GPGPU **2720** can additionally include an L3 data cache **2730**, which can be used to cache data accessed from the unified memory **2710** by the tensor accelerator **2723** and/or the compute elements within the compute blocks **2724A-2724N**. In one embodiment the L3 data cache **2730** includes shared local memory **2732** that can be shared by the compute elements within the compute blocks **2724A-2724N** and the tensor accelerator **2723**.

[0384] In one embodiment the GPGPU **2720** includes instruction handling logic, such as a fetch and decode unit **2721** and a scheduler controller **2722**. The fetch and decode unit **2721** includes a fetch unit and decode unit to fetch and decode instructions for execution by one or more of the compute blocks **2724A-2724N** or the tensor accelerator **2723**. The instructions can be scheduled to the appropriate functional unit within the compute block **2724A-2724N** or the tensor accelerator via the scheduler controller **2722**. In one embodiment the scheduler controller **2722** is an ASIC configurable to perform advanced scheduling operations. In one embodiment the scheduler controller **2722** is a micro-controller or a low energy-per-instruction processing core capable of executing scheduler instructions loaded from a firmware module.

[0385] In one embodiment some functions to be performed by the compute blocks **2724A-2724N** can be directly scheduled to or offloaded to the tensor accelerator **2723**. In various embodiments the tensor accelerator **2723** includes processing element logic configured to efficiently perform matrix compute operations, such as multiply and add operations and dot product operations used by 3D graphics or compute shader programs. In one embodiment the tensor accelerator **2723** can be configured to accelerate operations used by machine learning frameworks. In one embodiment the tensor accelerator **2723** is an application specific integrated circuit explicitly configured to perform a specific set of parallel matrix multiplication and/or addition operations. In one embodiment the tensor accelerator **2723** is a field programmable gate array (FPGA) that provides fixed function logic that can be updated between workloads. In one embodiment, the set of compute operations that can be performed by the tensor accelerator **2723** may be limited relative to the operations that can be performed by the compute block **2724A-2724N**. However, the tensor accelerator **2723** can perform parallel tensor operations at a significantly higher throughput relative to the compute block **2724A-2724N**.

[0386] FIG. **28A-28B** illustrate a matrix operation **2805** performed by an instruction pipeline **2800**, according to embodiments. FIG. **28A** illustrates the instruction pipeline **2800** when configured with a systolic array **2808** within the tensor accelerator **2723**. FIG. **28B** illustrates the instruction pipeline when configured with graphics processor cores **2810A-2810N** that each include matrix engines **2812A-2812N**.

[0387] As shown in FIG. **28A**, the instruction pipeline **2800** can be configured to perform a matrix operation **2805**, such as, but not limited to a dot product operation. The dot product of two vectors is a scalar value that is equal to sum of products of corresponding components of the vectors. The dot product can be calculated as shown in equation (1) below.

[00001]
$$\vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i = a_1 b_1 + \dots + a_n b_n \quad (1)$$

[0388] The dot product can be used in a convolution operation for a convolutional neural network (CNN). While 2D convolution is illustrated, N-dimensional convolution can be performed on an N-dimensional volume using N-dimensional filters. A receptive field tile **2802** highlights a portion of an input volume in an input volume buffer **2804**. The input volume buffer can be stored in memory **2830**. A dot product matrix operation **2805** can be performed between the data within the receptive field tile **2802** and a convolutional filter to generate a data point within output buffer **2806**, which can also be stored in memory **2830**. The memory **2830** can be any of the memory described herein, including system memory **2712**, GPGPU memory **2718**, cache **2727**, or L3 data cache **2730** as in FIG. **27**.

[0389] The combination of the data points within the output buffer **2806** represents an activation

map generated by the convolution operation. Each point within the activation map is generated by sliding the receptive field tile across the input volume buffer **2804**. The activation map data can be input to an activation function to determine an output activation value. In one embodiment, convolution of the input volume buffer **2804** can be defined within a framework as high-level matrix operation **2805**. The high-level matrix operations can be performed via primitive operations, such as a basic linear algebra subprogram (BLAS) operation. The primitive operations can be accelerated via hardware instructions executed by the instruction pipeline **2800**.

[0390] The instruction pipeline **2800** used to accelerate hardware instructions can include the instruction fetch and decode unit **2721**, which can fetch and decode hardware instructions, and the scheduler controller **2722** which can schedule decoded instructions to one or more processing resources within the compute blocks **2724A-2724N** and/or the tensor accelerator **2723**. In one embodiment, a hardware instruction can be scheduled to the compute blocks **2724A-2724N** and offloaded to the tensor accelerator **2723**. The one or more hardware instructions and associated data to perform the matrix operation **2805** can be stored in the memory **2830**. Output of the hardware instruction can also be stored in the memory **2830**.

[0391] In one embodiment, the tensor accelerator **2723** can execute one or more hardware instructions to perform the matrix operation **2805** using a systolic array **2808** of processing elements. The systolic array **2808** includes a combination of programmable and fixed function hardware that is configurable to perform matrix-matrix and matrix-vector dot product operations, as well as other operations, such as matrix-matrix and matrix-vector fused multiply-add operations.

[0392] In various embodiment, as an alternative or in addition to the tensor accelerator **2723**, matrix acceleration logic can also be included within the processing resources of the compute blocks **2724A-2724N**. For example, as shown in FIG. **28B**, in one embodiment each compute block (e.g., compute block **2724N**) includes an array of graphics cores **2810A-2810N**. Each graphics core in the array of graphics core **2810A-2810N** can include a matrix accelerator **2812A-2812N**. In one embodiment, the graphics cores **2810A-2810N** are graphics core **1815A-1815N** as in FIG. **18A** and the matrix accelerators **2812A-2812N** include a version of the matrix engine **1803** of FIG. **18C**. The scheduler controller **2722** can schedule matrix operations (dot products, fused multiply-adds, etc.) to available matrix accelerator **2812A-2812N** within the graphics cores **2810A-2810N** of the various compute blocks **2724A-2724N**.

[0393] While in one embodiment each of the compute blocks **2724A-2724N** include an array of graphics cores **2810A-2810N**, in another embodiment the compute blocks **2724A-2724N** share an architecture with the processing clusters **214A-214N** of the processing cluster array in FIG. **2A**. In such embodiment, the compute blocks **2724A-2724N** include multiple graphics multiprocessors **234** as in FIG. **2C**, which include internal components as illustrated in FIG. **2D**. Thus, the graphics multiprocessors within the compute blocks can include a load/store unit **266**, GPGPU cores **262**, and tensor/RT cores **263**. In one embodiment the compute blocks **2724A-2724N** can include multi-core group **365A-365N** of the GPU **380** of FIG. **3C** and include multiple sets of GFX cores **370**, tensor cores **371**, and ray tracing cores **372**. In such embodiment, the scheduler controller **2722** can schedule instructions to perform matrix operations to the tensor/RT cores **263** and/or tensor cores **371** within the compute blocks **2724A-2724N**. Accelerated matrix operations include dot product operations, matrix multiply operations, and/or fused multiply-add operations, which can be performed on integer or floating-point matrix elements and various levels of precision.

Additionally, in one embodiment the compute blocks **2724A-2724N** can include a variant of the compute units **1560A-1560N** of FIG. **15C**, where such variants include matrix acceleration logic as described herein (e.g., systolic array, tensor core, systolic tensor core) that can execute integer or floating-point matrix acceleration instructions.

[0394] FIG. **29** illustrates a systolic array **2900** including multiplier and adder circuits organized in a pipelined fashion. In one embodiment, systolic array **2900** is representative of the physical pipeline stages included in the accelerators **2812A-2812N** and includes capabilities described in

relation to that matrix engine **1803** of FIG. **18C**, including support for sparse and block sparse operations, and may additionally be configured to support structured sparsity within a vector of elements or across a set of channels. Inputs **2912A-2912H** for the first input matrix are represented by the data elements contained in the inputs labeled Src1 and Src1+1 through Src1+7. Inputs **2910A-2910H** correspond to the second input matrix and are labeled as Src2. Inputs **2902A-2902B**, which may include initial accumulator values, can be provided as Src0. An array of processing elements makes up the physical pipeline stages **2911A-2911H** of the systolic array **2900**. Matrix-Matrix or Matrix-Vector operations, including fused multiply-add and/or dot product operations, can be performed at each pipeline stage **2911A-2911H** during each clock cycle. On each cycle, every pipeline stage can receive a new Src2 input can be used by the processing elements of the pipeline stage to compute a value using either the new Src1 input or an older Src1 input that was previously read, although during initial startup it may take several cycles before all of the pipeline stages **2911A-2911H** become active as the initial set of computed values propagate through the stages.

[0395] Input **2902A** can provide a Src0 value to processing element of pipeline stage **2911A**, for use as an initial accumulator value. Alternatively, input **2902B** can provide the Src0 value to be added to the values computed by pipeline stage **2911H** of the systolic array, which enables partial pass operation for systolic array **2900** using the lower stages of the array while the unused upper stages are power gated. During operation, the data elements of a selected channel of the Src2 input are broadcast across all channels of the processing elements of the pipeline stages **2911A-2911H**, where each channel represents a vector of multiple elements. The number of elements per channel can vary based on the size of the elements. The processing elements of a stage then perform operations using the selected Src2 channel and all channels of a given Src1 input. A Src2 input operates with eight Src1 inputs (e.g., one Src1 input per stage). The data elements of a channel of the Src2 input are broadcast across all channels of processing elements **2911A-2911H**. The processing elements then operate the Src2 channel with all channels of a Src1 input. In a first clock cycle, a Src1 input is operated with data elements of the first channel of Src2. In the next cycle, a second Src1 (labeled as Src1+1) operates with the data elements of the second channel of Src2. This sequence repeats on the eight stages of the pipeline. Each stage adds its operation to the output of the previous stage. Across the pipeline stages, multiple Src2 inputs are operated in a pipelined fashion. As successive channels of a first Src2 input are pushed through the pipeline stages, a new Src2 input can be provided at the first stage.

[0396] Output **2922** from the final stage is labeled as Dst. Where d=the systolic depth and e=the number of data elements per channel, the output of a channel is described by equation (2) below:

$$[00002] \quad (2) \text{Dst}_i = \text{Src0}_i + \sum_{j=0}^d \text{Math.} \left(\sum_{k=0}^e (\text{Src1} + j)_{\text{element } k \text{ of channel } i} * \text{Src2}_{\text{element } k \text{ of channel } j} \right)$$

[0397] As shown in equation (2), each channel can include multiple data elements on which operations are performed in parallel. In one embodiment, each channel represents a four element data vector, although a different number of elements can be configured for each channel. In one embodiment, the number of data elements within a channel can vary based on the size of each data element. Dot products can be performed using, for example, four element vectors with 8-bit data types per element, two element vectors with 16-bit data types, eight element vectors with 4-bit data types (e.g., INT4), or 16 element vectors with 2-bit data types (e.g., INT2). The number of channels can be automatically adjusted depending on the datatype of Src1 and Src2. An instruction can also specify a required systolic depth to be used for the instruction.

[0398] In one embodiment the processing elements **2911A-2911H** may read inputs **2910A-2910H**, **2912A-2912H** directly from the general-purpose register file. In one embodiment systolic array **2900** includes logic to read inputs **2910A-2910H**, **2912A-2912H** from the general-purpose register file and store input data in registers, buffers, or memory that is internal to the systolic array. Internal logic can then feed the input data elements to the processing elements **2911A-2911H** for

processing. Output **2922** can be written to internal registers or memory of the systolic array **2900** and/or written directly to the general-purpose register file.

[0399] FIG. **30A-30B** illustrates the use of a systolic array **3000** that can be configured to execute operations at an arbitrary systolic depth. In the illustrated example, the systolic array **3000** has a physical depth of four, which corresponds with four physical pipeline stages. The systolic array can be configured to operate using an arbitrary number of logical stages, including four, eight, twelve, or sixteen logical stages, or other numbers of logical stages that are not divisible by the number of physical stages using partial-pass operations as in FIG. **31** described below. FIG. **30A** shows the array receiving Src0 inputs from an external source and processing the first four stages with Src1 and Src2 inputs. The output of this array is fed back into the second step shown in FIG. **30B**. FIG. **30B** shows that the next four stages are calculated using the loopback data that includes the already processed values and the Src1 and Src2 inputs.

[0400] As shown in FIG. **30A**, systolic array **3000** can accept input **2902**, as Src0 input, which is read (**3002**) via data selector **3004**. Data selector **3004** selects between the input **2902** and loopback input **3006**. Processing elements **2911A-2911D** can process inputs **2910A-2910D** and **2912A-2912D** in a similar manner as systolic array **2900**. If four stages are sufficient to complete an operation, pipeline stage **2911D** can write (**3022**) output **2922** to a specified Dst register or memory via data selector **3024**. Where further stages are required, data selector **3024** can write loopback output **3026**, which is provided as loopback input **3006** to processing elements of pipeline stage **2911A**.

[0401] As shown in FIG. **30B**, in one embodiment, loopback input **3006** can be further processed by processing elements **2911A-2911D**. Loopback input **3006** includes the already processed values. In one embodiment, loopback input **3006** can also include input **2910E-2910H**, input **2912E-2912H**, which can be pre-fetched while processing the first four stages. Data selector **3004** select loopback input **3006** for input by pipeline stage **2911A**. Processing elements of the pipeline stages **2911A-2911D** can then process inputs **2910E-2910H** and **2912E-2912H**. Data selector **3024** can then write (**3022**) the eighth stage result as output **2922** to the specified Dst register.

[0402] In one embodiment, the systolic array **3000** is modified to exclude the loopback output **3026** and loopback input **3006** and instead include intermediate storage **3025**, as shown in FIG. **30A-30B**. The intermediate storage **3025** may be a memory device or register that is internal to the systolic array **3000** or may be a register in a register file that is external to the systolic array **3000**. During the operations shown in FIG. **30A**, output from pipeline stage **2911D** can be stored in the intermediate storage **3025** instead of being output by loopback output **3026** and read by loopback input **3006** before the operations shown in FIG. **30B**. During the operations shown in FIG. **30B**, output from pipeline stage **2911D** can be added to the data stored in the intermediate storage **3025** and written to output **2922**. The systolic array **3000** can also be configured to perform multi-pass operations using at least one partial pass, as described below, to enable logical depths that are not divisible by the physical depth of the array.

Scalable Matrix Multiply Accelerator with Feedback Inputs

[0403] A second embodiment enables increased throughput using simultaneous instructions executed using parallel units. Several instances or paths of the multiply accelerator are run in parallel. These instances can share Src1, or they can have independent Src1 inputs. Each path will have their own Src2 and Src0 inputs. These instances will have their own src2 and src0 inputs. A version showing two paths with a depth of four stages is shown in FIG. **31**. Alternatively, a version using four paths of depth of two stages is shown in FIG. **32**.

[0404] FIG. **31** illustrates a two-path matrix multiply accelerator **3100** in which each path has a depth of four stages. The two-path matrix multiply accelerator **3100** includes input logic **3102A-3102B** for Src0 inputs, input buffers **3111A-3111B** to store data elements received from input logic **3110A-3110B**, and input buffers **3113A-3113B** to store data elements received from shared input logic **3112** for Src1. Each stage includes a pair of processing elements, which may operate in

parallel. Stage one includes processing elements **3131A-3131B**, stage two includes processing elements **3132A-3132B**, stage three includes processing elements **3133A-3133B**, stage four includes processing elements **3134A-3134B**. Hardware logic of each of the processing elements **3131A-3131B**, **3132A-3132B**, **3131A-3133B**, **3134A-3134B** can be the same as or similar to the hardware logic of processing elements of systolic array **2900** or systolic array **3000** and may be manufactured with the same process technology or a more advanced process technology. The processing elements of the two-path matrix multiply accelerator **3100** may also operate at a higher frequency relative to implementations of systolic array **2900**. The processing elements and may be manufactured using more advanced process technology.

[0405] Feedback may be implemented using data selectors that are the same as or similar to data selectors **3004**, **3024**. Depending on the configuration of the read logic, input data can be pre-fetched into the input buffer in advance or read from registers or a cache within the two-path matrix multiply accelerator **3100** one or more cycles before input into the processing elements **3131A-3131B**. Processing elements **3134A-3134B** of stage four can feed back into the corresponding processing elements **3131A-3131B** stage one. Dynamic logical depth may be enabled in multiples of four. After a configured number of logical stages, results may be written by output logic **3122A-3122B** to a specified destination.

[0406] FIG. **32** illustrates a four-path matrix multiply accelerator **3200** in which each path has a depth of two stages. Four-path matrix multiply accelerator **3200** includes the same number of processing elements as two-path matrix multiply accelerator **3100**, with the processing elements configured with twice as many paths, but each path is half as deep. Four-path matrix multiply accelerator **3200** includes input logic **3202A-3202D** for Src0, input buffers **3211A-3211D** to store input elements read by input logic **3210A-3210D** for Src2, and input buffers **3213A-3213D** to store input elements read by shared input logic **3212** for Src1. Processing elements **3231A-3231B** enable parallel processing for stage 1. Processing elements **3232A-3232B** enable parallel processing for stage 2. Stage 2 of each path can feed back into stage 1 or write results via output logic **3222A-3222D** to a specified destination. Processing elements **3231A-3231B**, **3232A-3232B** may include hardware logic similar to that of processing elements **3131A-3131B**, **3132A-3132B**, **3131A-3133B**, **3134A-3134B** and can implement loopback functionality using similar hardware logic.

[0407] The advantages of a two-path matrix multiply accelerator **3100** or a four-path matrix multiply accelerator **3200** include scalability, software compatibility, and throughput. The modular architecture of these accelerators enables more efficient scaling relative to an 8-deep systolic array. Different configurations of a matrix multiply accelerator can be tailored for different product needs or use cases without redesign. Additionally, the same software model that is used is independent of the hardware implementation. Algorithms designed for an instruction intended to be executed by a systolic pipeline of eight stages can be used in an implementation using a Matrix Multiply accelerator of four stages. Hardware will use feedback to simulate a pipeline of eight stages in a way that is transparent to the software. Multiple paths can be used in a design requiring high DPAS instruction throughput. Implementations with a greater number of paths can be coupled with higher bandwidth input logic and output logic. In one embodiment, the two-path matrix multiply accelerator **3100** and a four-path matrix multiply accelerator **3200** are configured to bypass inputs with block sparsity at a greater efficiency and/or finer granularity than possible with an 8-deep systolic array.

Sparse Multiplications on the Scalable Matrix Multiply Accelerator

[0408] A third embodiment facilitates increased instruction throughput when processing for data with irregular sparsity. Elements of Src1 and Src2 inputs can be individually selected via input multiplexer logic and processing can be performed using only non-zero values.

[0409] FIG. **33** illustrates a scalable sparse matrix multiply accelerator **3300** using systolic arrays with feedback inputs. Scalable sparse matrix multiply accelerator **3300** can include processing elements **3231A-3231D** as in four-path matrix multiply accelerator **3200**, or any other processing

elements described herein. Processing elements **3231A-3221B** at the beginning of each path include input logic for Src0. Each stage of each path of scalable sparse matrix multiply accelerator **3300** can receive any element of an independent or shared Src1 via input selectors **3312A-3312D**. Each stage of each path can also receive any element of a Src2. Independent Src2 inputs are provided via separate input element selectors (e.g., Src2A via input selector **3310A** and input selector **3311A**, Src2B via input selector **3310B** and input selector **3311B**). The separate Src2 input enables the separate paths to compute different instructions. Separate output logic **3322A-3322B** is present for each path to enable output for the different instructions.

[0410] FIG. **34** shows a scalable sparse matrix multiply accelerator **3400** using systolic arrays with feedback inputs and outputs on each stage. Scalable sparse matrix multiply accelerator **3400** includes similar hardware logic as scalable sparse matrix multiply accelerator **3300**, along with additional input and output logic to enable Src0 elements to be provided to each stage of each path and to provide separate outputs for each stage of each path. In addition to input selectors **3310A** and **3311A** to select Src2A elements for the first path and input selectors **3310A** and **3311B** to select Src2B input for the second path, an input splitter **3403A-3403B** is added for each path for Src0 input. Each input splitter **340A-3402B** can include a demultiplexer or similar hardware logic to enable Src0 input elements that are read by input logic **3402A-3402B** to be sent to each stage. Input selectors **3312A-3312D** are also included to enable Src1 input to be elected by each stage of each path. In addition to output logic **3322A-3322B** from the second stage of each path (processing element **3431C-3431D**), additional output logic **3422A-3422B** is provided to enable output from the first stage of each path (**3431A-3431B**). The processing elements **3431A-3431C** may be otherwise similar to other processing elements described herein.

[0411] During operation, scalable sparse matrix multiply accelerator **3400** is configurable to accept groups of only one element. Given Src2 input {B0, 0, B2, B3, 0, 0, 0, 0}, two groups ([B0,B2], [B3,0]) are made for the non-zero elements on Src2 for the third embodiment (e.g., scalable sparse matrix multiply accelerator **3300**), with the second group including a zero padding. The optimizations shown in FIG. **34** enable the groups to be formed as [B0,B2], [B3]. B0 and B2 will be assigned to the first and second stage of a path (e.g., either of a first set including of processing element **3431A** and processing element **3431C** or a second set including processing element **3431B** and processing element **3431D**). After the feedback, B3 will be assigned to the first stage of that path. As the first stage of a path can provide output (e.g., via either output logic **3422A** or **3422B**), there is no need to consume the second stage of the path (either of processing element **3431C** or processing element **3431D**). Moreover, the next Src2 input accepted for that path can start from the second stage, so a group of two elements will be assigned to the second and first stage respectively. Src0 for processing the new Src2 input can be assigned to the second stage of the path (e.g., via either output logic **3422A** or **3422B**)

[0412] In addition to the hardware logic of scalable sparse matrix multiply accelerator **3300** illustrated in FIG. **33** and scalable sparse matrix multiply accelerator **3400** illustrated FIG. **34**, some embodiments additionally include input and output hardware memory buffers. Input memory buffers can be used to store and have ready groups of Src0 and Src2 inputs, which reduces the need for high bandwidth input logic. The output buffer allows Dst outputs generated in a same cycle to be steadily written to memory at a slower rate, which reduces the need for high bandwidth output logic.

[0413] Additionally, some embodiments include a bypass for inputs in which all elements are zero. The bypass allows a direct write of Src0 as by output logic without passing through the systolic array. This bypass is used in concert with a data dependency strategy to prevent read-after-write (RAW) risks among instructions can damage the integrity of the data.

Matrix Accelerator Having a Dual Pipeline Parallel Systolic Array

[0414] FIG. **35** illustrates a dual pipeline parallel systolic array **3500** for a matrix accelerator, according to an embodiment. A matrix accelerator as described herein (e.g., tensor accelerator

2723, tensor/RT cores **263**, tensor cores **371**, matrix engine **1803**) can include a dual pipeline parallel systolic array **3500** that includes two systolic array pipelines (systolic pipeline **3502**, systolic pipeline **3504**) that operate in parallel to execute instructions. The dual pipeline parallel systolic array **3500** enables the row data that is provided as Src2 input to be partitioned, with the partitions being processed in parallel using a common Src1 input. Such configuration enables increased throughput for matrix operations without incurring the power and area costs associated with two separate and fully independent systolic arrays.

[0415] Input for matrix operations can be read from a register file (e.g., register file(s) **258**, **334A-334B**, **369**, vector registers **1561**, GRF **1821**, etc.) that is associated with the matrix accelerator.

The dual pipeline parallel systolic array **3500** includes an input **3521** for a Src1 operand that is shared between the two systolic array pipelines. The Src1 input inputs column data that is used by the two systolic array pipelines to perform matrix multiply operations in which two sets of matrix row data (Src2 input **3522A-3522B**) are multiplied by a single set of column data. A single Src2 register can store input for two stages of operation. For example, data from inputs **3522A-3522B** can be read in 64-bit blocks, with the lower 32-bits being used for operations at a stage of the systolic array and the upper 32-bits being used for operations at the next successive stage of the systolic array. As one Src2 read can be used for two operations on an array, the second cycle of a pair of Src2 read cycles can be used to read a new Src2 for the second array. The common input **3521** for Src1 data and the use of Src2 register data for multiple operations reduces read demand on the GRF relative to two fully independent systolic arrays. The reduce register read demand relative to the use of independent systolic arrays can reduce the potential negative impact on performance caused for other processing elements that share the register file with the systolic array when those processing elements are operating concurrently with the systolic arrays.

[0416] Separate inputs **3520A-3520B** are provided for Src0 (accumulator value) inputs. The data from inputs **3520A-3520B** is stored in a Src0 data buffer **3530A-3530B** and added to output from the systolic array pipelines, as opposed to being added at Stage **0** as in other systolic array designs. Output from each array can be stored in accumulator/adder circuits that include memory (e.g., an accumulator register) and an adder circuit. Accumulator/adder circuit **3532** can store output from systolic pipeline **3502** and add the output to data stored in Src0 data buffer **3530A**.

Accumulator/adder circuit **3534** can store output from systolic pipeline **3504** and add the output to data stored in Src0 data buffer **3530B**.

[0417] In one embodiment, multi-pass operation is enabled, such that the eight physical stages of the array operate as sixteen logical states. The eight stages of each of systolic pipeline **3502** and systolic pipeline **3504** can operate as sixteen logical stages by respectively storing the output of a first pass to the first accumulator/adder circuit **3532** and second accumulator/adder circuit **3534**. The values stored in the circuits can be accumulated with output generated by a second pass through each of systolic pipeline **3502** and systolic pipeline **3504**. For a given stage *i*, the stage operates as stage *i* during a first pass and stage *i*+8 during a second pass. The appropriate input data is provided to the arrays depending on whether the array is performing first pass operations or second pass operation. In one embodiment, operations for instructions of any number of logical stages may be supported via single pass and/or multiple or partial pass operation. A selector circuit **3536** enables data within the first accumulator/adder circuit **3532** and second accumulator/adder circuit **3534** to be output to a destination register.

[0418] FIG. **36** illustrates a stage pair **3600** for a channel of a systolic array. In one embodiment the physical pipeline stages for each array of the dual pipeline parallel systolic array **3500** of FIG. **35** are grouped as a stage pair **3600**. A stage pair **3600** for Stage **0** (**3610**) and Stage **1** (**3611**) is illustrated, with other pairs of stages (e.g., [2,3], [4,5], [6,7]) being configured similarly. Each channel of each stage includes a pair of multipliers (e.g., multipliers **3612A-3612B** for Stage **0**, multipliers **3613A-3613B** for Stage **1**) and a common adder **3604**. The accumulator input **3620** (Src0) is passed through to Src0 data buffer **3530A-3530B** shown in FIG. **35** and is not operated on

by the stage pair **3600**. The appropriate Src1 register data is provided as input to the appropriate stage. A single Src2 register read can store data for both stages in the stage pair **3600**.

[0419] FIG. **37** illustrates a systolic array **3700** including partial sum loopback and circuitry to accelerate sparse matrix multiply. In the systolic array **2808** described above, operands that include weight data may be stationary within the array and a partial sum is propagated throughout the array structure. While other details with respect to systolic array **2808** may be applicable, in systolic array **3700** a partial sum is recirculated instead of being propagated to a next systolic layer. In one embodiment a systolic array **3700** can be configured with M rows and N columns of processing elements (PE **3712AA**-PE **3712MN**). The processing elements can access registers storing input data in the form of row and column data for input matrices. The registers may be stored in a register file that is local to the systolic array **3700** or in a register file of a processing resource that is coupled with or includes the systolic array **3700**. The registers may store row elements of matrix A **3702A-3702M**, which are to be multiplied by column elements of matrix B **3701A-3702N**.

[0420] In one embodiment a fused multiply-add (FMA) can be performed at each processing element PE **3712AA**-PE **3712MN** each clock cycle. An element of matrix A is multiplied by a corresponding element of matrix B and then added to an accumulator value or, for the first cycle, an optional initial input value (e.g., SRC0). Partial sum loopback can be configured at each processing element. After each cycle, the accumulator value may be looped back within the processing element and used as input for the next cycle. Once operations are performed for an entire row, the result may be stored to a register file. Data movement between the processing elements PE **3712AA**-PE **3712MN** after a set of computational cycles can vary based on the instruction or macro-operation being performed.

Data Aware Sparsity with Compression

[0421] Embodiments described herein provide an encoding layout that enables sample blocks of sparse neural network data to be encoded in a reduced-bit format that reduces the amount of data that is required to be transmitted or stored when processing neural networks associated with the data. The number of non-zero values in a sample block is indicated in a header, followed by a significance map indicating a map of the non-zero values within the block. The non-zero values of the sample are encoded in order of appearance within the stream. In one embodiment, compression can be based on other values beyond zero values. For example, a specified value within a data set may be encoded and excluded from a compressed data stream, enabling compression based on ones, twos, or other specified values. In one embodiment compression is enabled based on near values. Values within a data set that are within a threshold of zero, or within a threshold of a specified value, may be compressed as though those values were zero or within a threshold of the specified value. Data aware sparsity with compression can be enabled via codec logic coupled with or within matrix accelerator logic.

[0422] FIG. **38A-38B** illustrate matrix acceleration circuitry including codecs to enable the reading of sparse data in a compressed format. FIG. **38A** illustrates a compute block **3800** including codec enabled disaggregated systolic logic. FIG. **38B** illustrates processing elements within a systolic array that are coupled with codecs to decompress input data.

[0423] As shown in FIG. **38A**, instead of including a systolic array **2808** in a separate tensor accelerator **2723**, as in FIG. **28A**, or including a matrix engine **1803A-1803N** in each graphics core **1815A-1815N**, a disaggregated set of systolic arrays **3812A-3812B** can be included in a compute block **3800** that is analogous to one of the compute blocks **2724A-2724N** of FIG. **27**. The compute block **3800** can include multiple interconnected processing resources (PR **3808A-3808O**), which may be similar to any processing resource architecture described herein, such as but not limited to processing resources described herein, including that may be similar to EU **1808A-1808N** or any other processing resource as described herein. In one embodiment the systolic arrays **3812A-3812B** include codecs **3824A-3824B** that enable the encoding and decoding of input and output data that is received for processing.

[0424] The systolic arrays **3812A-3812B** include a W wide and D deep network of data processing units that can be used to perform vector or other data-parallel operations in a systolic manner, similar to other systolic arrays described herein. In one embodiment the systolic arrays **3812A-3812B** can be configured to perform matrix operations, such as matrix dot product operations. In one embodiment the systolic arrays **3812A-3812B** support 16-bit floating point operations, as well as 8-bit and 4-bit integer operations. In one embodiment the systolic arrays **3812A-3812B** can be configured to accelerate machine learning operations. In such embodiments, the systolic arrays **3812A-3812B** can be configured with support for the bfloat 16-bit floating point format. By including systolic arrays **3812A-3812B** within the compute block **3800** but outside of the PRs **3808A-3808O**, the size and number of systolic arrays **3812A-3812B** can be scaled independently from the number of PRs **3808A-3808O**. Additionally, communication bandwidth within an PR that would otherwise be consumed by systolic array activity may be preserved. Furthermore, the systolic arrays **3812A-3812B** may be clock/power gated when matrix workloads are not being performed.

[0425] Communication between the systolic arrays **3812A-3812B** and the PRs **3808A-3808O** may be performed via a cache or shared local memory (cache/SLM **3810**) and/or a shared register file **3814**. In one embodiment, instead of a distinct shared register file **3814**, the cache/SLM **3810** may be partitioned for use as a shared register file. The shared register file **3814** may be structured similarly to other GPGPU register files, such as register file **369** as in FIG. 3C. The shared register file may also include a set of special purpose registers that are used to configure the interaction between the systolic arrays **3812A-3812B** and the PRs **3808A-3808O**. The cache/SLM **3810** may be an L1 cache, an L2 cache, and/or a block of explicitly addressable on-die memory.

[0426] Matrix data for processing by the systolic arrays **3812A-3812B** may be stored in the cache/SLM **3810**. Processing commands or instructions can be provided to the systolic arrays **3812A-3812B** via the shared register file **3814**. Processing results may be read from the cache/SLM **3810** by the PRs **3808A-3808O** or from destination/output registers within the shared register file. During operation, instead of consuming bus/fabric bandwidth within the PRs **3808A-3808O**, communication traffic may be localized to the systolic arrays **3812A-3812B**, the cache/SLM **3810**, and/or shared register file **3814**. Any of the PRs **3808A-3808O** within the compute block **3800** may offload a matrix workload to one or both systolic arrays **3812A-3812B**. A message may be sent from a PR to a systolic array with a command that specifies an operation to be performed and operands for the operation. The systolic arrays **3812A-3812B** can perform the requested operations (multiply/add, fused multiply/add, multiply/accumulate, dot product, etc.) and output the results to the shared register file **3814**. Input, intermediate and/or output data for requested operations may be stored in the cache/SLM **3810** and multiple dependent operations may be chained. In one embodiment when processing operations for training or inference for a neural network are performed, the systolic arrays **3828A-3828B** may also perform activation functions including but not limited to sigmoid, ReLU, and hyperbolic tangent (TanH) activations. In such embodiment, operations for neural networks may be offloaded to the systolic arrays **3812A-3812B** at coarse granularity.

[0427] The PRs **3808A-3808O** can provide input data to the systolic arrays **3812A-3812B** in a compressed format and the codecs **3824A-3824B** can be used to decompress the data. When output data is ready to be provided to the PRs **3808A-3808O**, the data may remain decompressed if the PRs will perform operations and the data and do not support the direct read of compressed data. If the PRs **3808A-3808O** support the reading of compressed data or will not perform additional operations on the data, the output data may be re-encoded. Zero-based encoding may be used and compression may be enabled or disabled based on the degree of data sparsity. Alternatively, other forms of encoding may be used based on the distribution of the data set to be processed or output. For example, the codecs **3824A-3824B** can be configured to decode sparse data that is encoded based on zero-based compression or using another form of compression described herein (e.g., one-

based, two-based, near-zero, near-one, near-two, etc.).

[0428] As shown in FIG. 38B, system 3850 illustrates processing elements of systolic array 3700, where the systolic array is configured to decode compressed sparse data. As described with respect to FIG. 37, each PE 3712AA-3713MN includes hardware logic to perform computations for matrix operations. A (A0, A1, through A.sub.M) and B (B0, B1, through B.sub.N) are elements of input matrices with associated with dot product, matrix multiply, multiply/add, or multiply accumulate operations. In one embodiment each PE 3712AA-3713MN is associated with codecs (3851a, 3851b, . . . , 3851m; 3852a, 3852b, . . . , 3852n) to decode compressed input operands associated with operations to be performed. The codecs can be configured to decode sparse data that is encoded based on zero-based compression or using another form of compression described herein.

[0429] Sparse neural network data can be encoded (e.g., compressed) using a variety of encoding techniques, such as but not limited to unique absolute value (UAV) table encoding, significance map (SM) encoding, table encoding (TE), unique value coordinate (UVC) encoding, and mean encoding (ME). Metadata for the encoded data indicates the type of encoding format used for the data. In one embodiment, specific encoding formats can be selected for specific types of data, such as kernel data or feature data. In one embodiment, statistical analysis is performed on the data prior to encoding to enable an appropriate encoder to be selected for each block of data. The encoding may be zero-based encoding, near-zero encoding or based on other values (ones, twos, etc.).

[0430] In one embodiment data generated during SM encoding can be used to facilitate provision of compressed data to a systolic tensor array. In zero-based SM encoding mode, only non-zero values in a block are encoded. The number of non-zero values in a sample block is indicated in the header, followed by a significance map indicating a map of the non-zero values within the block. The non-zero values of the sample are then encoded in order of appearance within the stream.

Temporally Amortized Supersampling Using Mixed Precision Convolutional Neural Network

[0431] Described herein are embodiments that provide a machine learning-based temporally amortized supersampling technique that replaces temporal anti-aliasing (TAA). A mixed low precision convolutional neural network is used that applied different computational precisions at different stages to enable the high performance generation of high quality images based on source images rendered at a relatively lower resolution than the target output resolution. The network model enables anti-aliasing and upscaling with support for multiple scale factors, including fractional scale factors such as, but not limited to 1.3×, 1.5×, 1.7×, 2×, or 2.2×. Other scale factors are also possible. Temporally stable upscaled output can be generated that has an image quality that is better than or equal to native rendering at the target resolution. In various embodiments, different versions are provided that can be implemented on a variety of different graphics processing architectures, including architectures with matrix acceleration hardware as described above in FIG. 28A through FIG. 34, as well as graphics processor architectures that lack dedicated matrix acceleration hardware.

[0432] FIG. 39 illustrates a conventional renderer 3900 with Temporal Anti-aliasing (TAA). The renderer within the rasterization and lighting stage 3910 can jitter (3905) the camera 3902 during rendering for every frame to sample different coordinates in screen space 3904. Different pixels can be sampled from different frames over time. The TAA stage 3916 accumulates these samples temporally to produce a supersampled image. A warping operation 3924 is applied to the previously accumulated frame (History 3923) using renderer generated velocity/motion vectors 3922 to align the previously accumulated frame with the current frame 3912 (frame N) before accumulation. Optional upscaling 3914 can be performed on the current frame before input to the TAA stage 3916, such that the current frame can be rendered at a lower resolution than the target resolution. The output frame can then be added to the history 3923 for use in processing the next frame. Post processing operations 3918 can then be performed at the upscaled target resolution. While applying upscaling with TAA can improve rendering performance, the output images are of lower quality than images rendered natively at the target resolution. Some TAA implementations

can use heuristics **3915** such as but not limited to neighborhood color clamping, object identifier comparisons, and depth value comparisons to detect mismatches between current and history frames and reject the history pixels. However, these heuristics often fail and produce a noticeable amount of ghosting, over-blurring and/or flickering.

[0433] FIG. **40** illustrates a renderer **4000** that replaces the TAA stage with a temporally amortized supersampling stage, according to embodiments provided herein. Renderer **4000** differs from renderer **3900** of FIG. **39** in that, in renderer **4000**, temporally amortized supersampling is performed using a convolutional neural network model **4050**. The convolutional neural network model **4050** is a mixed, low-precision model that replaces the TAA stage in the game renderer, achieving significantly better image quality than conventional TAA-based techniques, as well as providing a performance boost by enabling rendering to be performed at lower resolution. The renderer **4000** can render the current frame **3912** at a lower than target resolution. An upscaling filter **4014** is applied to the rendered image to upscale the image to the target resolution. In one embodiment, the upscaling filter **4014** is applied by the renderer **4000** before the current frame **3912** is provided to the supersampling stage. In one embodiment, the upscaling filter is performed by the neural network model **4050** during pre-processing operations. The upscaling filter **4014** can include optimizations to enhance the image quality of temporal stability of images that result from the processing performed by the neural network model **4050**. Warping operations **4024** on the history **3923** can be performed by an input block of the neural network model **4050**. In one embodiment the history **3923** is a multi-frame history that includes data from multiple previous frames.

[0434] The mixed, low-precision convolutional neural network is implemented via a neural network model **4050** that consists of multiple convolution layers, as well as other operations that are performed at low precisions, such as INT8, mixed with operations performed at a higher precision, such as FP16. The mix of precisions enable the network to achieve a fast computational speed while generating high quality output images. The lower precision values are not limited to INT8 and different low-precision data formats (e.g., INT4, binary, bipolar binary, ternary, etc.) can be used for variations. The majority of the neural network model **4050** and the operations associated with the neural network model are performed at the lower precision to enable high inference performance. A computationally smaller part is performed at a relatively higher precision to preserve output quality. In addition to using FP16 for higher precision operations, other floating-point precisions may also be used, such as FP8, BF16 or TF32. Additionally, the majority of the neural network model **4050** is also in a reduced spatial dimension to provide fast inference performance by shuffling input pixels from the spatial (width, height) dimension to a depth or feature map channel dimension with no pixel information loss. The spatial dimension is shuffled back from the channel dimension during generating an output image.

[0435] Temporally amortized supersampling is performed by combining the current frame and the previous output frame warped with the current motion vectors. The neural network model **4050** determines the manner in which to combine the upscaled current frame **3912** and the history **3923**. In various embodiments, multiple different approaches are applied to preserve output quality. In one embodiment, a high precision combination of the upscaled current frame **3912** and the history is generated using 1×1 or 3×3 output convolution. In another embodiment, pixel prediction and high precision filtering of the upscaled image is performed to generate a high-quality upscaled image. The neural network model **4050** is used to generate input that is provided to the kernel prediction and filtering operations.

[0436] During training of the neural network model **4050**, both perceptual and temporal loss functions are optimized to enhance both the image quality and the temporal stability of the upsampling and anti-aliasing. In one embodiment, generalized training is sufficient to enable high quality output across a variety of games without requiring extensive per-game, per-upscale factor, or per-target resolution training.

[0437] FIG. 41 illustrates an implementation of a neural network model **4100**, according to an embodiment. The neural network model **4100** is an implementation of the neural network model **4050** of FIG. 40. In one embodiment, the neural network model **4100** is composed of three components: an input block **4108**, a feature extraction network **4110**, and an output block **4120**. Lower precision (e.g., Integer) operations are used for the majority of the neural network model to achieve fast inference performance. Output of the neural network model is generated using higher precision (e.g., floating-point) operations to enable the generation of high-quality output images. For example, the encoders (encoder block 1 through encoder block N), bottleneck block, and decoder blocks (decoder block 1 through decoder block N) in the feature extraction network **4110** are executed with relatively lower precision (e.g., INT8) compared to the output block **4120**, which is executed at a relatively higher precision (e.g., FP16). Utilizing lower precision in the feature extraction network **4110** significantly reduces the complexity of computation and improves memory bandwidth for fast inference performance. Utilizing higher precision in the output block **4120** enables the generation of output images having an image quality that is as good as, or in some cases better than images that are natively rendered at the target resolution. As noted above, other precisions or data types in addition to INT8 and FP16 can be used, such as but not limited to INT4 for lower precision operations and BF 16 or TF32 for higher precision operations.

[0438] The input block **4108** receives, as input, history data **4102**, velocity data **4104**, the current frame **4106**, and a jitter offset **4107** for the camera. The history data **4102** includes previously generated output. The previously generated output includes at least the immediate previous frame (frame N-1), which is warped using the velocity data **4104** to align the frame with the current frame **4106** for temporal accumulation. In various embodiments, in addition to the previous frame, the history data **4102** can also include one or more additional frames of previous generated output (e.g., frame N-2, etc.), which can also be provided as input to the feature extraction network **4110**. The jitter offset **4107** is the camera offset that is applied to jitter the scene, with different jitter values being used for successive frames. The jitter offset **4107**, in one embodiment, is a sub-pixel offset. The input block generates both lower and higher precision tensors. Lower precision tensors are provided to the feature extraction network **4110**. Higher precision tensors are provided to the output block **4120**. Further details on the input block **4108** are shown in FIG. 42.

[0439] The feature extraction network **4110** is built upon a U-shaped network architecture, such as, for example, the U-net architecture. The feature extraction network **4110** differs from the conventional U-net architecture in that the feature extraction network **4110** includes an asymmetric structure in the encoder **4112** and decoder **4116**. The encoder **4112** of the feature extraction network **4110** includes a series of encoder blocks that downsample the spatial dimension of an input tensor while increasing the number of channels (depth or feature maps) until the production of a latent representation **4114** at a bottleneck block in the middle of the network. The latent representation **4114** is the abstract multi-dimensional space that encodes the meaningful features of the input data. The decoder blocks of the decoder **4116** reverse this process by upsampling spatial dimension and decreasing the number of channels. The encoder blocks have a skip connection to a corresponding decoder block, which enables high-frequency details to be relayed between the encoder **4112** and the decoder **4116**. Output from encoder block 1 is provided to decoder block 2 to be processed in conjunction with output from decoder block 3. Output from encoder block 2 is provided to encoder block 3 to be processed in conjunction with output from the previous decoder block in the network. The input for encoder block N is provided to decoder block N. Decoder block 1, the final decoder block, receives input from the input block **4108** and decoder block 2. The decoder **4116**, from decoder block 1, provides data to the output block **4120** in either a higher precision format or a lower precision format depending on the implementation approach used for the output block **4120**. Further details on the output block are shown in FIG. 43A and FIG. 43B.

[0440] FIG. 42 illustrates further details for the input block **4108** of the neural network model **4100**, according to embodiments. The input block **4108** receives input including history data **4102**,

velocity data **4104**, the current frame **4106**, and the jitter offset **4107**. The input block **4108** includes a warping unit **4202** to warp the previous output within the history data **4102** using motion vectors within the velocity data **4104**. The input block **4108** also include an upscaling unit **4203** to upscale the current frame **4106**. In one embodiment, the upscaling filter applied by the upscaling unit **4203** is an adaptive filter that adjusts the upscaling based on the jitter offset **4107**. A space to channel/depth shuffle unit **4204** shuffles pixels from the spatial dimension (width, height) to a channel (e.g., feature map) or depth dimension, which facilitates high performance inferencing via reduction of numerical precision and spatial dimension during feature extraction. For example, for an input image having (channel, height, width) pixels of data in the spatial dimension, the pixel data can be shuffled to

$$[00003](\text{channel} \times r^2, \frac{\text{height}}{r}, \frac{\text{width}}{r}),$$

which reduces the spatial dimension in which the feature extraction is performed, which improves the performance of the feature extraction network **4110**. The input block **4108** generates both lower precision (e.g., INT8) and higher precision (e.g., FP16) tensors. The lower precision tensors are provided as input to the feature extraction network **4110**, while the higher precision tensors are passed to the output block **4120**, **4320A-4320B**. The input block **4108** also include an optional convolution/activation layer **4206** that can be applied before data is output to the feature extraction network.

[0441] FIG. **43A-43B** illustrates output block variants for the neural network model, according to embodiments. FIG. **43A** illustrates a decoder block **4320** and a variant of the output block **4320A** that is configured to perform direct generation of pixel data for the output image. FIG. **43B** illustrates a decoder block **4320** and a variant of the output block **4320B** that is configured as a kernel prediction network that applies kernel pixel prediction and filtering to generate the output image. In FIG. **43A-43B**, a decoder block **4320** (decoder block **1**) is shown as an example. While each encoder block of the encoder **4112** includes a downsample block and one or more convolution/activation layers that facilitate feature extraction, each decoder block of the decoder **4116** includes an upsample block **4322** to increase spatial dimension and one or more convolution/activation layer(s) **4324**, **4326** to restore features. Decoder block **1** receives data from decoder block **2** as well as skip connection data from the input block. For the output block **4320A-4320B**, two different approaches can be taken to preserve quality with higher precision. One embodiment provides an output block **4320A**, as shown in FIG. **43A**, which configures the neural network model **4100** to operate as a direct reconstruction network. One embodiment provides an output block **4320B**, as shown in FIG. **43B**, which configures the neural network model **4100** to operate as a kernel prediction network.

[0442] For output block **4320A** of FIG. **43A**, data from the input block **4108** and the feature extraction network **4110** is combined using a 1×1 or 3×3 output convolution layer **4330** to directly generate data for the output image. The output convolution layer **4330** receives, as input, higher precision (e.g., FP16) output from the convolution/activation layer(s) **4326** of the final decoder block **4320**, as well as higher precision input from the input block **4108**. Data generated by the output convolution layer **4330** is provided to the depth/channel to space shuffle unit **4332**, which shuffles the data back into the spatial dimension to generate an output image **4340**. The output image **4340** can be output via a display or further post-processed before output via the display.

[0443] For output block **4320B** of FIG. **43B**, kernel prediction and filtering are performed. Instead of directly generating an output image, per-pixel kernel values (e.g., weights) are predicted by a kernel prediction layer **4334**. Lower precision (INT8) tensors are output by the decoder block **4320** for use by the kernel prediction layer **4334**, which uses the lower precision tensors in combination with the higher precision tensors provided by the input block **4108**. The depth/channel to space shuffle unit **4332** shuffles frame data back into the spatial dimension to generate an intermediate output image. The intermediate output image is then filtered by the filter/blend layer **4346** using the per-pixel kernel values generated by the kernel prediction layer **4334** and blending with the

previous output using blend weights generated by the kernel prediction layer **4334**. The filtered and blended image is then provided as the output image **4340**.

[0444] FIG. **44** illustrates a method **4400** to perform temporally amortized supersampling. The method **4400** includes to receive, at an input block of a neural network model described herein (e.g., neural network model **4050**), history data, velocity data, and current frame data (**4402**). The history data includes one or more previously generated frames. The velocity data includes renderer generated motion vectors that are used to align the one or more previously generated frames with the pixel data of the current frame. The current frame data includes a frame of a 3D graphics program, such as a 3D game application, that is output by a raster and lighting stage of the render pipeline of the graphics processor. In one embodiment, the current frame is an upscaled frame that has been upscaled by an upscaling filter from an initial rendering resolution to a target resolution. In one embodiment, the current frame is upscaled to the target resolution during pre-processing. The input block provides output at multiple precisions, with a first set of output being provided to the output block at high precision and a second set of output being provided to the feature extraction network at a relatively lower precision. In one embodiment, the first set of output is provided as floating-point data (e.g., FP16, BF16), while the second set of output is provided as integer data (e.g., INT4, INT8).

[0445] The neural network model can then pre-process the history data, velocity data, and current frame data at the input block and provide the pre-processed data to a feature extraction network (**4404**). The pre-processed data that is provided to the feature extraction network includes aligned history data and current frame data. The history data is warped using the velocity data to generate warped history data. The warped history data is then aligned with the current frame data to generate aligned history data. The aligned history data provides additional sample data that can be used to generate a supersampled anti-aliased output image via temporal accumulation. In one embodiment, the pre-processing includes upscaling the current frame data from the resolution output by the raster and lighting stage to the target resolution.

[0446] The neural network model processes the pre-processed data at the feature extraction network via one or more encoder stages and one or more decoder stages (**4406**). The encoder stages reduce the spatial resolution of the input data and extracts the most salient features within the input data. The spatial resolution is then expanded via the decoder stages to generate tensor data that is used to process the current upscaled frame in view of the aligned history to generate a high quality upscaled frame that has an image quality that is, at the least, equal to an image that is natively rendered at the target resolution. The features extracted are used to determine an optimized combination of the current and previous frames during temporal accumulation.

[0447] The neural network model can then generate an output frame via an output block of the neural network model via temporal accumulation using direct reconstruction or kernel prediction (**4408**). The output frame is an anti-aliased image that has a higher resolution than the rendering resolution of the render pipeline, with additionally generated pixels to enhance the image quality beyond that of the originally upscaled image. In one embodiment, the neural network model is configured as a direct reconstruction network which, via one or more convolution layers, generates a high-quality output image for display. When configured as a direct reconstruction network, the feature extraction network provides higher precision tensors (e.g., FP16, BF16) as input to the output block. The output block uses the higher precision output from the feature extraction network in combination with the higher precision output from the input block to generate the output image. In one embodiment, the neural network model is configured as a kernel prediction network that generates per-pixel kernel values that applied to a high-precision filter. When configured as a kernel prediction network, the feature extraction network provides lower precision tensors (e.g., INT4, INT8, FP8) to the output block. The output block uses the lower precision output from the feature extraction network in combination with the higher precision output from the input block to predict the pre-pixel kernels/blend weights used to filter the upscaled input and blend the filtered

input with the previous output.

[0448] FIG. **45** illustrates exemplary rendering performance comparisons for multiple rendering techniques described herein. Rendering time for a low-quality rendering **4505**, for example, at 1080p resolution, is significantly lower than the rendering time for a high-quality rendering **4501**, for example, at 4K resolution. Traditional upscaling **4504** (TAA Upsampling, Temporal Super Resolution, FidelityFX Super Resolution) renders frames at low resolution and the low-resolution image is upsampled to the target display resolution to achieve performance boost and potentially an image quality improvement over low-quality rendering **4505**.

[0449] One implementation of temporally amortized supersampling using a mixed precision convolutional neural network is X.sup.e SS provided by Intel® Incorporated. X.sup.e SS can be performed on hardware that includes a matrix accelerator (e.g., tensor accelerator **2723**) via the use of Intel X.sup.e Matrix Extensions (XMX). Rendering via X.sup.e SS+XMX **4502** can produce an image that is significantly higher quality than low-quality rendering **4505** or traditional upscaling **4504** and with significantly lower rendering times than high-quality rendering **4501** at native 4K resolutions. Rendering via X.sup.e SS+DP4a **4503** replaces XMX with a dot product instruction (DP4a) that can be executed by a variety of graphics processor architectures from a variety of vendors and results in a high-quality image and a rendering time that is still significantly lower than high-quality rendering **4501** at native 4K resolutions. In one embodiment, X.sup.e SS+XMX **4502** is performed using direct reconstruction via output block **4320A** of FIG. **43A**, while X.sup.e SS+DP4a **4503** is performed using kernel prediction and filtering via output block **4320B** of FIG. **43B**.

Motion Vector Refinement Via Learned Residual Optical Flow

[0450] The temporally amortized supersampling techniques described herein can be configured make use of motion vectors provided by the 3D rendering engine of a 3D application that is processed on a graphics processing architecture described herein. A rendered 3D scene has an associated motion field that reflects the 3D motion of the objects within the scene. An optical flow field is a 2D projection of the motion field, such that a 2D optical flow vector is representative of the 3D motion vector when translated into a 2D image. Pixel-by-pixel optical flow vectors may be substituted with block-based motion vectors, which provide an estimate or approximation of the optical flow between frames. The motion vectors are used during temporally amortized supersampling to warp and align a previous frame with a current frame. However, motion vectors may not be supplied for some portions of a frame. For example, animated textures, mirrored/reflected objects, and/or moving objects without motion information may reside within a frame. Motion vectors may not be provided for those regions within the frame.

[0451] Described herein is a technique to refine the motion vectors used to warp previous frames via learned residual optical flow. A residual network is trained to predict a set of residual motion vectors based on an existing set of motion vectors, an upsampled current frame, and a previous frame. The residual motion vectors provide additional motion data for portions of the frame for which motion vectors are not provided. The set of residual motion vectors are combined with the existing set of motion vectors to generate a set of refined motion vectors. The refined motion vectors can be used to enable a higher quality transformation (e.g., warping) of the previous frame.

[0452] FIG. **46** illustrates data flow for image reconstruction and upscaling, according to an embodiment. In one embodiment, the data flow for image reconstruction and upscaling is implemented using the neural network model **4100** of FIG. **41**, as executed using graphics and/or compute accelerator hardware described herein. A current frame **4602** that is rendered below a target resolution is received for processing. Depth data **4613**, jittered samples **4614**, and motion vectors **4612** for the current frame **4602** are provided as input to circuitry **4616** that is configured to upsample the motion vectors. The motion vectors **4612** can be generated by and received from a game engine. For scenarios in which a game engine is not configured to provide motion vectors, the graphics processor may be configured to generate at least a partial or sparse set of motion

vectors using video encode hardware (e.g., within video codec engine **1606**). Color data for the current frame **4602** is provided to circuitry **4604** that is configured to upsample the color data to the target resolution. The upsampled motion vectors and a previous output frame **4622** are provided to circuitry **4618** that is configured to warp the previous output frame **4622** to generate a warped previous output frame **4620**. The warped previous output frame **4620** and the upsampled color data are provided to an image reconstruction network **4606** described herein. The image reconstruction network **4606** is configured to either 1) directly reconstruct an anti-aliased image at the target resolution or 2) predict per-pixel kernels/blend weights to use to blend the upsampled color data and the warped previous output frame. During training operations, a reference frame **4608** that is rendered at the target resolution is used to generate an output loss **4610**. Stochastic gradient descent and backpropagation may be used to minimize the output loss **4610** during training.

[0453] FIG. **47** illustrates an extended flow that includes a residual motion refinement network **4724** and warp loss optimization, according to an embodiment. The residual motion refinement network **4724** is a neural network that is trained to predict residual motion vectors for portions of an image without motion vectors based on the motion vectors that are provided for other parts of the image. A warp loss **4722** is optimized during joint training of the residual motion refinement network **4724** and the image reconstruction network **4606**. In this extended flow, the previous output frame **4622**, the upsampled motion vectors generated by circuitry **4616**, and the upsampled color data generated by circuitry **4604**, are provided to the residual motion refinement network **4724** that is executed via the graphics and/or compute acceleration hardware described herein. The residual motion refinement network **4724** is trained to predict a set of residual motion vectors **4726** that provide additional motion data for portions of the frame for which motion vectors are not provided or available. For example, in some configurations, motion vectors are not generated by the game engine for animated textures and mirrored/reflected objects. Depending on the engine, other types of moving objects may not have associated motion information.

[0454] The graphics processor includes circuitry **4702** that is configured to combine the upsampled motion vectors and the residual motion vectors **4726** to generate refined motion vectors **4712**. Circuitry **4718** is configured to use the refined motion vectors **4712** to warp the previous output frame **4622** to generate a warped previous output frame **4720**. The warped previous output frame **4720** is provided to the image reconstruction network **4606**, which generates an output frame based on the warped previous output frame **4720** and the upsampled color data for the current frame **4602**. During training, circuitry **4718** is configured to output warp loss **4722** that is used to train the residual motion refinement network **4724**.

[0455] In some embodiments, the residual motion refinement network **4724** can be used to enable temporally amortized supersampling in scenarios where motion vectors are not provided by the game engine. In one embodiment, the trained residual motion refinement network **4724** may be configured to generate at least a partial set of motion vectors based on the upsampled color data generated by circuitry **4604** and the previous output frame **4622**. The residual motion refinement network **4724** may also be configured to refine a set of sparse motion vectors that are generated by processing the upsampled color data and previous output frame **4622** via video encoder hardware. The video encoder hardware (e.g., video codec engine **1606**) may be configured to rapidly generate a sparse set of motion vectors that are less detailed than the motion vectors generally used during video encoding. Those sparse motion vectors may be refined using the residual motion refinement network **4724** to generate refined motion vectors **4712** that may be used by circuitry **4718** to generate the warped previous output frame **4720**. The refined motion vectors **4712** can be used to generate more precisely warped frames, which improves the sharpness of the final image.

[0456] The residual motion refinement network **4724** may be implemented using a variety of neural network architectures. In one embodiment, the residual motion refinement network **4724** is implemented as a custom convolutional neural network. In one embodiment, the residual motion refinement network **4724** can be implemented using a version of an existing neural network

architecture including but not limited to the Res-Net or U-Net architectures. In one embodiment, the residual motion refinement network **4724** is a version of the neural network model **4100** of FIG. **41**.

[0457] FIG. **48** illustrates a method **4800** of motion vector refinement via a residual motion refinement network, according to an embodiment. The method **4800** can be implemented in part by temporally amortized supersampling logic executed by and/or associated with a graphics processor having an architecture described herein, such as but not limited to the GPGPU **2720** of the data processing system **2700** of FIG. **27**. Operations of the method may be accelerated using matrix processing hardware described herein. The method **4800** enables operations to generate additional motion vectors to improve the quality of previous frame warping and alignment during temporally amortized supersampling, which enables the generation of a higher quality output image.

[0458] In one embodiment, the method **4800** includes to perform operations to train a neural network (e.g., residual motion refinement network **4724**) to generate residual motion vectors based on an existing set of motion vectors, a current frame, and a previous frame (**4802**). The neural network may be a second neural network that is trained jointly with a first neural network (e.g., image reconstruction network **4606**). The training can be performed to optimize or minimize a warp loss that is generated by image warping logic. The image warping logic can generate the warp loss based on warping and alignment operations that are performed to warp the previous frame and align the warped previous frame with the current frame. The training process enables the neural network to predict residual motion vectors for portions of the scene for which motion vectors are not provided. The training process can be performed prior to the deployment of neural network. The neural network can be deployed as a portion of a graphics driver, software development kit, or runtime library to enable or facilitate temporally amortized supersampling as described herein.

[0459] During rendering of a current frame of a rendered scene, temporally amortized supersampling logic can provide the current frame, the previous frame, and a first set of motion vectors to the neural network (**4804**). The temporally amortized supersampling logic can then generate, via the neural network, a second set of motion vectors (**4806**). The second set of motion vectors augment the first set of motion vectors by providing motion information for portions of the scene that are not covered by the first set of motion vectors, such as but not limited to animated textures or reflected objects. The temporally amortized supersampling logic can then combine the second set of motion vectors with the first set of motion vectors to generate a combined set of motion vectors (**4808**). The temporally amortized supersampling can then warp the previous frame via the combined set of motion vectors (**4810**). In one embodiment, the previous frame can be warped to generate history data, which in one embodiment can include multiple warped previous frames. The warped history data can be aligned with the current frame and provided as input to the image reconstruction network.

Additional Exemplary Computing Device

[0460] FIG. **49** is a block diagram of a computing device **4900** including a graphics processor **4904**, according to an embodiment. Versions of the computing device **4900** may be or be included within a communication device such as a set-top box (e.g., Internet-based cable television set-top boxes, etc.), global positioning system (GPS)-based devices, etc. The computing device **4900** may also be or be included within mobile computing devices such as cellular phones, smartphones, personal digital assistants (PDAs), tablet computers, laptop computers, e-readers, smart televisions, television platforms, wearable devices (e.g., glasses, watches, bracelets, smartcards, jewelry, clothing items, etc.), media players, etc. For example, in one embodiment, the computing device **4900** includes a mobile computing device employing an integrated circuit (“IC”), such as system on a chip (“SoC” or “SOC”), integrating various hardware and/or software components of computing device **4900** on a single chip. The computing device **4900** can be a computing device including components illustrated in the data processing system **2700** as in of FIG. **27**.

[0461] The computing device **4900** includes a graphics processor **4904**. The graphics processor

4904 represents any graphics processor described herein. In one embodiment, the graphics processor **4904** includes a cache **4914**, which can be a single cache or divided into multiple segments of cache memory, including but not limited to any number of L1, L2, L3, or L4 caches, render caches, depth caches, sampler caches, and/or shader unit caches. In one embodiment the cache **4914** may be a last level cache that is shared with the application processor **4906**.

[0462] In one embodiment the graphics processor **4904** includes a graphics microcontroller that implements control and scheduling logic for the graphics processor. The control and scheduling logic can be firmware executed by the graphics microcontroller **4915**. The firmware may be loaded at boot by the graphics driver logic **4922**. The firmware may also be programmed to an electronically erasable programmable read only memory or loaded from a flash memory device within the graphics microcontroller **4915**. The firmware may enable a GPU OS **4916** that includes device management logic **4917** and driver logic **4918**, and a scheduler **4919**. The GPU OS **4916** may also include a graphics memory manager **4920** that can supplement or replace the graphics memory manager **4921** within the graphics driver logic **4922**.

[0463] The graphics processor **4904** also includes a GPGPU engine **4944** that includes one or more graphics engine(s), graphics processor cores, and other graphics execution resources as described herein. Such graphics execution resources can be presented in the forms including but not limited to execution units, shader engines, fragment processors, vertex processors, streaming multiprocessors, graphics processor clusters, or any collection of computing resources suitable for the processing of graphics resources or image resources or performing general purpose computational operations in a heterogeneous processor. The processing resources of the GPGPU engine **4944** can be included within multiple tiles of hardware logic connected to a substrate, as illustrated in FIG. 24B-24D. The GPGPU engine **4944** can include GPU tiles **4945** that include graphics processing and execution resources, caches, samplers, etc. The GPU tiles **4945** may also include local volatile memory or can be coupled with one or more memory tiles, such as memory tiles **1626A-1626D** as in FIG. 16B-16C.

[0464] The GPGPU engine **4944** can also include and one or more special tiles **4946** that include, for example, a non-volatile memory tile **4956**, a network processor tile **4957**, and/or a general-purpose compute tile **4958**. The GPGPU engine **4944** also includes a matrix multiply accelerator **4960**. The general-purpose compute tile **4958** may also include logic to accelerate matrix multiplication operations. The non-volatile memory tile **4956** can include non-volatile memory cells and controller logic. The controller logic of the non-volatile memory tile **4956** may be managed by one of device management logic **4917** or driver logic **4918**. The network processor tile **4957** can include network processing resources that are coupled to a physical interface within the input/output (I/O) sources **4910** of the computing device **4900**. The network processor tile **4957** may be managed by one or more of device management logic **4917** or driver logic **4918**.

[0465] In one embodiment, the matrix multiply accelerator **4960** is a modular scalable sparse matrix multiply accelerator. The matrix multiply accelerator **4960** can include multiple processing paths, with each processing path including multiple pipeline stages. Each processing path can execute a separate instruction. In various embodiments, the matrix multiply accelerator **4960** can have architectural features of any one of more of the matrix multiply accelerators described herein. For example, in one embodiment, the matrix multiply accelerator **4960** is a systolic array **3000** that is configurable to operate with a multiple of four number of logical stages (e.g., four, eight, twelve, sixteen, etc.). In one embodiment the matrix multiply accelerator **4960** includes one or more instances of a two-path matrix multiply accelerator **3100** with a four-stage pipeline or a four-path matrix multiply accelerator **3200** with a two-stage pipeline. In one embodiment the matrix multiply accelerator **4960** includes processing elements configured as a scalable sparse matrix multiply accelerator. The matrix multiply accelerator **4960** can be used to accelerate matrix operations performed via XMX extensions, or another compute library that facilitates the acceleration of matrix compute operations. For example, the matrix multiply accelerator **4960** can perform tensor

computations for training or inference of the neural network models **4050**, **4100** described herein. [0466] As illustrated, in one embodiment, and in addition to the graphics processor **4904**, the computing device **4900** may further include any number and type of hardware components and/or software components, including, but not limited to an application processor **4906**, memory **4908**, and input/output (I/O) sources **4910**. The application processor **4906** can interact with a hardware graphics pipeline to share graphics pipeline functionality. Processed data is stored in a buffer in the hardware graphics pipeline and state information is stored in memory **4908**. The resulting data can be transferred to a display controller for output via a display device. The display device may be of various types, such as Cathode Ray Tube (CRT), Thin Film Transistor (TFT), Liquid Crystal Display (LCD), Organic Light Emitting Diode (OLED) array, etc., and may be configured to display information to a user via a graphical user interface.

[0467] The application processor **4906** can include one or processors, such as processor(s) **102** of FIG. **1** and may be the central processing unit (CPU) that is used at least in part to execute an operating system (OS) **4902** for the computing device **4900**. The OS **4902** can serve as an interface between hardware and/or physical resources of the computing device **4900** and one or more users. The OS **4902** can include driver logic for various hardware devices in the computing device **4900**. The driver logic can include graphics driver logic **4922**, which can include the user mode graphics driver **2326** and/or kernel mode graphics driver **2329** of FIG. **23**. The graphics driver logic can include a graphics memory manager **4921** to manage a virtual memory address space for the graphics processor **4904**. The graphics memory manager **4921** can facilitate a unified virtual address space that may be accessed by the application processor **4906** and the graphics processor **4904**.

[0468] It is contemplated that in some embodiments the graphics processor **4904** may exist as part of the application processor **4906** (such as part of a physical CPU package) in which case, at least a portion of the memory **4908** may be shared by the application processor **4906** and graphics processor **4904**, although at least a portion of the memory **4908** may be exclusive to the graphics processor **4904**, or the graphics processor **4904** may have a separate store of memory. The memory **4908** may comprise a pre-allocated region of a buffer (e.g., framebuffer); however, it should be understood by one of ordinary skill in the art that the embodiments are not so limited, and that any memory accessible to the lower graphics pipeline may be used. The memory **4908** may include various forms of random-access memory (RAM) (e.g., SDRAM, SRAM, etc.) comprising an application that makes use of the graphics processor **4904** to render a desktop or 3D graphics scene. A memory controller hub, such as memory controller **1416** of FIG. **14**, may access data in the memory **4908** and forward it to graphics processor **4904** for graphics pipeline processing. The memory **4908** may be made available to other components within the computing device **4900**. For example, any data (e.g., input graphics data) received from various I/O sources **4910** of the computing device **4900** can be temporarily queued into memory **4908** prior to their being operated upon by one or more processor(s) (e.g., application processor **4906**) in the implementation of a software program or application. Similarly, data that a software program determines should be sent from the computing device **4900** to an outside entity through one of the computing system interfaces, or stored into an internal storage element, is often temporarily queued in memory **4908** prior to its being transmitted or stored.

[0469] The I/O sources can include devices such as touchscreens, touch panels, touch pads, virtual or regular keyboards, virtual or regular mice, ports, connectors, network devices, or the like, and can attach via a platform controller hub **1430** as referenced in FIG. **14**. Additionally, the I/O sources **4910** may include one or more I/O devices that are implemented for transferring data to and/or from the computing device **4900** (e.g., a networking adapter); or, for a large-scale non-volatile storage within the computing device **4900** (e.g., SSD/HDD). User input devices, including alphanumeric and other keys, may be used to communicate information and command selections to graphics processor **4904**. Another type of user input device is cursor control, such as a mouse, a

trackball, a touchscreen, a touchpad, or cursor direction keys to communicate direction information and command selections to GPU and to control cursor movement on the display device. Camera and microphone arrays of the computing device **4900** may be employed to observe gestures, record audio and video and to receive and transmit visual and audio commands.

[0470] The I/O sources **4910** can include one or more network interfaces. The network interfaces may include associated network processing logic and/or be coupled with the network processor tile **4957**. The one or more network interface can provide access to a LAN, a wide area network (WAN), a metropolitan area network (MAN), a personal area network (PAN), Bluetooth, a cloud network, a cellular or mobile network (e.g., 3.sup.rd Generation (3G), 4.sup.th Generation (4G), 5.sup.th Generation (5G), etc.), an intranet, the Internet, etc. Network interface(s) may include, for example, a wireless network interface having one or more antenna (e). Network interface(s) may also include, for example, a wired network interface to communicate with remote devices via network cable, which may be, for example, an Ethernet cable, a coaxial cable, a fiber optic cable, a serial cable, or a parallel cable.

[0471] Network interface(s) may provide access to a LAN, for example, by conforming to IEEE 802.11 standards, and/or the wireless network interface may provide access to a personal area network, for example, by conforming to Bluetooth standards. Other wireless network interfaces and/or protocols, including previous and subsequent versions of the standards, may also be supported. In addition to, or instead of, communication via the wireless LAN standards, network interface(s) may provide wireless communication using, for example, Time Division, Multiple Access (TDM A) protocols, Global Systems for Mobile Communications (GSM) protocols, Code Division, Multiple Access (CDMA) protocols, and/or any other type of wireless communications protocols.

[0472] It is to be appreciated that a lesser or more equipped system than the example described above may be preferred for certain implementations. Therefore, the configuration of the computing devices described herein may vary from implementation to implementation depending upon numerous factors, such as price constraints, performance requirements, technological improvements, or other circumstances. Examples include (without limitation) a mobile device, a personal digital assistant, a mobile computing device, a smartphone, a cellular telephone, a handset, a one-way pager, a two-way pager, a messaging device, a computer, a personal computer (PC), a desktop computer, a laptop computer, a notebook computer, a handheld computer, a tablet computer, a server, a server array or server farm, a web server, a network server, an Internet server, a work station, a mini-computer, a main frame computer, a supercomputer, a network appliance, a web appliance, a distributed computing system, multiprocessor systems, processor-based systems, consumer electronics, programmable consumer electronics, television, digital television, set top box, wireless access point, base station, subscriber station, mobile subscriber center, radio network controller, router, hub, gateway, bridge, switch, machine, or combinations thereof.

[0473] The techniques described herein relate to a graphics processor including a set of processing resources configured to perform a supersampling anti-aliasing operation via a mixed precision convolutional neural network. The set of processing resources include circuitry configured to receive, at an input block of a neural network model, a set of data including previous frame data, current frame data, and velocity data. The previous frame data includes one or more previously generated output frames, the current frame data includes output of a raster and lighting stage of a render pipeline, and the velocity data includes motion vectors generated by the render pipeline. The circuitry can pre-process the set of data to generate pre-processed data. To pre-process the set of data includes to generate residual motion vectors via a second neural network model. The residual motion vectors are generated for areas of the previous frame data that do not have associated motion vectors that indicate how to warp those regions. The circuitry can then provide first pre-processed data to a feature extraction network of the first neural network model and second-processed data to an output block of the first neural network model, the first pre-processed data at a

first precision and the second pre-processed data at a second precision that is higher than the first precision. The circuitry can then process the pre-processed data at the feature extraction network via one or more encoder stages and one or more decoder stages, output tensor data from the feature extraction network to the output block, and generate an output frame via the output block based on the second pre-processed data from the input block and the tensor data output from the feature extraction network. The output frame is an anti-aliased frame. The circuitry can include a matrix accelerator that is configured to perform matrix operations for the neural network model. The matrix accelerator can include a systolic array.

[0474] In one embodiment, to generate the pre-processed data includes to generate the residual motion vectors via the second neural network based on the current frame data, previous frame data, and the motion vectors, combine the motion vectors and the residual motion vectors to create refined motion vectors, warp the previous frame data based on the refined motion vectors to generate warped history data, align the warped history data with the current frame data to generate aligned history data, and shuffle the aligned history data and current frame data from a spatial dimension to a channel dimension. The channel dimension can include a depth channel or multiple feature map channels. To generate an output frame via the output block includes to generate the output frame via one or more convolution layers of the output block. The generate an output frame via the output block includes to predict a set of per-pixel kernel values and blend weights, filter the current frame data via the per-pixel kernel values, and blend the aligned history data with filtered current frame data. The circuitry is configured to upscale the output of the raster and lighting stage from a first resolution to a second resolution that is higher than the first resolution before the current frame data is provided to the input block of the first neural network model. In one embodiment, to generate the pre-processed data includes to upscale the output of the raster and lighting stage from a first resolution to a second resolution that is higher than the first resolution. In one embodiment, to generate the pre-processed data additionally includes to apply a convolution and an activation function to shuffled aligned history data and current frame data.

[0475] An additional embodiment provides a method to perform the operations of the graphics processor described above. A further embodiment provides a non-transitory machine-readable medium that stores instructions to perform the operations of the graphics processor described above. A further embodiment provides a data processing system including the graphics processor described above.

[0476] The foregoing description and drawings are to be regarded in an illustrative rather than a restrictive sense. Persons skilled in the art will understand that various modifications and changes may be made to the embodiments described herein without departing from the broader spirit and scope of the features set forth in the appended claims.

Claims

1. A graphics processing unit, comprising: memory interface; and a graphics processing cluster including a plurality of streaming multiprocessors coupled via an interconnect, a streaming multiprocessor of the plurality of streaming multiprocessors including a tensor core configured to perform mixed precision matrix operations, wherein the streaming multiprocessor is configured to: receive a set of input data for a first frame used by the streaming multiprocessor to generate multiple additional frames, the set of input data including first color data and motion vector data; execute first matrix operations to upsample the first color data from a first resolution to a second resolution to generate upsampled color data; and execute second matrix operations to generate at least one of the multiple additional frames based at least in part on the upsampled color data and an optical flow.
2. The graphics processing unit of claim 1, wherein the interconnect is configured to enable exchange of data between the plurality of streaming multiprocessors.

3. The graphics processing unit of claim 1, wherein the set of input data for the first frame includes depth data and the streaming multiprocessor is configured to generate at least one of the multiple additional frames based at least in part on the depth data.
4. The graphics processing unit of claim 1, wherein the first color data includes sample data with a sub-pixel jitter.
5. The graphics processing unit of claim 1, wherein the multiple additional frames comprise at least a second frame.
6. The graphics processing unit of claim 1, wherein the optical flow corresponds to one or more residual motion vectors.
7. The graphics processing unit of claim 6, wherein the motion vector data is to be received from a renderer.
8. The graphics processing unit of claim 7, wherein the one or more residual motion vectors are to provide motion data for a portion of at least one frame for which motion vector data is not provided.
9. The graphics processing unit of claim 8, wherein the streaming multiprocessor is configured to execute third matrix operations to generate the one or more residual motion vectors based on the motion vector data and at least one frame.
10. The graphics processing unit of claim 9, wherein the first matrix operations are associated with a first neural network, the second matrix operations are associated with a second neural network, the third matrix operations are associated with a third neural network, and the third neural network is an image reconstruction network.
11. A method comprising: receiving, at a streaming multiprocessor of a graphics processing unit, a set of input data for a first frame to generate multiple additional frames, the set of input data including first color data and motion vector data; executing, using a tensor core of the streaming multiprocessor, first matrix operations to upsample the first color data from a first resolution to a second resolution to generate upsampled color data; and executing, using the tensor core, second matrix operations to generate at least one of the multiple additional frames based at least in part on the upsampled color data and an optical flow.
12. The method of claim 11, wherein the set of input data for the first frame includes depth data and the streaming multiprocessor is configured to generate at least one of the multiple additional frames based at least in part on the depth data.
13. The method of claim 11, wherein the first color data includes sample data with a sub-pixel jitter.
14. The method of claim 11, wherein the multiple additional frames comprise at least a second frame.
15. The method of claim 11, wherein the optical flow corresponds to one or more residual motion vectors, the motion vector data is to be received from a renderer, and the one or more residual motion vectors are to provide motion data for a portion of at least one frame for which motion vector data is not provided.
16. A graphics processing system, comprising: a memory device; and a graphics processing unit coupled with the memory device, the graphics processing unit including memory interface coupled with the memory device and a graphics processing cluster including a plurality of streaming multiprocessors coupled via an interconnect, wherein the interconnect is configured to enable exchange of data between the plurality of streaming multiprocessors, a streaming multiprocessor of the plurality of streaming multiprocessors including a tensor core configured to perform mixed precision matrix operations, wherein the streaming multiprocessor is configured to: receive a set of input data for a first frame used by the streaming multiprocessor to generate multiple additional frames, the set of input data including first color data and motion vector data; execute first matrix operations to upsample the first color data from a first resolution to a second resolution to generate upsampled color data; and execute second matrix operations to generate at least one of the multiple additional frames based at least in part on the upsampled color data and an optical flow.

- 17.** The graphics processing system of claim 16, wherein the set of input data for the first frame includes depth data and the streaming multiprocessor is configured to generate at least one of the multiple additional frames based at least in part on the depth data.
- 18.** The graphics processing system of claim 16, wherein the first color data includes sample data with a sub-pixel jitter.
- 19.** The graphics processing system of claim 16, wherein the multiple additional frames comprise at least a second frame.
- 20.** The graphics processing system of claim 16, wherein the optical flow corresponds to one or more residual motion vectors, the motion vector data is to be received from a renderer, and the one or more residual motion vectors are to provide motion data for a portion of at least one frame for which motion vector data is not provided.
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